

Handouts for S022 (Data Science in Education)

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Preface

This “book” is basically a set of handouts generated for a Data Science in Education course taught at the Harvard Graduate School of Education by Luke Miratrix. They are loosely organized, but the primary purpose of this “book” is to make all the handouts easy to access and find.

The handouts were generated in response to student questions, and aim to short-circuit staring at the possibly confusing world of Stack Overflow. They are curated in the sense that these activities tend to come up over and over in the course of fundamental data science work.

I hope you find this resource useful. Please send feedback to lmiratrix@g.harvard.edu.

A note on duplicated material

Some of this material is taken from a sister “textbook” for my MLM course (S043/Stat151). That textbook is [here](#). In particular, you will find entire chapters of that textbook replicated here.

About Statistical Programming

Statistical programming entails using a computer to analyze data in a way where you design and control the analysis and sometimes aspects of data generation (simulation). This is a step (or possibly several steps) beyond simply running built-in commands and interpreting output. It includes cleaning and transforming data to suit your needs, designing visualizations and plots to illustrate points you want to make, and conducting analyses that are tailor-made to your questions of interest.

S022 will make extensive use of the statistical software package R, which runs on both PCs and Macs. The software is [free and available online](#). R is straightforward to learn, but is sufficiently powerful and versatile to be useful for real projects that you might carry out after this course. It is used widely in many other statistics courses and also in research in such fields as education, psychology, economics, medical research, epidemiology, public health, and political science. See below for further argument as to why you should learn R.

We highly recommend using RStudio, which makes using R easier. RStudio is an Integrated Development Environment (IDE) that structures your experience, helps keep things organized, and offers multiple time-saving features to make your programming experience better. You might also consider R Markdown. R Markdown allows for generating documents with embedded R code and R output in a clean format, which can greatly help report generation.

We will teach R and the associated tools as part of this course. We assume no background knowledge with statistical programming or any other programming.

The R Mystique

Many people seem to believe that R is particularly technically challenging and difficult to master. This probably stems from its extreme flexibility; it is a fully functional programming language as well as a statistical analysis package. R can do things that many other software packages (I'm looking at you, Stata) essentially can't. These more involved things are frequently hard to do because they require you to think like a programmer rather than a data analyst. As a result, R is perceived as a “hard” language to use.

However, for straightforward, off-the-shelf analyses, R is arguably easier to learn and use in many ways than programs such as Stata or SAS. This is especially true when R is combined with RStudio for a better programming experience.

Don't believe me? Take S022, have us teach you some R, and find out for yourself.

Do I take S-022? A self-assessment

I received numerous inquiries as to whether a given background is enough to take S022. Let me elaborate at length. The stated prerequisite to this course is S040 or equivalent. Ideally, you would have taken S052 as well, but enrolling in it concurrently is just as good. Technically, if you have *just* taken S040 or equivalent, you could take this course. Having gone a bit beyond means this course is a bit more manageable with more of these foundational skills. More specifically, we greatly prefer you to have the following skills:

- You can calculate summary statistics and visualizations for data (the mean, standard deviation, scatterplots, histograms) using some sort of statistical software.
- You can load data into some sort of statistical software, use that software to fit a regression model, and then interpret the results of the model.
- You know how to include categorical covariates in a linear regression.
- You know what an interaction term in a linear regression model is, and ideally can add one to a model you are fitting.
- You can interpret confidence intervals and p -values.
- You can write down a regression equation and identify what the covariates and parameters are in a presented regression equation.

You will have an easier time if you have some of the following skills and experiences as well:

- You have some experience with doing things in R (even a little helps here).
- You know what logistic regression is, and know how to fit a logistic regression model using some sort of statistical software.
- You have some experience doing quantitative research with real data in almost any capacity.

If you have a strong mathematical background, or strong comfort with math, or if you have a strong computer programming background, or strong comfort with that, then you can likely get a lot out of this course even if you do not have some of the above skills and knowledge.

People from many different backgrounds take S022. In general, the more background experience you have, the easier the course. Even if you don't have as solid a background, you can

still get a lot out of this course if you are willing to sign on for an intense experience. But *please* don't be surprised if you sign up for an intense experience... and it is really intense!

To get a very rough sense of what S022 might feel like, take the quiz below.

A self-assessment quiz

Trying to figure out how S022 will be for you? Add up the points across the following categories:

Work Experience

- +0 I have no work experience with quantitative data, or
- +1 I have some work experience with quantitative data, or
- +2 I have substantial work experience with quantitative data

Stat Courses

- 2 I have no prior stat experience under my belt, or
- +1 I have taken a very intro stat course (e.g., S012) or have a little stat knowledge, or
- +3 I have taken a linear regression course (e.g., S030, S040), or
- +5 I have taken an intermediate or advanced stats course (e.g., S052 or beyond)
- +1 bonus if concurrently enrolling in S052
- +1 bonus if prior quantitative classes were comfortable for you

Programming

- +0 I have basically no experience doing computer programming, or
- +1 I have some experience doing computer programming, or
- +2 I have a lot of experience doing computer programming

R Skills

- +0 I have no real experience with R, or
- +1 I have a small bit of experience with R (e.g., ran scripts of it in S040), or
- +2 I have some experience with R (e.g., played around with scripts a bit in S040), or
- +3 I have substantial experience with R (e.g., write my own R code for my work)
- +1 bonus if learning R has been comfortable for you
- +1 bonus if you are comfortable with something like STATA

Math

- +0 I don't recall much math from my past education, or
- +1 I have taken (and somewhat remember) Calculus, or
- +2 I have taken classes beyond Calculus
- +1 bonus if classes were comfortable for you

Other

- +2 I am content with getting a B or taking the class SAT/UNSAT

Total the above

A VERY ROUGH recommendation is:

- < 4: **Danger!** Please email the instructor to talk about how this might go for you.
- 4-6: **It will be really hard!** You could take this course, but will likely find it will take much more time than a typical course. We would support you through this, but be warned that this could feel like a lot to take on.
- 7-8: **It will be hard!** There are lots of folks like you who are taking this course. The course will likely be a fair bit of work and could feel confusing/overwhelming at times. We would support you through this. At the end you will have learned a lot if you stick with it.
- 9-12: **It will probably feel like a normal course.** No reservations. You can either work a reasonable amount and learn a lot about data science, or dig deeper to really go far with the skills we cover.

13+: **It will probably be a cake-walk.** You will learn some concepts but not have to work particularly hard in this course. We would still love to have you!

Students have historically said this course teaches you a lot; the question is just whether you have the time to allocate for the course. This quiz helps assess the time.

Assignment Formatting Guidelines

The following describes our expectations on what turned in assignments should look like. Please read this document carefully before your first assignment. Note that many of these requirements will be automatically fulfilled if you use R Markdown.

Guiding Principle

Your submitted work should present your work clearly and without undue clutter, be easy to navigate, and adhere to basic publication norms.

The following are some specific details that further the guiding principle. It is not an exhaustive list.

Formatting

- Start each question on a new page.
- Provide at least a brief title for the question or sub-question along with the question number. E.g. “(a). Association between graduation rates and school type.” You can copy the entire prompt for your future reference if you want, but abbreviated prompts are fine with us!
- Use headings or other means to highlight problem titles (e.g., bold, italic, etc.).
- Make sure you answer all parts of each question, and do so under the proper labeled subpart. For each question or sub-question, include any R code, R output and answer.
- Use different fonts/formatting for your R code, R output and answers and use font formatting consistently throughout each assignment.
- Use single line spacing and normal 1-inch margins.
- Include page numbers.
- *Let people say of your work: “There is no bombast, no similes, flowers, digressions, or unnecessary descriptions. Everything tends directly to the catastrophe.”* —Horace Walpole, The Castle of Otranto

Numbers

- Round your final answers, not intermediary steps.

- Put a zero in front of a decimal place (e.g. 0.2 instead of .2). This is optional for bounded numbers (e.g., p-values or proportions).
- Round to the nearest meaningful digit. What this means is a little hard to say, and different people can have different standards. However, if you have a statistic with a standard error of 1, it's not meaningful to report decimals because the estimate simply isn't precise enough to estimate numbers that small. Similarly, if you're reporting average salaries, it's usually not meaningful to report past the hundreds place regardless of your precision, because tens of dollars are too small to matter. Because this is so imprecise, we'll give a lot of lee-way, but spurious precision will annoy. If you're not sure what this means in practice, feel free to ask a TF.
- Format your p-values! Round p-values to 3 places, and never report a p-value of 0. Instead say, for example, " $p < 0.001$ " or " $p < 10^{-r}$ " for some r .
- *"Numbers have life; they're not just symbols on paper."* —Shakuntala Devi

Plots

- Make sure your plots are well labelled, including title, axis, legends and any other elements you choose to include.
- Your plots should be self-explanatory.
- Include notes and captions as necessary.
- Try to make plots easier to compare when you have multiple plots. For example, it is nice to have the same X -axis bounds if giving two histograms.
- Do not include best fit lines unless you have some reason, e.g. a significant p -value or a scientific basis for understanding an association.
- *"Above all else show the data."* —Edward Tufte

Avoiding bias

- Use plural phrases, nouns or pronouns, e.g. "children and their toys" for "a child and his toy." You may use the singular "they" pronoun ("a child and their toy"), but be warned that broader academic communities are still in flux regarding this usage.
- Try to avoid biased forms of language concerning race, gender, disability and sexuality. Walden University has [a reasonable list of case studies to consider on this topic](#). The APA also has [a guide for writing about race](#).
- *"I have yet to see a piece of writing, political or non-political, that does not have a slant. All writing slants the way a writer leans, and no man is born perpendicular."* — E.B. White

Conciseness

- “*Brevity is the soul of wit.*” —Hamlet

Self Grading Instructions

To self-grade an assignment, go through the solutions and compare the answer to your own work.

Grade each piece as following

- “+” - 1 point, basically nailed it.
- “check” - Half point, did at least part of it, but had some problems.
- “X” - zero points, didn’t get it.

A “piece” is each part of a problem, usually given a letter. So if you had 1(a), 1(b), and 1(c), that would be three pieces for a maximum total of three points for Problem 1.

As you go through the solutions, see if you can figure out where you went wrong, if you did so. If you still don’t understand a problem, make a note as to problem number.

Then, for the next ready-for-class check-in you will be asked to enter your score (this score just helps me understand how folks are doing with these mini assignments), how you feel about it, and some further commentary. In your commentary please include a sentence or two on what was tripping you up (list the items), if you got things wrong. Also note whether things are not making sense, even with the solutions.

For reporting your score, **report total number of points, not percent correct.** For example, if there are two questions, with the first having 3 points and the second 2, you would report a score between 0 and 5.

We grade miniassignments on completion, meaning you will get full credit if you (1) do the mini-assignment and (2) self-grade the miniassignment (regardless of the score you give yourself as long as your work shows reasonable effort).

Acknowledgement:

This idea of self-grading was inspired by Cora Wigger. She used to have some stuff on twitter, but she appears to have removed her account.

1 Tidy Projects for Analysis in R

Organizing Projects for Research & Collaboration

S022 Final Projects

In addition to submitting the slides for your final S022 project, you will also submit a zipped project folder containing all of the data, code, and analysis used to create your presentation. Below, we outline some best practices for keeping your projects organized as they develop, with the ultimate goal of sharing a clean, replicable project to the teaching team at the end of the semester.

Organizational File Structure

Often, complex projects involve several scripts that take a variety inputs and generate an array of outputs. Ultimately, an intentional file structure that is specific to your project can be extremely helpful in keeping your code organized in a way that facilitates collaboration and replication. As a starting point, consider using something like the basic file structure below:

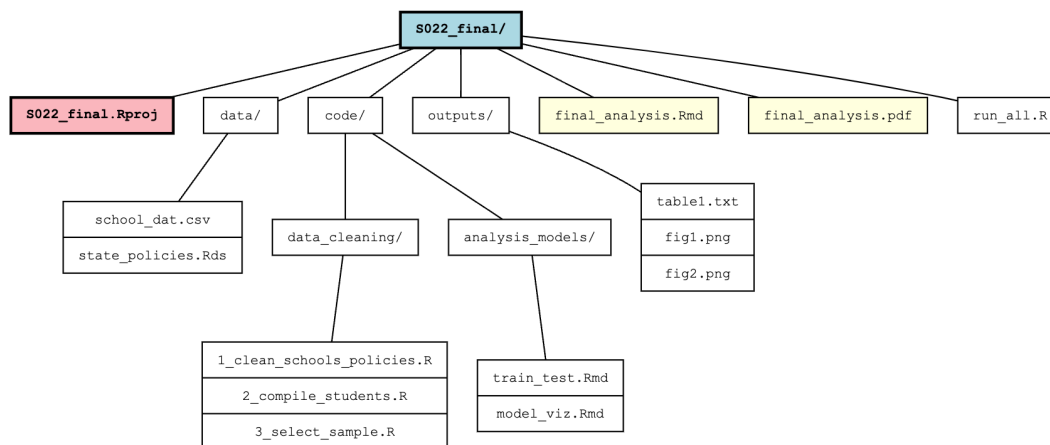


Figure 1.1: Figure 1: An organized research project

An organized project contains:

- All data files associated with the final analysis
- The cleaning scripts to prepare, merge, and process code for the final analysis
- All tables and outputs associated with the final analysis
- A markdown file (ex. `final_analysis.Rmd`) and rendered PDF (ex. `final_analysis.pdf`) report with the code and products for your research project
- An `.Rproj` file, which bundles the code together and manages relative file paths within the project.

The goal of your project's structure is to be clearly understood by collaborators or your future self.

Using R Projects

R Projects are an extremely useful tool for organizing code and keeping the variables and data in your environment contained across separate research projects. You can read more about Projects in [R for Data Science \(2e\), Chapter 6](#).

The main thing to know is that R Projects are one of the most effective ways to format code for collaborators to share across different computers and file locations.

The `here` package

In working with R files, you have probably come across the `getwd()` and `setwd()` functions. However, R Projects obviate the need for manually setting a working directory by identifying the location of your R project (and the associated `.Rproj` file) as the root directory that all of your code will reference. The `here` package leverages this feature by automatically detecting your project root, allowing you to build file paths that will work consistently across different computers.

For example, the code below will find the project root associated with this Quarto handout!

```
library(here)
here::here()
```

```
[1] "/Users/lmiratrix/Dropbox/S022 S2025/S022-textbook"
```

Relative paths with `here()`

By centering your code around the `.Rproj` file, you can write code that seamlessly runs on multiple collaborators' computers, regardless of the Project's location on those computers.

Good:

Uses the `here()` function to set relative file paths to locations in the project directory. Note that this example creates separate sub-directories for raw and cleaned data. A nice way to take it easy on your collaborators and your future self!

```
school_dat <- read_csv(here::here("data/raw_data/school_data_2025.csv"))
...
# Cleaning code here
saveRDS(clean_dat, here::here("data/clean_data/clean_school_data.Rds"))
```

Bad:

Imports data from far-flung locations, uses absolute file paths that will break on others' computers or if the project directory changes.

```
school_dat <- read_csv("/Users/noradelaney/Downloads/school_data_2025.csv")
...
# Cleaning code here
saveRDS(clean_dat,
        "/Users/noradelaney/Documents/s022_final_proj/data/clean_school_data.Rds")
```


Part I

Getting Started

2 Getting Started with R

To get started with R, first download and install R and RStudio (the IDE, or integrated development environment, that makes using R much, much more pleasant). See links at [this HGSE software support page](#).

2.1 Some Resources and Directions

The following are ways of getting help with R.

Handouts

This textbook has many handouts written by the teaching team that illustrates how to use R for a variety of tasks. Also see the Resources page on Canvas for further commentary and organization.

Class Code

Each class has a R script that shows how to do the stuff from that class. See the Packets on Canvas for this code.

A Very Important Online Textbook

The online textbook [R for Data Science](#). provides important information on wrangling data, making plots, and doing statistical programming. It is full of examples and code snippets you can steal. There are two versions:

- [The First Edition](#) – this what we often refer to in these materials. But we are slowly updating to...
- [The Second Edition](#) – this one has a different organization, less weird modeling, and covers recent changes to the “tidyverse.”

Class Sections

Sections will typically have a hands-on component which will give you a chance to try things out yourself. These sections will also publish the final R code for future reference. See the section pages on Canvas to get this information.

Office Hours

Office hours are fine time to get help troubleshooting a specific or script you are working on.

GSE Stat Help Desk

Education students can write to stathelp@gse.harvard.edu to get help getting started with R. If groups of 3 or more want some tutorials, they can ask for them as well.

The Help Desk sometimes has Intro to R workshops. For example, grab some old workshop materials here: <http://its.gse.harvard.edu/gentle-introduction-r>

3 Resources on Learning R

So, where can you get help with R?

When looking for help, first try to stick with this textbook as there are many ways of doing things in R, and if you leap into a broad internet search you can be led into strange lands that look very different from the land of S022. So first, check the table of contents of this book and look for things that can help you. Or search the book itself!

But sometimes you have to go further afield, or want a different kind of help. This index is designed to steer you away from getting lost by pointing you to various resources that are more tailored to the course.

3.1 Live Class Code and Sections

The labs and classes usually have R scripts that shows how to do the stuff from that class. Find the class by looking at the table of contents on Canvas, and then open the packet page for that week. The packets should provide all code used.

Sections will also typically have a hands-on component which will give you a chance to try things out yourself. These sections will also publish the final R code for future reference. This will also be on the packet page for the specified week.

3.2 R for Data Science: The Very Important Online Textbook

[R for Data Science \(2e\)](#) has the main coding concepts we use in our work. This textbook provides important information on wrangling data, making plots, and doing statistical programming. It is full of examples and code snippets you can steal. Note that we use the second edition of this textbook (2e).

There are also [solutions to this guide](#). It can be quite useful to do the exercises in this text!

A supplementary textbook Also check out Modern Dive’s [“Statistical Inference via Data Science”](#), especially for some beginning R information in Chapter 1. It is a more gentle introduction to R, which can help get ready for R for DS.

3.3 Office Hours

Office hours are fine time to get help troubleshooting a specific or script you are working on.

3.4 GSE Stat Help Desk

Education students can write to stathelp@gse.harvard.edu to get help getting started with R. If groups of 3 or more want some tutorials, they can ask for them as well.

The Stat Help Desk sometimes have intro to R workshops. For example, grab some old workshop materials here: <http://its.gse.harvard.edu/gentle-introduction-r>

3.5 R Cheat Sheets

R Cheat Sheets are one page sheets that give lots of R commands for specific tasks. VERY USEFUL! See <https://rstudio.com/resources/cheatsheets/> for a full compendium of cheat sheets for work in RStudio, such as the ggplot sheet.

We've covered/touched on a lot of the cheat sheet topics in class or section like using the dplyr package, R-Markdown, or RegEx, but there are also things that we haven't discussed, such as creating survival plots in ggplot or developing a web-based app.

3.6 Further information on using R Markdown

In addition to `?@sec-intro__markdown` (which has some initial thoughts and how-tos tailored to the course), here are a few additional external resources on using R Markdown:

- [rmarkdown-cheatsheet-2.0.pdf](#)
- [rmarkdown-reference.pdf](#)
- See the async lecture on configuring code chunks in Week 1, on Canvas, as well.

3.7 Useful links to materials on the web

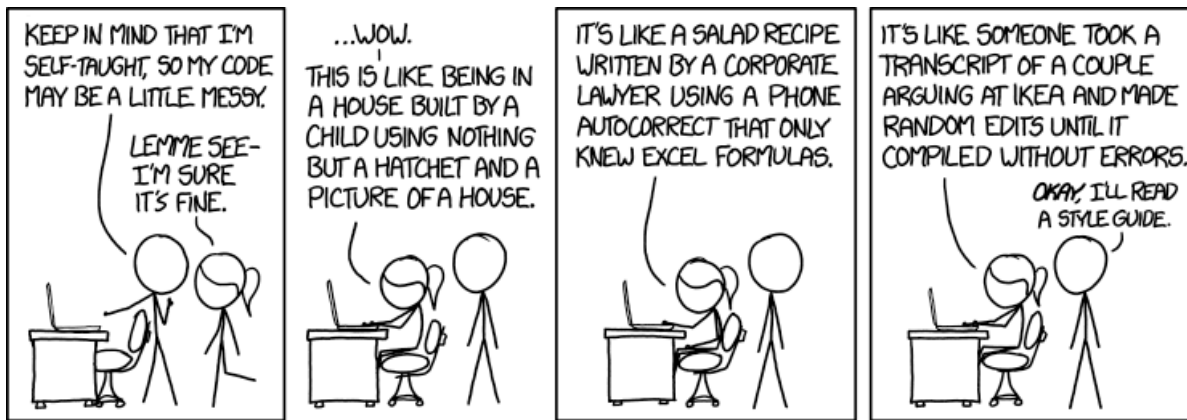
- R: <https://www.r-project.org/>
- RStudio: <https://www.rstudio.com/>
- Columbia RStudio overview: [here](#)
- TryR code school: <http://tryr.codeschool.com>
- Data Camp: <https://www.datacamp.com/courses/free-introduction-to-r>

- Princeton R tutorials: <http://data.princeton.edu/R/>
- D-Lab R training: <https://github.com/dlab-berkeley/R-for-Data-Science>
- GGPlot2: <http://ggplot2.org/>
- Google/stackoverflow

3.8 Writing your own math

- **Handout: math reference** – This handout, from my handout textbook for S043 (multilevel modeling) has the Latex you need to write equations easily for your reports. But what you really want is the Rmd version that has the code to make this handout, which is linked in the handout itself.

4 An R Code Style Guide (Miratrix version)



4.1 Why have coding style?

Many style decisions in coding are arbitrary, so why bother adhering to them? The primary reasons are that good style makes your code more readable, allows you to focus on writing quality code rather than worrying about inconsistencies, and helps you avoid being judged negatively for poor style.

More importantly, if you write in a clean style, you will be able to read your own code a few weeks later, when you have forgotten what you have been thinking. It also makes it *a lot* easier for the teaching team to help you understand possible problems in your code.

In this chapter, I lay out some tips for writing clean code. Much of this was initially taken from the Tidyverse style guide at <http://style.tidyverse.org>; we are primarily focusing on Chapters 2 and 4. But I will note that I depart substantially from some of their recommendations.

The cartoon is [xkcd](#); read those if you want to be an awesome nerd.

For further reading about coding practice, also see [Code and Data for the Social Sciences: A Practitioner's Guide](#), Gentzkow and Shapiro. Worth a skim!

4.2 The naming of things

“There are only two hard things in Computer Science: [cache invalidation](#) and naming things.” —Phil Karlton

- Variable and function names should be lowercase.
- Use an underscore to separate words within a name.
- Generally, variable names should be nouns and function names should be verbs.
- Ideally, your names should be self-explanatory and your code should be “self-documenting.”
- However, avoid making your variables too long (less than about 15 characters), unless you love typing.
- Avoid using numbers to store versions of a data frame (e.g., `df1`, `df2`, `df3`, ...).
- Names for variables, functions, files, etc. should generally consist of complete words.
- Naming is hard. more art than science.

Good

```
day_one  
first_day
```

Bad

```
first_day_of_the_month  
DayOne  
dayone  
djm1
```

4.2.1 Don't use common names

If R uses a name already, just avoid it.

Bad

```
TRUE <- FALSE  
pi <- 10  
mean <- function(x) sum(x)
```

4.2.2 Name constants for clarity

Good

```
winsor_upper <- 0.99  
winsor_lower <- 0.01  
diamonds <-  
  diamonds %>%
```



```
mutate(y_winsor = winsorize(y, probs = c(winsor_lower, winsor_upper)))

# Mediocre
diamonds_clean <-
  diamonds %>%
  mutate(y = winsorize(y, probs = c(0.01, 0.99)))
```

4.3 Syntax

4.3.1 Appropriate spaces make code readable

- Put a space before and after = when naming arguments in function calls.
- Always put a space after a comma, and never before (just like in regular English).

```
# Good
average <- mean(x, na.rm = TRUE)

# Also good
average <- mean( x, na.rm = TRUE )

# Bad
average<-mean(x, na.rm = TRUE)
average <- mean(x ,na.rm = TRUE)
```

4.3.2 Judicious spaces group elements to increase readability

- Most infix operators (==, +, -, <-, etc.) should be surrounded by spaces.
- The exception are those with relatively [high precedence](#): ^, :, ::, and ::: (“High precedence” means that these operators are evaluated first, like multiplication goes before addition.)

```
# Good
height <- (feet * 12) + inches
sqrt(x^2 + y^2)
x <- 1:10
base::get

# Bad
height<-feet*12 + inches
sqrt(x ^ 2 + y ^ 2)
```

```
x <- 1 : 10
base :: get
```

4.3.3 Lots of spaces to line things up is nice

Extra spacing (i.e., more than one space in a row) is ok if it improves alignment of equal signs or assignments (<-).

```
# Good
list(
  total = a + b + c,
  mean  = (a + b + c) / n
)

# Less good, but livable
list(
  total = a + b + c,
  mean  = (a + b + c) / n
)
```

4.3.4 Including names when calling functions

A function typically takes several arguments, namely the **data** to compute on and further arguments giving the **details** of the desired computation.

Omit names of common arguments (e.g. **data**, **aes**). However you can include these to make your code more readable to others.

If you override the default value of an argument, use the full name:

```
# Good
mean(1:10, na.rm = TRUE)

# Also Good
mean(x = 1:10, na.rm = TRUE)

# Bad
mean(x = 1:10, , FALSE)
mean(, TRUE, x = c(1:10, NA))
```

4.3.5 No line of code should be longer than 80 characters

- use one line each for the function name, each argument, and the closing ‘)’.
- use indentation to make it clear that the arguments are part of the function call.

Good

```
do_something_very_complicated(  
  something = "that",  
  requires = many,  
  arguments = "some of which may be long"  
)
```

Very bad

```
do_something_very_complicated(something = "that", requires = many, arguments = "some of which
```

Still bad

```
do_something_very_complicated(  
  something = "that", requires = many,  
  arguments = "some of which may be long"  
)
```

Yup, still bad

```
do_something_very_complicated(  
  something = "that", requires = many, arguments =  
  "some of which may be long"  
)
```

Exception to the above: short unnamed arguments can also go on the same line as the function name, even if the whole function call spans multiple lines.

Good

```
map(x, f,  
  extra_argument_a = 10,  
  extra_argument_b = c(1, 43, 390, 210209)  
)
```

4.3.6 Use <- for assignment (if you are prissy)

Use <-, not =, for assignment.

Good

```
x <- 5
```

```
# Upsets the nitpickers
x = 5
```

4.3.7 Double quotes are standard

Use `"`, not `'`, for quoting text. The only exception is when the text already contains double quotes and no single quotes.

```
# Good
"Text"
'Text with "quotes"'
'<a href="http://style.tidyverse.org">A link</a>'

# Bad
"Text"
'Text with "double" and \'single\' quotes'
```

4.3.8 Comments are good

Comments can really help with code, and help a reader assess if the code is achieving the desired effect. Comments should give the purpose of the code, not just restate the code in English.

That said, comments can be dangerous because if you change your code without changing the comments, you can cause confusion. This happens a lot, potentially making comments a source of confusion rather than clarity.

4.4 Pipes (`%>%`) with `magrittr`

Use `%>%` (or `|>`, if you are modern) to emphasize a sequence of actions, rather than the object that the actions are being performed on.

Avoid using the pipe when:

- You need to manipulate more than one object at a time. Reserve pipes for a sequence of steps applied to one primary object.
- There are meaningful intermediate objects that could be given informative names (cf rule 2.9).

4.4.1 Surround the pipe with whitespace

- `%>%` should always have a space before it, and should usually be followed by a new line.
- After the first step, each line should be indented by two spaces.

Adhering to the above structure makes it easier to add new steps (or rearrange existing steps) and makes it harder to overlook a step.

```
# Good
iris %>%
  group_by(Species) %>%
  summarize_if(is.numeric, mean) %>%
  ungroup() %>%
  gather(measure, value, -Species) %>%
  arrange(value)
```

```
# Bad
iris %>% group_by(Species) %>% summarize_all(mean) %>%
ungroup() %>% gather(measure, value, -Species) %>%
arrange(value)
```

4.4.2 Formatting short pipes

It is ok to keep a one-step pipe in one line, if you want. But not doing so is also fine.

```
# Good
iris %>% arrange(Species)
```

```
# Also good
iris %>%
  arrange(Species)
```

```
arrange(iris, Species)
```

4.4.3 Do not pack pipes inside of other pipes

This is brain-hurty:

```
# Bad
x %>%
  select(a, b, w) %>%
  left_join(
```

```

    y %>% filter(!u) %>% gather(a, v, -b) %>% select(a, b, v),
    by = c("a", "b")
  )

```

Instead, separate the pipes into their own lines:

Good

```

x_join <- x %>%
  select(a, b, w)
y_join <- y %>%
  filter(!u) %>%
  gather(a, v, -b) %>%
  select(a, b, v)
left_join(x_join, y_join, by = c("a", "b"))

```

Generally when you are doing something to two datasets, put both inside the parenthesis, rather than having one in and one piped in.

4.4.4 Keep your parenthesis even if you have no arguments

magrittr allows you to omit () on functions that don't have arguments. Avoid this. This way data objects never have parentheses and functions always do.

Good

```

x %>%
  unique() %>%
  sort()

```

Bad

```

x %>%
  unique %>%
  sort

```

4.4.5 Rule 4.6: Assignment

The first line can have the assignment and also the starting dataframe of a pipe sequence.

Good

```

iris_long <- iris %>%
  gather(measure, value, -Species) %>%
  arrange(-value)

```

4.5 A few final remarks

- Style is awesome. Save a future researcher from spending months trying to disentangle your spaghetti!
- You don't need to memorize these rules! Just as you have spell check and grammarly on your computer for prose, there is a package `styler` to help you follow the code style guide.
- Just as you still need to learn to spell (since spell checker doesn't capture everything), you need to learn these rules as well.
- If you get tired of rules and typing, try using the [Tidyverse's 'styler' package](<https://www.tidyverse.org/blog/2017/12/styler-1.0.0/>). The package links to the “Addins” widget in RStudio and is a fast way to clean up your code according to Tidyverse conventions.

In closing:

“Good coding style is like correct punctuation: you can manage without it, but it sure makes things easier to read.” —Hadley Wickham

Part II

On markdown

5 Intro to R Markdown

5.1 Overview

This document will show you how to use R Markdown to create documents which draw directly on your data to produce reports.

[R Markdown](#) (and its newer cousin [Quarto](#)) is a simple but powerful markdown language which you can use to create documents with inline R code and results. This makes it much easier for you to complete homework assignments and reports; makes it much less likely that your work will include errors; and makes your work much easier to reproduce. For example, if you find you have to drop cases from your dataset, you can simply add that line of code to your document, and recompile your document. Any text that's drawn directly from your analyses will be automatically updated.

To get started with R Markdown, watch Miratrix's video on RMarkdown. You can find it on [this page](#), under the “Do it for Lab” tab.

To use a markdown document, you generally open it and work inside of R Studio. To compile (or “knit”) the document—meaning have the computer run all the code in the document, and then generate a new document with the results of the code and all the formatting specified carried out—click on the button that says ‘Knit’. You can make a new document by saying “new markdown document” and then immediately “knit” it to see what it does.

Other R packages, such as **Sweave** and **knitr**, allow you to do the same things, but R Markdown has the added advantage of being relatively simple to use.

5.2 Getting started

Every R Markdown document starts with a header, called the YAML header. YAML headers look like this:

```
---
title: "My perfect homework"
author: "R master"
output: pdf_document
---
```

A header can contain more or less information, as you see fit. Your computer needs to have a copy of LaTeX installed in order to output .pdf documents. If you don't, you should change `output: pdf_document` to `output: html_document` or `output: word_document`.

You can specify a lot of options in the header. The most important ones are the title of your document, the author, and the output format.

In the main part of your document, you identify sections of the document using hashtags; more hashtags indicate less important sections.

For example, this:

```
# A big section
```

produces a big header (large font, etc.)

while this

```
## A small section
```

produces a smaller header (still a large font, but less large).

Also, if your document includes a table of contents, the sections get used to automatically generate the table of contents.

You can *italicize* words by writing `*italicize*` or `_italicize_`. You can **bold** words with `**bold**` or `__bold__`.

You can add superscripts ($E=mc^2$) by writing `E=mc^2^`.

You can create unordered lists:

- Item 1
- Item 2
- Item 3

to get

- Item 1
- Item 2
- Item 3

Or ordered lists:

1. Item 1
2. Item 2
3. Item 3

to get

1. Item 1
2. Item 2
3. Item 3

To start a new page, just type `\newpage` (not relevant for HTML output).

As you may have noticed, one of the driving ideas behind R Markdown is that the text should be interpretable even if it's not compiled. A person should be able to read the initial text file and understand the basic organization and what all of the symbols denote.

You can also add links and images, and do many other things beyond what we'll show you in this class. There are many resources out there, but [here's](#) one place you can start.

We also note that newer versions of Markdown and Quarto have a [visual editor](#) that allows you to format things in the usual way, e.g., control-B for bold. Some people prefer to take that approach.

Regardless, to compile or knit the document, click on the button that says **Knit** or **Render**, or Shift + Ctrl/Cmd + K.

5.3 Embedding R code

The power of R Markdown is you can include R code and that code will get run and the output included in your report automatically. There are two main ways to embed R code in R Markdown, code chunks (this is the primary way) or inline (this is if you want to get fancy).

5.3.1 Code chunks

To insert a code chunk click on **Insert** on the top right corner of your R Markdown file and select **R**. Or use keyboard shortcuts: Ctrl + Alt + I for PC and Cmd + Option + I for Mac.

A code chunk starts with three backticks and the letter **r** in curly braces, and ends with three more backticks on their own line. In between these lines you put the R code.

Code chunks have a number of different options you can put after the **r** in the opening line. The most important ones for us are:

- `eval = TRUE`, which means every time you knit the file, the code inside the R code chunk will get evaluated. This is the default.
- `echo = TRUE`, which means every time you knit the file, the code inside the R code chunk will be rendered, and you can see both the code itself and the results from evaluating the code.

For class, you should keep `echo = TRUE`, so that we can see your code and be able to tell what went wrong, if something did. You can set `echo = FALSE` for code chunks that load and manipulate data in the setup of your analysis, if you don't want people to look at that part of your work.

Other code chunks options you may see in class are:

- `warnings = FALSE`, which means warning messages generated by the code will not be displayed.
- `results = 'asis'`, which means results will not be reformatted when the file is compiled (useful if results return raw HTML).
- `fig.height` and `fig.width`, which specify the height and width (in inches) of plots created by the chunk.

Instead of specifying code chunks options every time, you can specify them globally in the setup chunk by using

```
knitr::opts_chunk$set(echo = TRUE, eval = TRUE)
```

You can then add additional options only to relevant chunks. For example, you want to exclude specific chunks, you can re-set `echo = FALSE` and `eval = FALSE` for those specific chunks.

Running code chunks: A good practice for writing your code is to run individual code chunks to make sure they are doing what you want them to do as you write them. You can do this by executing individual lines of code, or whole chunks. Go to **Run** in the upper right corner and select what chunks to execute, e.g. **Run Current Chunk**, **Run Next Chunk**, etc.

5.3.2 Inline code

Code results can also be inserted directly in the text of your R Markdown file. This is particularly useful when you are extracting and interpreting model parameters. You can extract the coefficient from the model and use inline code to report it. If the data or model change, *the text will also change* when you knit the document.

To add inline code, enclose it in ``r ``. For example, to report the mean reading score, you can use

```
-0.0443549
```

Which will produce -0.0443549. That's a few too many decimals, let's round it off, using

-0.04

which produces "-0.04."

Here we used two commands: `round` and `mean`. You can use more commands and write more complex inline code, depending on what you want to report.

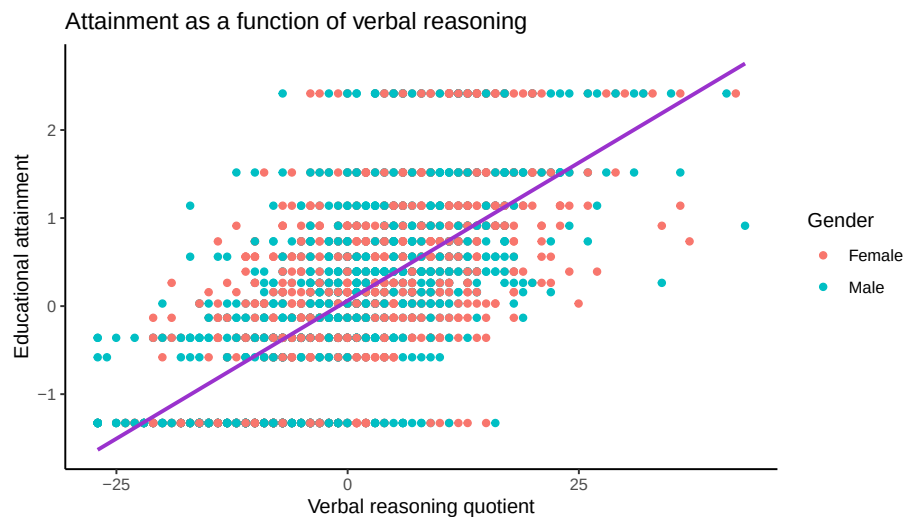
5.4 Embedding plots

Plots are easy to embed. For example,

```
library(ggplot2)

dat$male <- factor(dat$male, levels = c(0, 1), labels = c("Female", "Male"))

ggplot(data=dat, aes(p7vrq, attain, colour=male)) +
  geom_point() +
  labs(title="Attainment as a function of verbal reasoning",
       x = "Verbal reasoning quotient",
       y = "Educational attainment", colour="Gender") +
  geom_smooth(method="lm", formula = y ~ x, se=FALSE, colour="darkorchid3")
```



Girls are rendered as coral, boys are rendered in turquoise, and the line of best fit is drawn in `darkorchid3` (because why not). Just because you have a lot of colors and plotting characters to work with doesn't mean you need to use them all. In the options, I specified `fig.width = 7` and `fig.height = 7`. Notice that this command draws on `dat`, which we loaded in a previous chunk. When knitting the document, code chunks get executed in order and the results persist throughout the R Markdown document.

For the purposes of class, we want to see both your plot code and the plot itself. It's not uncommon to use wrong code to create a plot that looks correct (at least visually).

5.5 Embedding tables

You can directly render tables in R Markdown. The idea is, inside an R chunk, you call a command that prints out a table. The report then takes this printout and integrates it into your overall report. There are many different packages to make tables, but in class we'll mostly use `knitr`, `texreg`, `stargazer`, and the `tab_model()` function in `sjPlot`.

You can use these packages to create a descriptive table. For example:

```
head( dat ) %>%
  knitr::kable( digits = 2 )
```

neighid	schid	attain	p7vrq	p7read	dadocc	dadunempdaded	momed		male	deprive
675	0	0.74	21.97	12.13	2.32	0	0	0	Male	-0.18
647	0	0.26	-7.03	-12.87	16.20	0	0	1	Female	0.21
650	0	-1.33	-11.03	-31.87	-23.45	1	0	0	Male	0.53
650	0	0.74	3.97	3.13	2.32	0	0	0	Male	0.53
648	0	-0.13	-2.03	0.13	-3.45	0	0	0	Female	0.19
648	0	0.56	-5.03	-0.87	-3.45	0	0	0	Female	0.19

See Chapter 8 for more on making various tables.

We can also use `texreg` or `stargazer` to create a taxonomy of regression models.

For example:

```
library(texreg)

# fit some models
m1 <- lm(attain ~ male, data=dat)
m2 <- lm(attain ~ male + momed, data=dat)
m3 <- lm(attain ~ male + momed + daded, data=dat)
```

```
screenreg(list(m1,m2,m3),
           custom.coef.names=c("Intercept", "Male",
                               "Maternal education", "Paternal education"))
```

	Model 1	Model 2	Model 3
Intercept	0.15 *** (0.03)	0.03 (0.03)	-0.02 (0.03)
Male	-0.12 ** (0.04)	-0.12 ** (0.04)	-0.12 ** (0.04)
Maternal education		0.49 *** (0.05)	0.24 *** (0.05)
Paternal education			0.54 *** (0.06)
R ²	0.00	0.05	0.09
Adj. R ²	0.00	0.05	0.08
Num. obs.	2310	2310	2310

*** p < 0.001; ** p < 0.01; * p < 0.05

Both packages include a lot of options and make it easy to produce publication-quality tables with little effort.

5.6 Embedding math

We'll be writing some mathematical models in class. R Markdown can use LaTeX style math-writing to display mathematical script. Similar to code chunks and inline code, you can use LaTeX for single or multiple equations, or for individual parameters embedded in the text.

For example, the following statement

```
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$
```

compiles to

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

And the following statement `$$\mu$` compiles to μ .

5.7 Help! R Markdown report generation doesn't work

If your markdown will not render, first try restarting R and then selecting run all chunks from the menu. If that does not work, then try fixing your code.

If that does work, but knitting still does not work, then something stranger is going on. First, check to see if you have a “View()” command in your Markdown file. This will cause your document to not be able to knit.

Also watch for the `skim()` command—it can crash report generation as well.

If you can't knit PDFs you need to install latex (tex). Once you do, reboot your computer. If things don't work, then knit to Microsoft word (or, failing that, html as a last resort), print to pdf, and turn that in. But then ask a teaching fellow to help get things set up, since PDFs make for much more readable reports.

6 Creating Reports in Quarto

6.1 What is Quarto?

Quarto is the “next-generation” of RMarkdown. Most tutorials on Quarto are intended for people who have experience with RMarkdown, so a great place to start is to read the prior chapter before reading this one.

Once you’re familiar with RMarkdown, the tricks you know will almost always work in Quarto. The major difference is that Quarto will render *inside* RStudio, as you write your document, which makes it a little nicer to work with. There are other new (and very cool) features, but they aren’t essential for anything in this course.

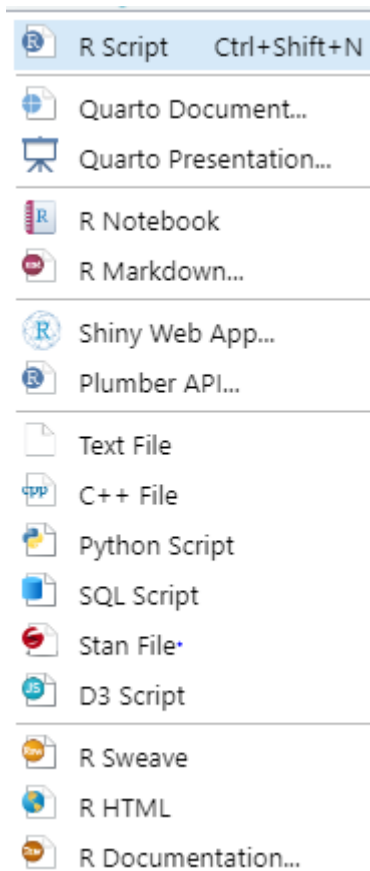
6.2 Using Visual Editor with Rmd

An obvious, cool thing about Quarto is the fancy visual editor, but you can turn this feature on for regular old RMarkdown files, too. Just click on this , and click “use visual editor.” You can also add `editor: visual` to the yaml (top-matter) of an rmd file and it’ll open in the visual editor by default.

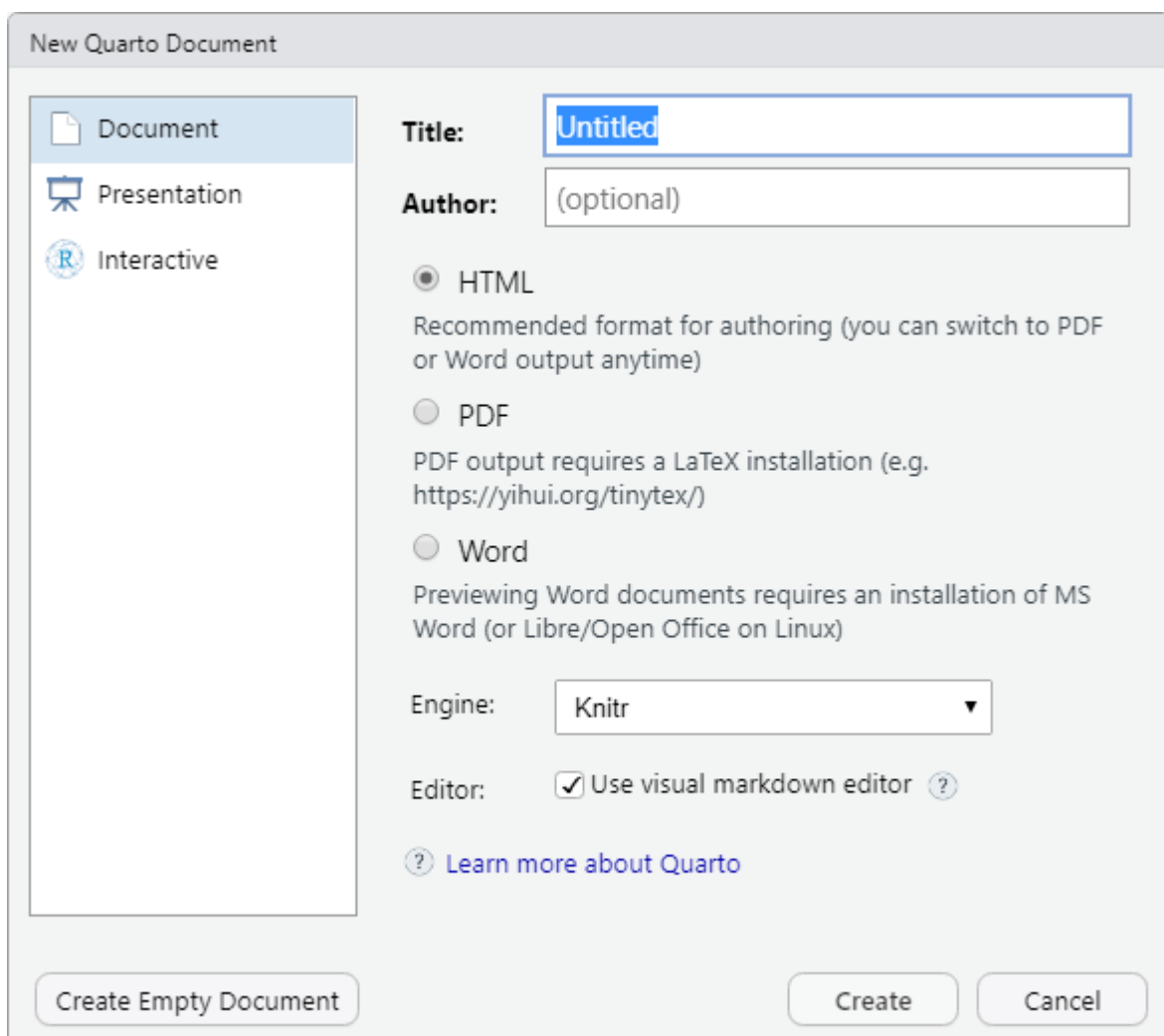
6.3 Creating a Quarto Report

In RStudio, click this icon in the upper left: 

That’ll give you the following drop-down menu, where you can select “Quarto Document”:



When you click on “Quarto Document...” RStudio might take a few seconds to load. Then you’ll see this pop-up:



Fill out the document title and author (just like for RMarkdown). You can always change the title and author later. You'll want to render your reports as PDFs, so select that option. Finally, hit the "Create" button at the bottom.

RStudio will load up a new Quarto doc (with some boilerplate markdown in it). From here, you can treat it like an RMarkdown file.

If you want a more thorough introduction to Quarto, check out [this tutorial](#).

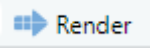
6.4 Rendering to PDF

To render your report to PDF, you'll need to have an installation of LaTeX. You can set this up from within RStudio.

Down by your console, there's a tab called "Terminal." Click on it to open the terminal. Inside the terminal type the following:

```
quarto install tool tinytex
```

I recommend restarting your computer after this.

Now you should be able to click  and get a pdf version of your report. By default, a copy of the pdf will be saved in the same folder as your Quarto document.

6.5 A Couple Quick Tricks

6.5.0.1 Making Code Chunks

Use the keyboard shortcut `ctrl+alt+i` or `command + option + i` to create a new R code chunk.

6.5.0.2 Adding Images

You can copy-paste or drag-and-drop images into a Quarto doc. That's how I put the above screenshots into this document.

6.6 Additional Quarto Resources

Here are a couple more links if you'd like to learn more:

- The [RMarkdown Cheatsheet](#) has great information. Remember, most things that work in RMarkdown work in Quarto. Also, there are a lot of other great cheatsheets at this site as well.
- A [Quarto Intro Video](#) that walks through using Quarto for the first time.
- A [In Depth Quarto Intro](#) video that covers many amazing new features (like making interactive html reports).

7 Working with Rmarkdown chunks

When you write R Markdown (or Quarto) reports, you are going to have a lot of “chunks” of code. These are the things that start with “`{r chunk_name, blah}`” or “`{r, blah}`.” When you render your report, these are run and the output is then taken and put in your report depending on how the chunk is configured.

This file gives some options for how to control these chunks.

7.1 Markdown chunks

7.1.1 Options for including/suppressing code and output

include: Should chunk be included in knit file? Defaults to **TRUE**. If **FALSE**, code chunk is run, but chunk and any output is not included in the knit file.

eval: Should chunk be evaluated by R? Defaults to **TRUE**. If **FALSE**, code chunk is included in the knit file, but not run.

echo: Should the code from this chunk be included in knit file along with output? Defaults to **TRUE**. If **FALSE**, the output from the chunk is included, but the code that created it is not. Most useful for plots.

7.1.2 Options for including/suppressing R messages

R has “errors,” meaning it could not run your code, “warnings,” meaning that the code was wrong, but there are some potential issues with it, and “messages,” which are simply information about what your code ran. You can include or suppress each of these types of message.

error: Should R continue knitting if code produces an error? Defaults to **FALSE**. Generally don’t want to change this because it means you can miss serious issues with your code.

warning: Should R include warnings in knit file? Defaults to **TRUE**.

message: Should R include informational messages in knit file? Defaults to **TRUE**. Easy way to clean up your markdowns.

For example, the following code chunk produces an error, but because the header is `error = TRUE`, `warning=FALSE`, `message=FALSE`, only the error is shown:

```
#This code produces an error
dat %>%
  filter(dest = 1)
```

```
Error in `filter()`:
! We detected a named input.
i This usually means that you've used `=` instead of `==`.
i Did you mean `dest == 1`?
```

```
#Example warning
parse_number(c("1", "$3432", "tomato"))
```

```
[1]    1 3432  NA
attr(,"problems")
# A tibble: 1 x 4
   row   col expected actual
  <int> <int> <chr>    <chr>
1     3    NA a number tomato
```

```
#Example message
library(gridExtra)
```

7.1.3 Options for modifying figure outputs

`out.width`: What percentage of the page width should output take?

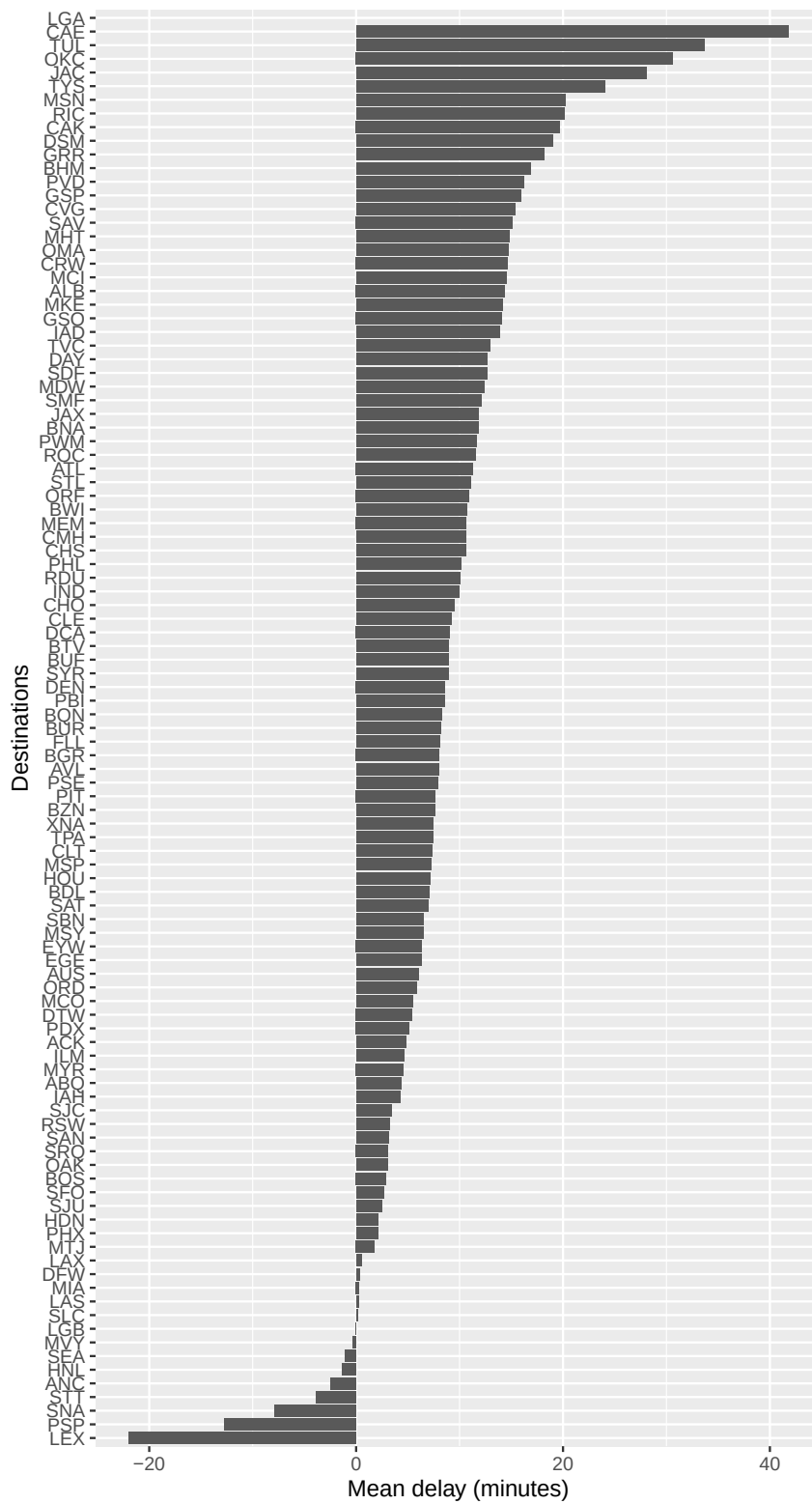
`fig.height`: What should be the height of figures?

`fig.width`: What should be the width of figures?

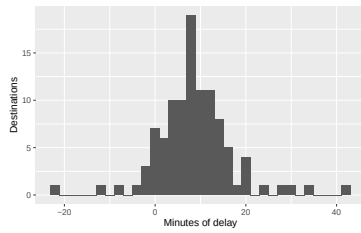
`fig.asp`: What should be the aspect ratio of figures?

`fig.align`: How should figures be aligned?

We might want a bigger plot for this:



And a smaller plot for this:



7.2 Changing your defaults

At the beginning of your code, you can set custom defaults so all your chunks will render the same way (unless you override by specifically adding arguments to a chunk itself). This is handy in that you will then not need to repeat the custom arguments in each code chunk. For example, you can set a default figure size.

Here is an example:

```
knitr::opts_chunk$set(echo = TRUE,  
                        fig.width = 5,  
                        fig.height = 3,  
                        out.width = "5in",  
                        out.height = "3in", fig.align = "center")
```


8 Making tables in Markdown

When writing reports you may, from time to time, need to include a table. You should probably make a chart instead, but every so often a table actually is a nice thing to have. This chapter focuses on two key aspects: creating the table itself, and formatting it for a report or presentation.

Many table-making packages and functions in R produce basic tables that display nicely in a monospace font on the screen. This is a good starting point, but you'll often need additional formatting to make the table publication-ready. R offers several excellent packages to help with this, catering to different needs. Some are particularly suited for HTML documents (such as websites), while others are better for PDF documents (such as reports and papers). Finding the right package for your specific use case can take some trial and error.

To illustrate these concepts, let's start with some fake data.

```
library( tidyverse )
dat = tibble( G = sample( LETTERS[1:5], 100, replace=TRUE ),
              X = rnorm( 100 ),
              rp = sample( letters[1:3], 100, replace=TRUE ),
              Z = sample( c("tx", "co"), 100, replace=TRUE ),
              Y = rnorm( 100 ) )
```

We can make summary of it by our grouping variable:

```
sdat <- dat %>% group_by( G ) %>%
  summarise( EY = mean( Y ),
            pT = mean( Z == "tx" ),
            sdY = sd( Y ) )
```

Our intermediate results:

```
sdat
```

```
# A tibble: 5 x 4
  G      EY    pT   sdY
<chr> <dbl> <dbl> <dbl>
1 A    -0.314  0.368 0.893
2 B    -0.104  0.368 1.01
```

```

3 C      0.258   0.522 1.09
4 D     -0.00202 0.4   1.18
5 E      0.155   0.667 1.04

```

We can print this out in a much cleaner form using the `kable()` method from the `knitr` package:

```
knitr::kable( sdat, digits = 2 )
```

G	EY	pT	sdY
A	-0.31	0.37	0.89
B	-0.10	0.37	1.01
C	0.26	0.52	1.09
D	0.00	0.40	1.18
E	0.16	0.67	1.04

Say our grouping variable is a set of codes for something more special. We can merge in better names by first making a small “cross-walk” of the ID codes to the full names, and then merging them to our results:

```

names = tribble( ~ G, ~ name,
                  "A", "fred",
                  "B", "doug",
                  "C", "xiao",
                  "D", "lily",
                  "E", "unknown" )

```

```
names
```

```
# A tibble: 5 x 2
```

```

  G      name
  <chr> <chr>
1 A      fred
2 B      doug
3 C      xiao
4 D      lily
5 E      unknown

```

```

sdat = left_join( sdat, names, by = "G" ) %>%
  relocate( name )

```

Again, the easiest way to make a nice clean table is with the `kable` command.

```
knitr::kable( sdat, digits=2 )
```

name	G	EY	pT	sdY
fred	A	-0.31	0.37	0.89
doug	B	-0.10	0.37	1.01
xiao	C	0.26	0.52	1.09
lily	D	0.00	0.40	1.18
unknown	E	0.16	0.67	1.04

`kable()` is a great workhorse table-making tool! There are expansion R packages as well, e.g. `kableExtra`, which can do lots of fancy customization stuff.

8.1 Making a “table one”

The “table one” is the first table in a lot of papers that show general means of different variables for different groups. Perhaps not surprisingly, the `tableone` package is useful for making such tables:

```
library(tableone)
```

```
# sample mean
```

```
CreateTableOne(data = dat,
               vars = c("G", "Z", "X"))
```

```

              Overall
n              100
G (%)
  A             19 (19.0)
  B             19 (19.0)
  C             23 (23.0)
  D             15 (15.0)
  E             24 (24.0)
Z = tx (%)      48 (48.0)
X (mean (SD)) -0.02 (0.90)
```

```
# you can also stratify by a variables of interest
```

```
tb <- CreateTableOne(data = dat,
                    vars = c("X", "G", "Y"),
                    strata = c("Z"))
```

```
tb
```

	Stratified by Z			
	co	tx	p	test
n	52	48		
X (mean (SD))	-0.10 (1.00)	0.06 (0.77)	0.393	
G (%)			0.215	
A	12 (23.1)	7 (14.6)		
B	12 (23.1)	7 (14.6)		
C	11 (21.2)	12 (25.0)		
D	9 (17.3)	6 (12.5)		
E	8 (15.4)	16 (33.3)		
Y (mean (SD))	-0.12 (1.00)	0.17 (1.07)	0.167	

You can then use `kable` on your table as so:

```
print(tb$ContTable, printToggle = FALSE) %>%
  knitr::kable()
```

	co	tx	p	test
n	52	48		
X (mean (SD))	-0.10 (1.00)	0.06 (0.77)	0.393	
Y (mean (SD))	-0.12 (1.00)	0.17 (1.07)	0.167	

8.2 The stargazer package

You can easily make pretty tables using the **stargazer** package. You need to ensure the data is a `data.frame`, not `tibble`, because **stargazer** is old school. It appears to only do continuous variables. Stargazer is probably best known for making regression tables, but it can make other kinds of tables as well, such as data summaries.

When using **stargazer** to summarize a dataset, you can specify that it should include only some of the variables and you can omit stats that are not of interest:

```
# to include only variables of interest
stargazer(as.data.frame(dat), header=FALSE,
  omit.summary.stat = c("p25", "p75", "min", "max"),
  # to omit percentiles
  title = "Table 1: Descriptive statistics",
  type = "text")
```

Table 1: Descriptive statistics

Statistic	N	Mean	St. Dev.
X	100	-0.024	0.895
Y	100	0.017	1.039

See the `stargazer` help file for how to set/change more of the options: <https://cran.r-project.org/web/packages/stargazer/stargazer.pdf>

Warning: `stargazer` does not work well with tibbles (the data frames you get from tidyverse commands), so you need to convert your data to a `data.frame` before using it. In particular, you have to “cast” your data to a `data.frame` to make it work:

```
library(stargazer)

# to include all variables
stargazer( as.data.frame(dat), header = FALSE, type="text")
```

To use `stargazer` in a PDF or HTML report, you will want the report to format the table so it doesn’t look like raw output. To do so, you would not set `type="text"` but rather `type="latex"` or `type="html"`, and then in the markdown chunk header (the thing that encloses all your R code) you would say “results=‘asis’” in your code chunk header like so:

```
::: {cell layout-align="center" messages='false'}

:::
```

This will ensure the output of `stargazer` gets formatted properly in your R Markdown.

Unfortunately, it is hard to dynamically make a report that can render to either html or a pdf, so you will have to choose one or the other. If you are making a PDF, you will want to use `type="latex"` and if you are making an HTML report, you will want to use `type="html"`.

8.3 The xtable package

The `xtable` package is another great package for making tables. It is particularly good for LaTeX documents. It is a bit more complicated to use than `stargazer`, but it is very powerful. Here is an example of how to use it:

```
library(xtable)
xtable(sdat, caption = "A table of fake data" )
```

Here you would again use the “results=‘asis’” in the chunk header to get the table to render properly in your R Markdown document.

Part III

On Plotting

9 Intro to ggplot

This chapter demonstrates some powerful features of `ggplot2`, the plotting package that we use most in this course.

`ggplot` can initially seem like a nightmare to some, but once you wrestle it to the ground it is one of the most powerful visualization tools you might have in your toolbox. Happily, it is fairly easy to get some basics up and running once you start looking at the world the way it does. Let's start doing that.

First, `ggplot` thinks of a plot as a collection of layers stacked on top of each other. The way this looks in code is a bunch of weird function calls connected together with `+`. You read this series of calls left to right. The first call is always a statement saying what data you are plotting and what variables you care about. So before you can even plot, you need to make sure your data are in a nice, tidy data frame.

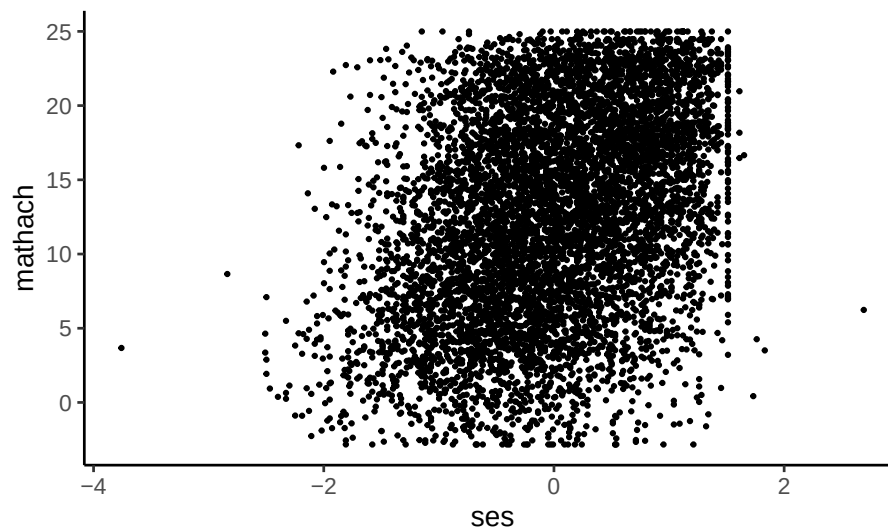
Happily, when you load data, it usually is. For example:

```
dat <- read_dta("data/hsb.dta")
head( dat )

# A tibble: 6 x 26
  minority female      ses mathach  size sector pracad disclim himinty schoolid
    <dbl>   <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
1      0     1 -1.53     5.88   842     0  0.350   1.60     0    1224
2      0     1 -0.588   19.7   842     0  0.350   1.60     0    1224
3      0     0 -0.528   20.3   842     0  0.350   1.60     0    1224
4      0     0 -0.668    8.78   842     0  0.350   1.60     0    1224
5      0     0 -0.158   17.9   842     0  0.350   1.60     0    1224
6      0     0  0.0220    4.58   842     0  0.350   1.60     0    1224
# i 16 more variables: mean <dbl>, sd <dbl>, sdalt <dbl>, junk <dbl>,
#   sdalt2 <dbl>, num <dbl>, se <dbl>, sealt <dbl>, sealt2 <dbl>, t2 <dbl>,
#   t2alt <dbl>, pickone <dbl>, mmses <dbl>, mnses <dbl>, xb <dbl>, resid <dbl>
```

The easiest full plot to make has two elements. The first gives what your variables are, and the second says how to plot them:


```
ggplot(dat, aes(y = mathach, x = ses)) +
  geom_point( cex=0.5)
```

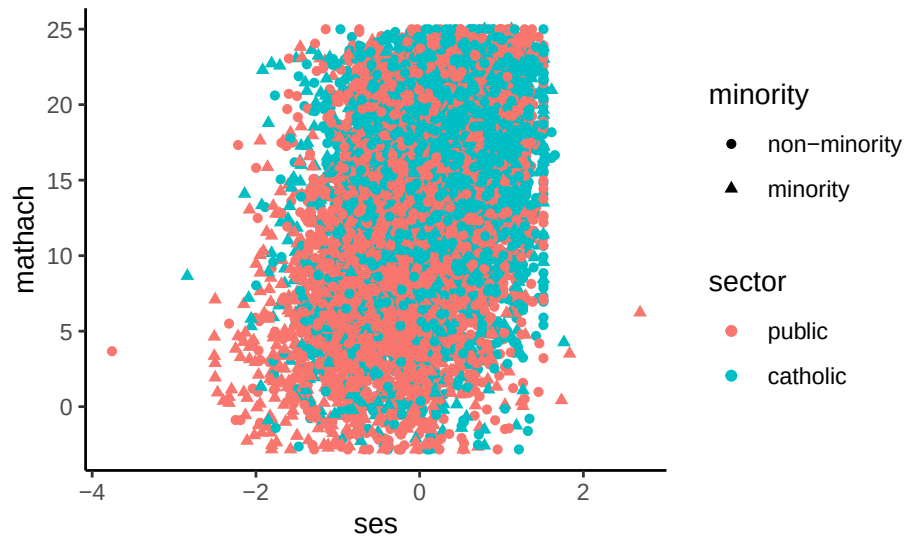


So far, nothing too scary, right? The `ggplot( dat, aes(x=mathach, y=ses) )` says “My plot is going to use `dat` for my data, and my *y*-axis is the `mathach` variable and my *x*-axis is `ses`.” The `aes()` bit is “aesthetics”—it is a way of tying variables to different kinds of things you could have on your plot: *x* location, *y* location, color, plotting symbol, and a few other things.

For example:

```
dat <- dat |>
  mutate(sector = factor(sector, levels=c(0,1), labels=c("public","catholic")),
         minority = factor(minority, labels=c("non-minority","minority")))

ggplot( dat, aes(y=mathach, x=ses, col=sector, pch=minority) ) +
  geom_point()
```

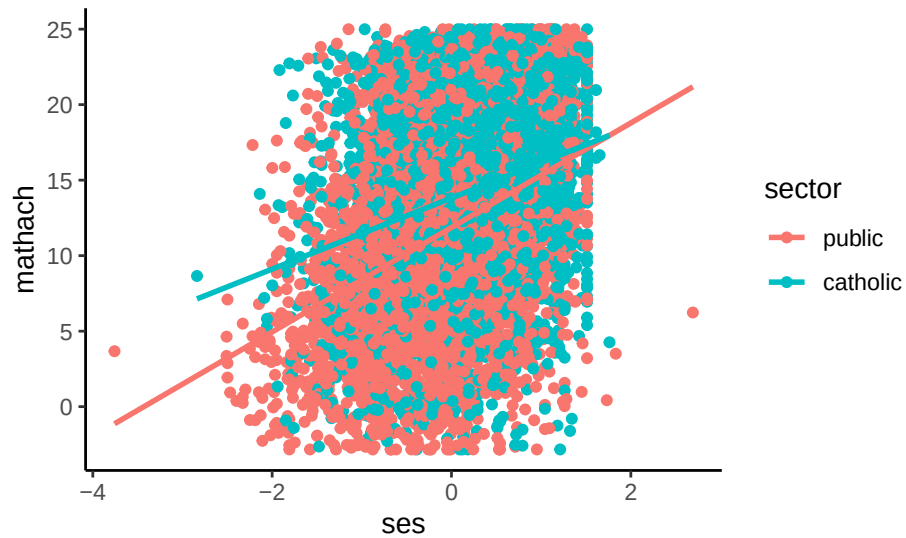


Note that `ggplot` wants the data frame to be neatly put together, including that categorical variables are listed as factors. This is why we convert the dummy `sector` to a factor above. Once you do this, however, it will label things in a nice way.

9.1 Summarizing

You can also automatically add various statistical summaries, such as simple regression lines:

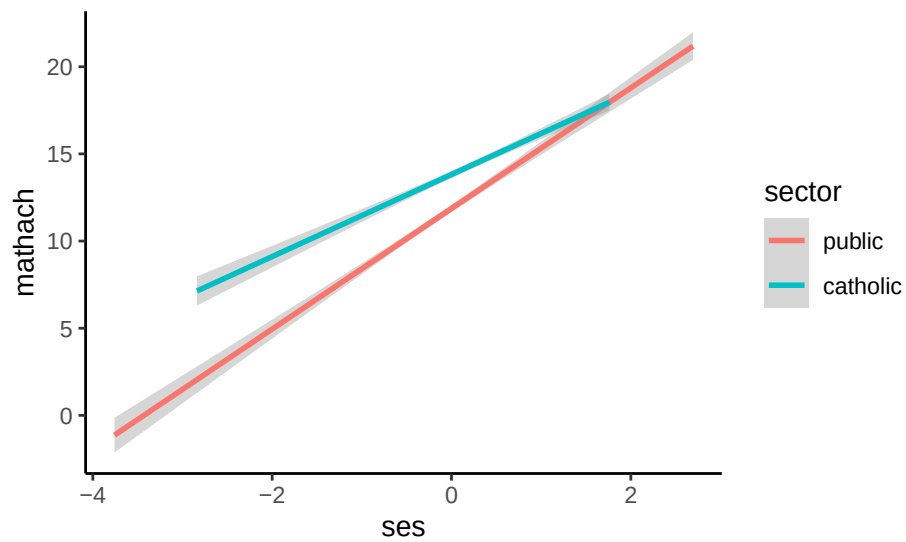
```
ggplot( dat, aes(y=mathach, x=ses, col=sector ) ) +  
  geom_point() +  
  stat_smooth( method="lm", se = FALSE )
```



Notice how it automatically realized you have two subgroups of data defined by sector. It gives you a regression line for each group.

The elements of the plot are stacked, and if you remove one of the elements, it will not appear:

```
ggplot( dat, aes(y=mathach, x=ses, col=sector ) ) +
  stat_smooth( method="lm" )
```



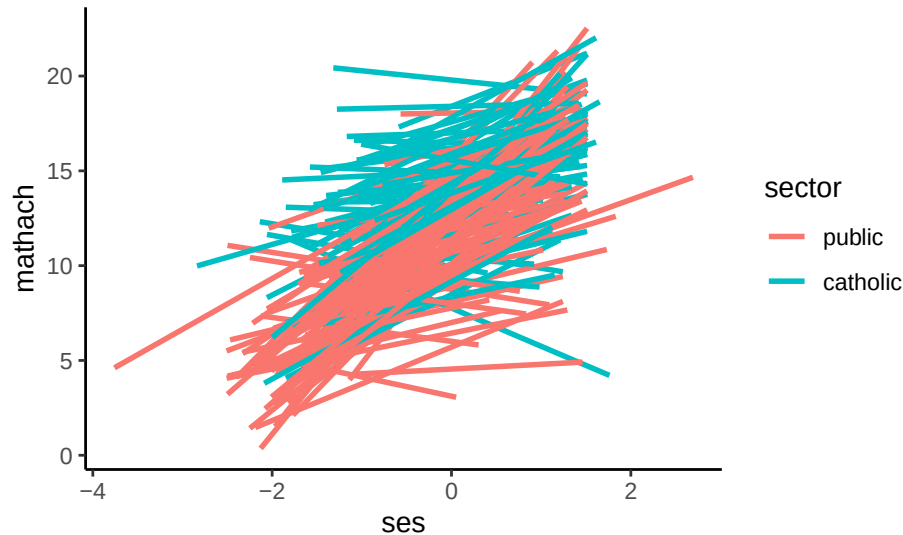
Here we also added some uncertainty bars around the regression lines by not saying `se = FALSE`. (Including uncertainty is the default; this uncertainty is not to be trusted, especially in this course, as it is not taking clustering into account.)

9.2 Grouping

Combining these ideas we can make a trend line for each school:

```
my.plot = ggplot( dat, aes(y=mathach, x=ses, col=sector, group=schoolid ) ) +  
  stat_smooth( method="lm", alpha=0.5, se = FALSE )
```

```
my.plot
```



The trendlines automatically extend to the limits of the data they are run on, hence the different lengths.

Also, notice we “saved” the plot in the variable `my.plot`. Only when we “print” the plot will the plot appear on your display. When we type the name of a variable, it prints. Once you have a plot stored in a variable you can augment it very easily.

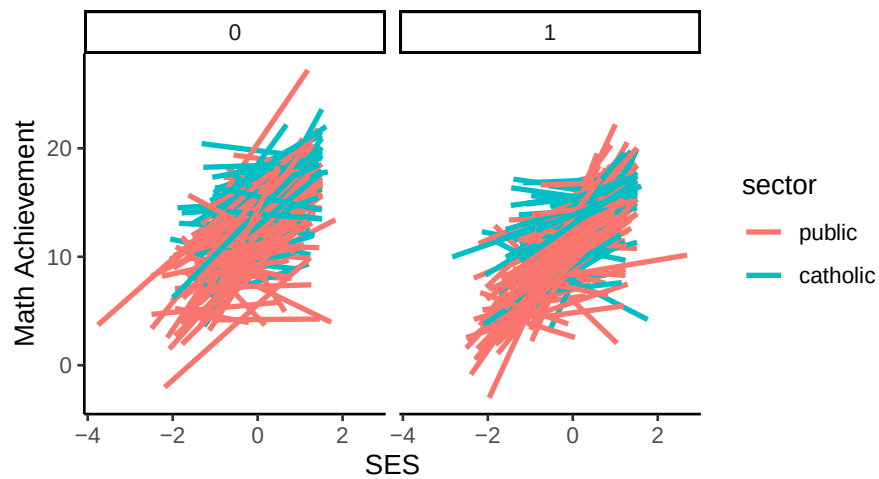
As you may now realize, `ggplot2` is very, very powerful.

9.3 Customization

We next show some other things you can do. For example, you can make lots of little plots:

```
my.plot +  
  facet_grid( ~ female ) +  
  ggtitle("School-level trend lines for their male and female students") +  
  labs(x="SES", y="Math Achievement")
```

School-level trend lines for their male and female studen



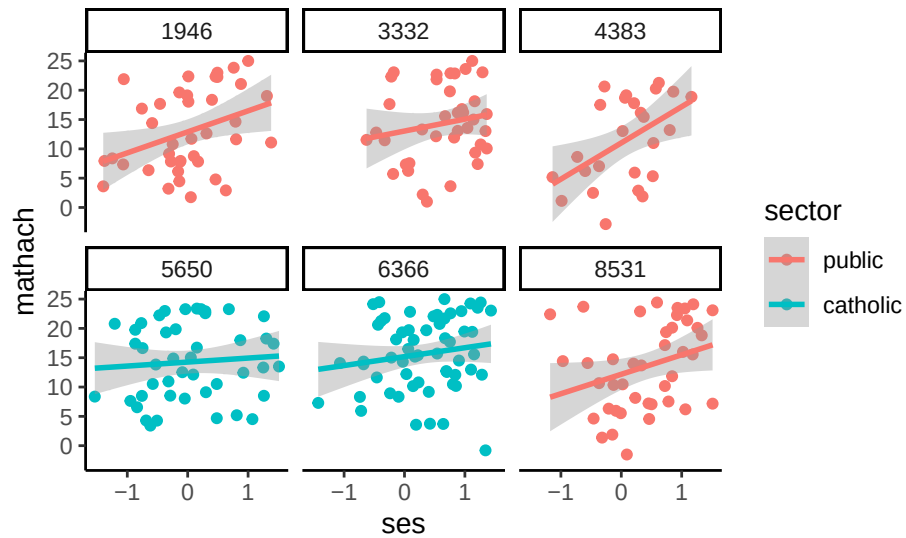
Or,

```
# random subset of schoolid
sch <- sample( unique( dat$schoolid ), 6 )

# pipe into ggplot
sch.six <- dat |>
  filter(schoolid %in% sch)

my.six.plot <- ggplot( sch.six, aes(y=mathach, x=ses, col=sector ) ) +
  facet_wrap( ~ schoolid, ncol=3 ) +
  geom_point() + stat_smooth( method="lm" )

my.six.plot
```



Also shown in the above are adding titles.

9.4 Themes

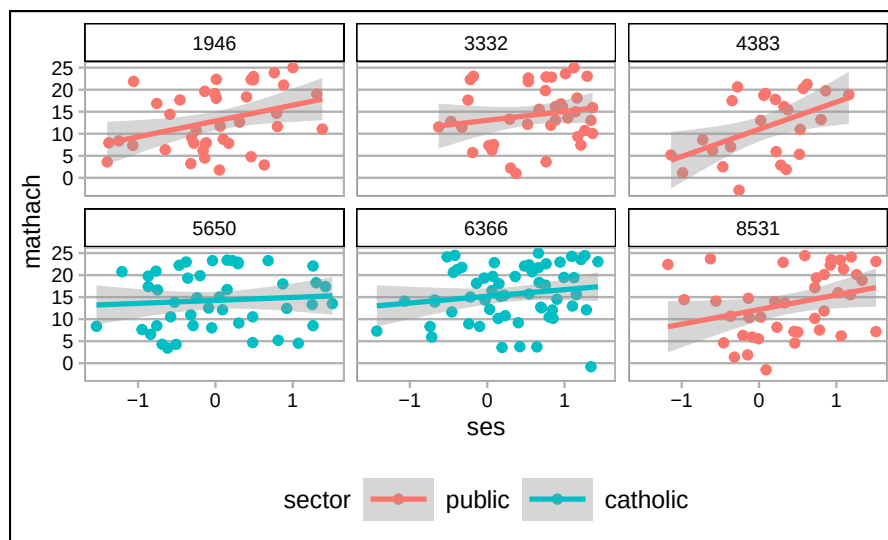
You can very quickly change the entire presentation of your plot using **themes**. There are pre-packaged ones, and you can make your own that you use over and over. Here we set up a theme to be used moving forward

```
library( ggthemes )
my_t = theme_calc() + theme( legend.position="bottom",
                             legend.direction="horizontal",
                             legend.key.width=unit(1,"cm") )

theme_set( my_t )
```

Compare the same plot from above, now with a new theme.

```
my.six.plot
```



Cool, no?

9.5 Next steps

There is a lot of information out there on `ggplot` and my best advice is to find code examples, and then modify them as needed. There are tutorials and blogs that walk through building plots (search for “ggplot tutorial” for example), but seeing examples seems to be the best way to learn the stuff. For example, you could use the above code for your project one quite readily. And don’t be afraid to ask how to modify plots on Piazza!

In particular, check out the excellent “[R for Data Science](#)” textbook. It extensively uses `ggplot`, starting [here](#).

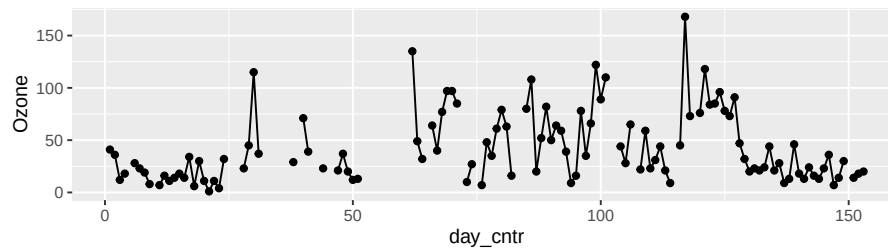
10 Some tips for making plots

This document provides several simple plotting tips for using RStudio to make plots (primarily using the `ggplot2` package in `tidyverse`). These are all very simple tricks that can radically enhance the legibility of a plot. This document also covers how to save your plots so you can use them in presentations and other documents, and how to control the size of your plot in a R Markdown document.

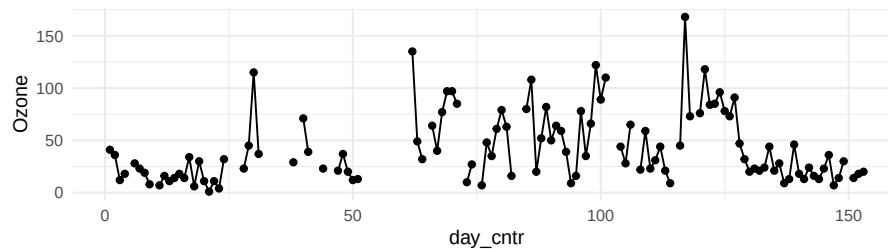
11 Setting a theme

Themes in ggplot give you a bunch of default options for plotting. For example, you can get rid of the grey background of plots with the `theme_minimal()` command.

```
airquality$day_cntr = 1:nrow(airquality)
ggplot( airquality, aes( day_cntr, Ozone ) ) +
  geom_point(na.rm=TRUE) + geom_line()
```

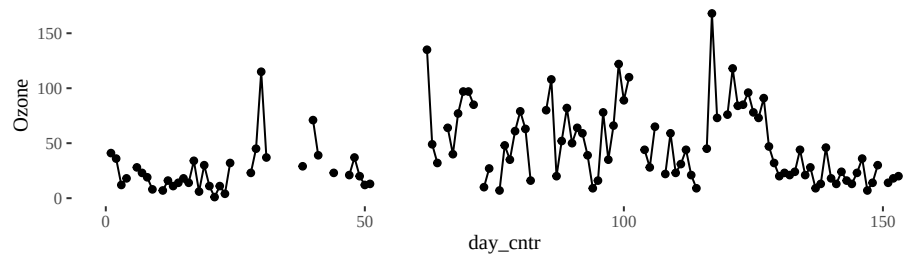


```
ggplot( airquality, aes( day_cntr, Ozone ) ) +
  geom_point(na.rm=TRUE) + geom_line() +
  theme_minimal()
```



I also quite like Tufte theme in the `ggthemes` bonus library (this library is a library of various themes with styles good and bad):

```
ggplot( airquality, aes( day_cntr, Ozone ) ) +
  geom_point(na.rm=TRUE) + geom_line() +
  ggthemes::theme_tufte()
```



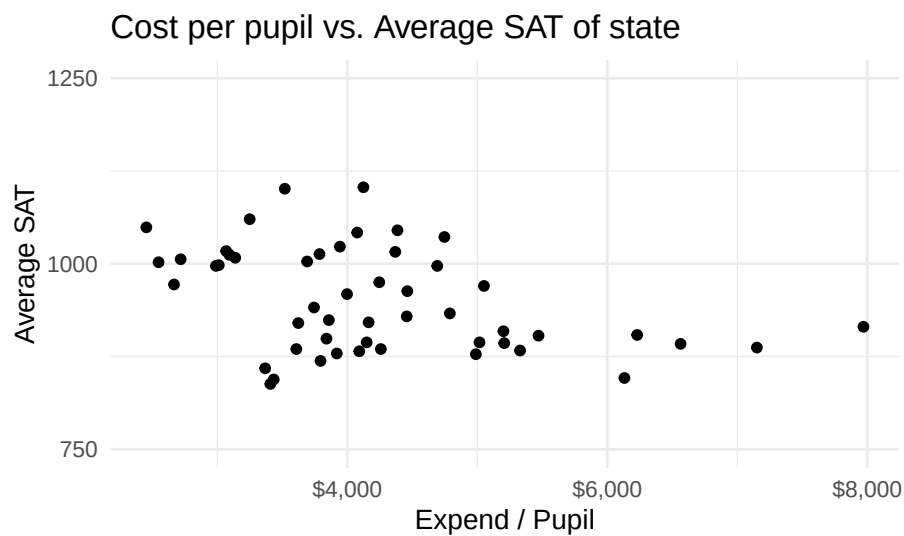
You can also set the theme globally:

```
theme_set( theme_minimal() )
```

12 Formatting axes and making labels

The `labs()` command is critical for getting labels for your plot. The `scale_x_continuous()` (or `y`) allow you to control the scales on each axis. Also, there is a nice package, `scales` that will format your x and y-axes. Witness!

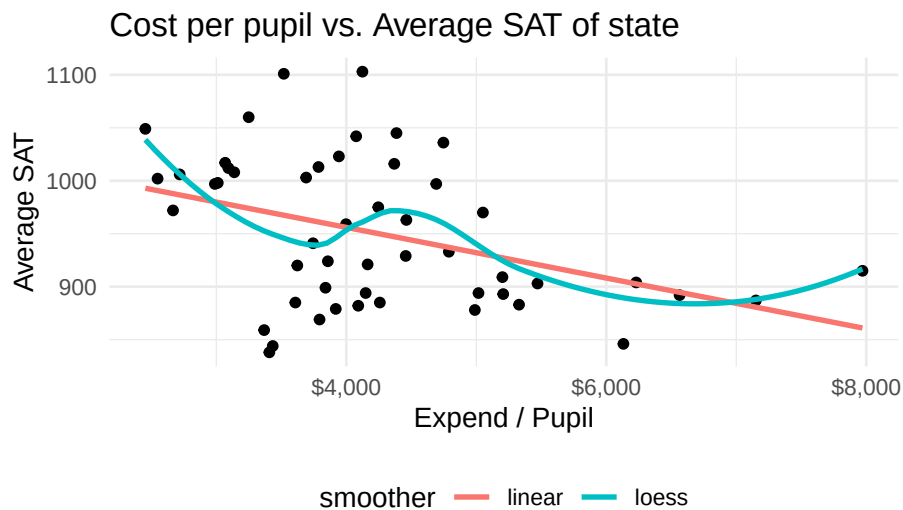
```
library( scales ) # for label_dollar(), below
ggplot( sat93, aes( expend, tot_sat ) ) +
  geom_point() +
  scale_x_continuous( labels = label_dollar() ) +
  labs( title = "Cost per pupil vs. Average SAT of state",
        y = "Average SAT",
        x = "Expend / Pupil" ) +
  theme_minimal() +
  scale_y_continuous( limits = c( 750, 1250 ),
                      breaks = c( 750, 1000, 1250 ) )
```



13 Making legends using aes()

If you are making a bunch of lines and want to color them, you can do so by giving them all names and then letting R pick the colors for you. This also lets you have a nice legend telling you which color is which. Also note how you can put a name for your legend inside `labs()`. Witness!

```
library( scales ) # for label_dollar(), below
ggplot( sat93, aes( expend, tot_sat ) ) +
  geom_point() +
  scale_x_continuous( labels = label_dollar() ) +
  labs( title = "Cost per pupil vs. Average SAT of state",
        y = "Average SAT",
        x = "Expend / Pupil",
        color = "smoother" ) +
  geom_smooth( aes( col = "linear" ), method="lm", se=FALSE ) +
  geom_smooth( aes( col = "loess" ), se=FALSE ) +
  theme_minimal() +
  theme(legend.position = "bottom" )
```

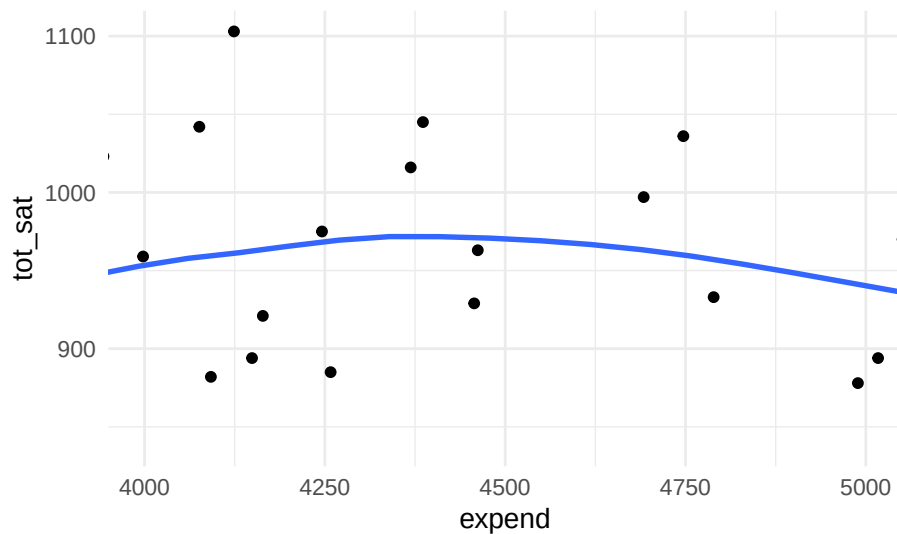


14 Cropping vs. zooming in

If you set the scale of your plot via `scale_x_continuous()` you will drop data from your analysis. If you use `coord_cartesian()` you will zoom in on the specified window, but it will be using all the data for smoothers, etc. This can be an important difference:

```
ggplot( sat93, aes( expend, tot_sat ) ) +  
  geom_point() +  
  geom_smooth( se=FALSE ) +  
  coord_cartesian( xlim=c(4000, 5000) )
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

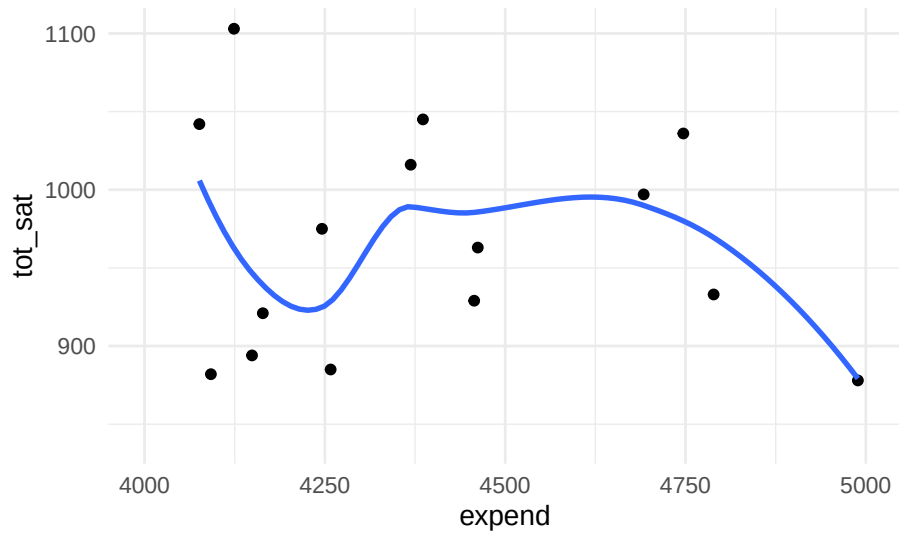


```
ggplot( sat93, aes( expend, tot_sat ) ) +  
  geom_point() +  
  geom_smooth( se=FALSE ) +  
  scale_x_continuous( limits = c(4000, 5000) )
```

``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

Warning: Removed 36 rows containing non-finite outside the scale range
(`stat\_smooth()`).

Warning: Removed 36 rows containing missing values or values outside the scale range
(`geom\_point()`).



15 Controlling size in markdown files

You can add `fig.height=2`, `fig.width=7` into the header of a chunk. For example, the chunk above had:

```
{r,r my_figure, fig.height=2, fig.width=7}
```

You can also set overall size in your setup chunk via

```
knitr::opts_chunk$set(echo = TRUE,  
  fig.width = 5,  
  fig.height = 3,  
  out.width = "75%",  
  fig.align = "center")
```

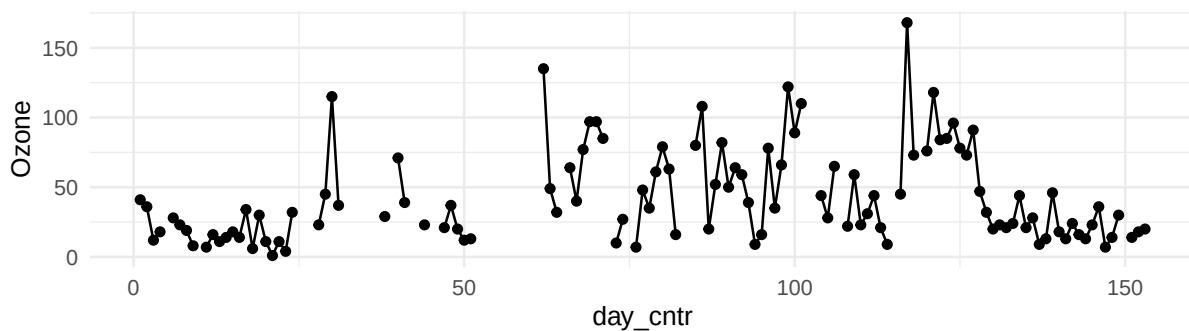
The `fig.width` controls how big the figure is when plotting. The figure is then rescaled to fit into the area dictated by `out.width`. The “75%” can be replaced with, e.g., “4in” for 4 inches. The percent is the percent of text width of the page.

16 Making the fonts of labels, etc., larger

If you make the figure smaller, the axes labels and line thicknesses and everything get larger, relatively speaking. Compare the following, with the header chunks printed out for clarity:

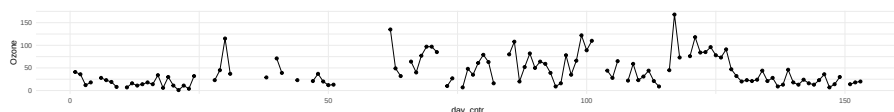
```
{r,r, fig.height=2, fig.width=7, out.width="100%", out.height="100%"}
```

```
ggplot( airquality, aes( day_cntr, Ozone ) ) +  
  geom_point(na.rm=TRUE) + geom_line(na.rm=TRUE)
```



```
{r,r, fig.height=2, fig.width=16}
```

```
ggplot( airquality, aes( day_cntr, Ozone ) ) +  
  geom_point(na.rm=TRUE) + geom_line()
```



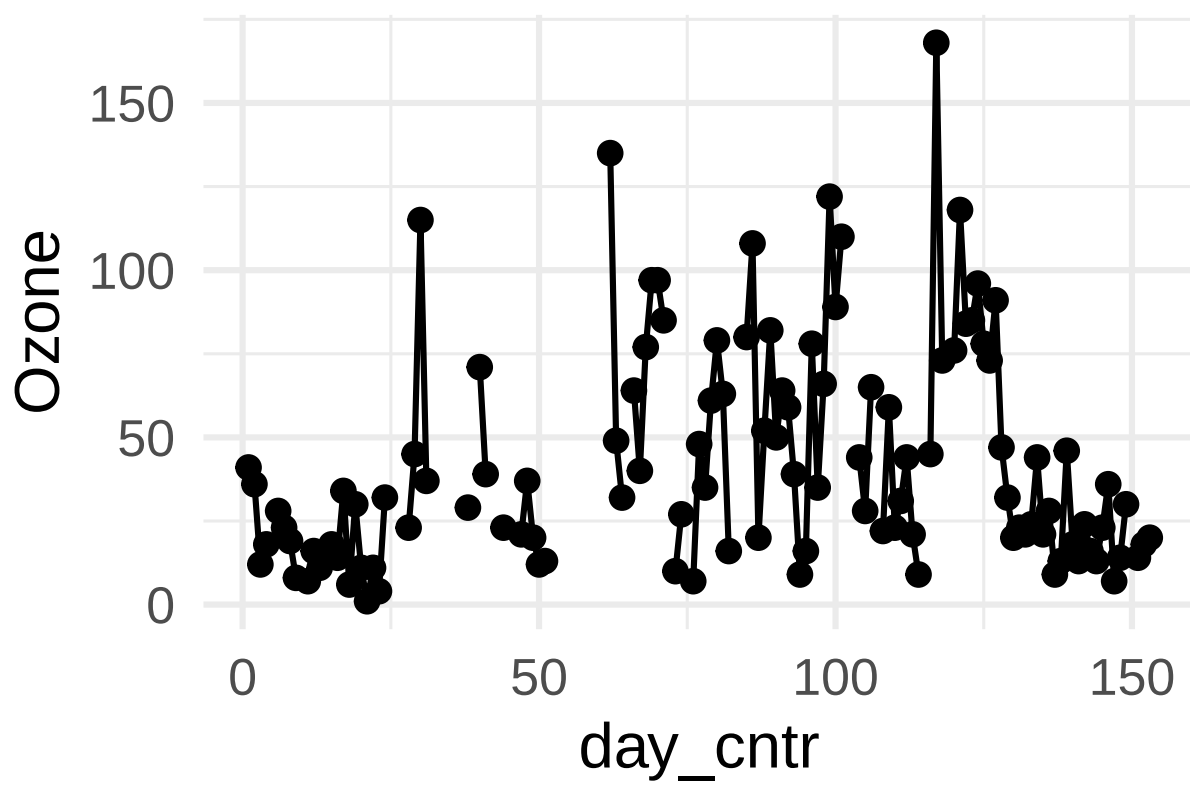
See how the second figure is a bigger figure rescaled to fit into a smaller space (width-wise). Everything thus looks tiny. The other plots in this document are `fig.height=2`, `fig.width=7`.

If the plot is small, but `out.width` is large, it will not scale up but instead keep to the desired size. The `out.width` only ensures plots are no larger than given. E.g. `fig.width=3`, `fig.height=2`, `out.width="100%"` gives:

```
{r,r, warning = FALSE, fig.width=3, fig.height=2, out.width="100%"}
```



```
ggplot( airquality, aes( day_cntr, Ozone ) ) +  
  geom_point(na.rm=TRUE) + geom_line()
```



17 Saving high resolution plots

There are three ways to export plots from RStudio. You can click on the export button on RStudio. If you say “Copy to Clipboard” you will get a bitmap plot (basically a digital photo of your plot), which will be fuzzy if you blow it up in your report. One way around that is specify a larger width and height in the dialog box before you save. “Save as Image” has the same concern.

The “Save to PDF” will save a vector image of your plot. Vector images remember the dots and lines used for your plot, and will be crisp if you make them large or small.

You can also use `ggsave` as follows:

```
ggplot( airquality, aes( day_cntr, Ozone ) ) +  
  geom_point(na.rm=TRUE) + geom_line()  
ggsave( "my_ac_plot.pdf", width = 8, height=2 )
```

This allows you to specify the width and height and then, if you want to re-make your plot, it will save the exact same size. This is also good if you have a series of plots you want to make the same size.

I recommend saving to PDF. The image bitmap plots always look a bit lousy.

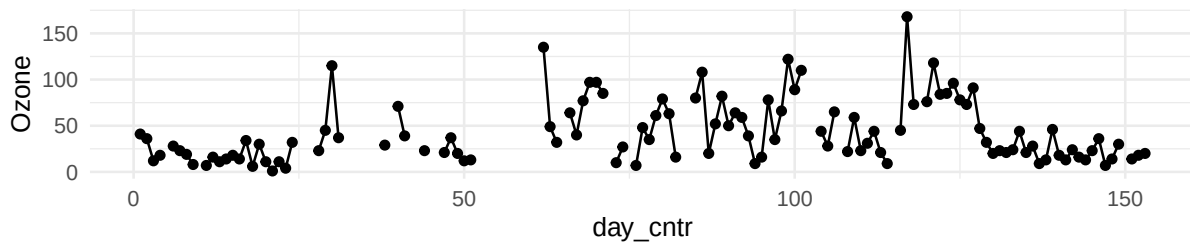
18 Think about the height of a plot.

Very often, making your plot *shorter* will make it better. Big vertical distance amplifies variation and makes it hard to follow trends.

Compare

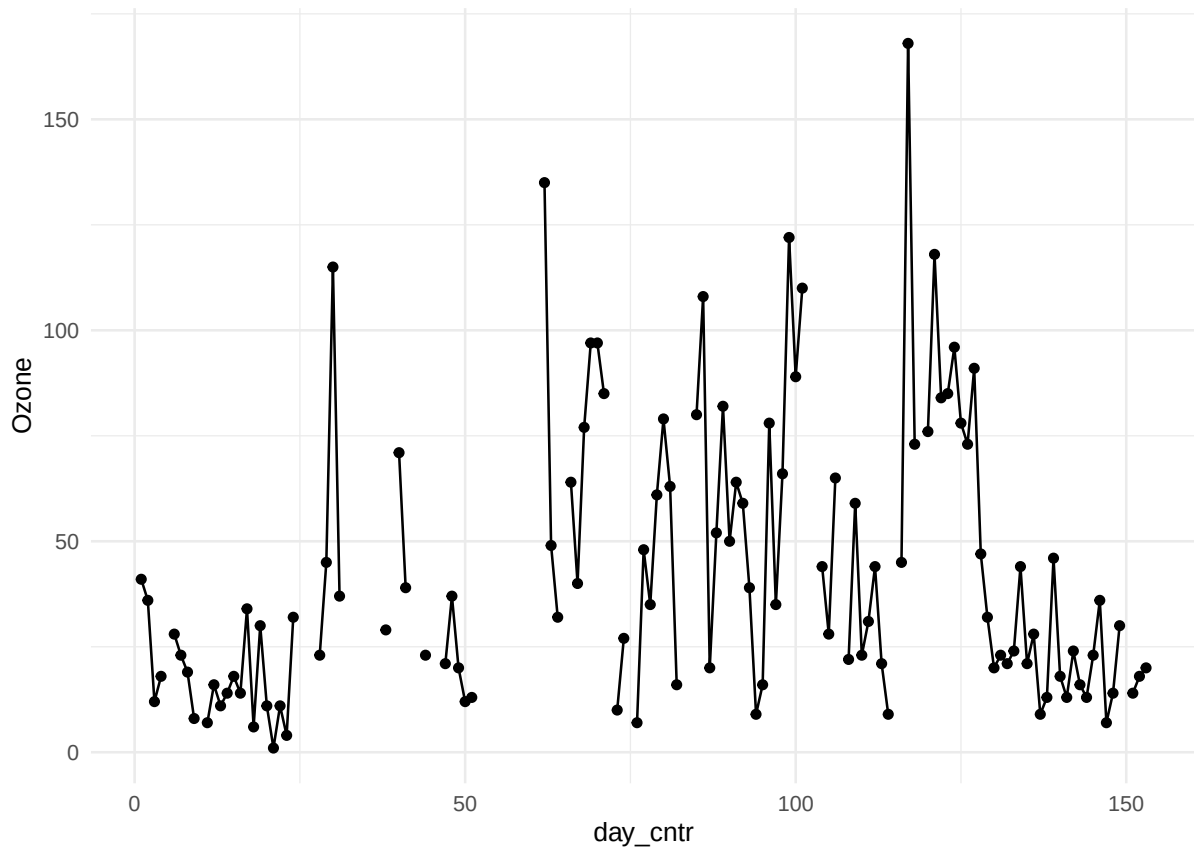
```
{r,r, fig.height=1.5, fig.width=7, out.width="100%", out.height="100%"}
```

```
ggplot( airquality, aes( day_cntr, Ozone ) ) +  
  geom_point(na.rm=TRUE) + geom_line(na.rm=TRUE)
```



```
{r,r, fig.height=5, fig.width=7, out.width="100%", out.height="100%"}
```

```
ggplot( airquality, aes( day_cntr, Ozone ) ) +  
  geom_point(na.rm=TRUE) + geom_line(na.rm=TRUE)
```

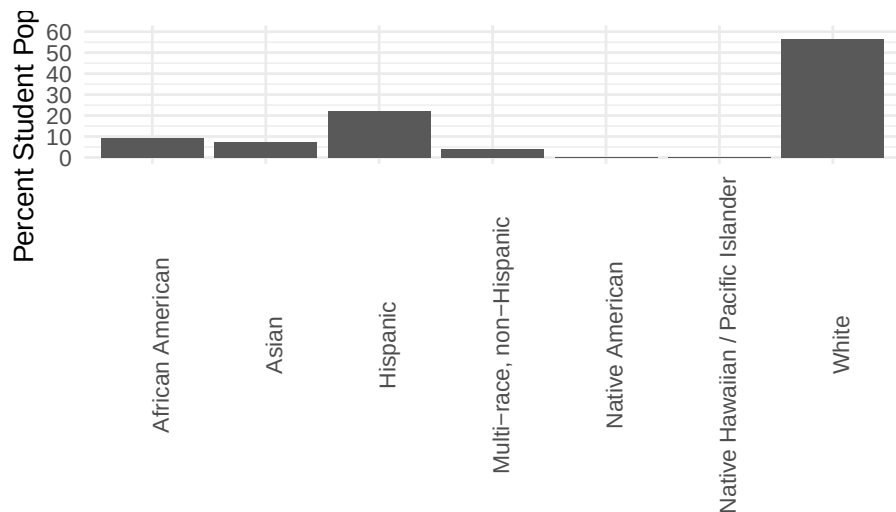


19 Think about the x and y axis of your plot

Similarly, swapping the x and y axes of a plot can make labels *a lot* easier to read.

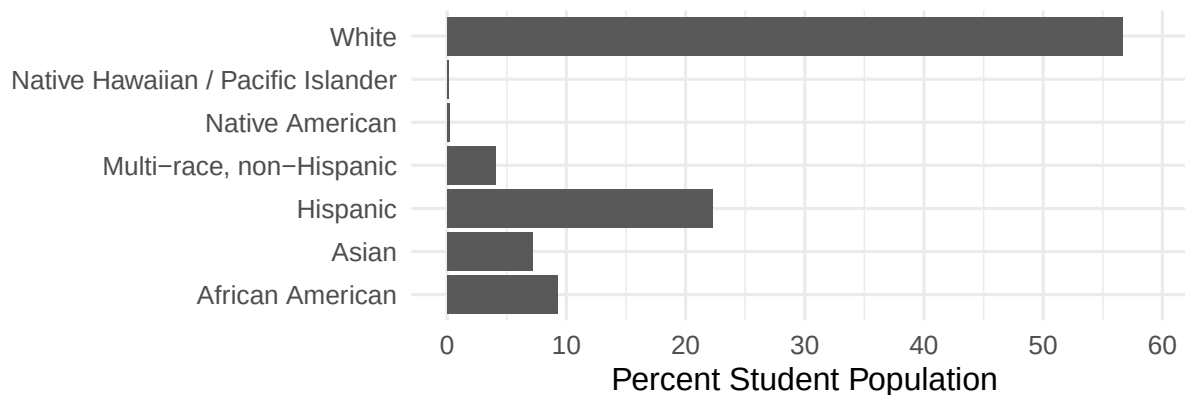
Consider the following two plots:

```
ggplot( dat, aes( x = Category, y=per_students ) ) +  
  geom_bar(stat="identity") +  
  theme(axis.text.x = element_text(angle = 90)) +  
  scale_y_continuous( limits = c(0,60), breaks = c(0,10,20,30,40,50,60) ) +  
  labs( y = "Percent Student Population", x="" )
```



```
{r,r, fig.height=2, fig.width = 6, out.width="100%"}
```

```
ggplot( dat, aes( x = Category, y=per_students ) ) +  
  geom_bar(stat="identity") +  
  theme(axis.text.x = element_text(angle = 90)) +  
  labs( y = "Percent Student Population", x="" ) +  
  scale_y_continuous( limits = c(0,60), breaks = c(0,10,20,30,40,50,60) ) +  
  coord_flip() +  
  theme(axis.text.x = element_text(angle = 0))
```



Flipping is easy to do. Just add `coord_flip()` to your `ggplot` command! The choice of vertical height of the plot also matters here. Make your bars not too thick, but still a nice amount thick so your labels are well spaced.

For the first plot, we had to rotate the labels so they didn't overlap. Other choices are likely possible.

Also notice the rotating of the axes labels. With `coord_flip` be careful as to what element you are controlling. Usually fiddling around will get it right in no time!

Note: these data are from the Massachusetts Department of Elementary and Secondary Education, school year 2019-20, for the state.

20 Controlling colors, etc

For this, read the “Graphics for Communication” chapter of R for DS: <https://r4ds.had.co.nz/graphics-for-communication.html>

21 Tennessee Star Handout

This handout demonstrates a step-by-step worked example of creating a publication-ready plot, along with improvements and stylistic enhancements. Additionally, you will find detailed explanations and use cases for several useful functions, such as `grid.arrange()`, `annotate()`, and `geom_text()`, to help you refine your visualizations effectively.

```
# Load in required libraries
library(tidyverse) #this automatically loads ggplot2

# Note: you will likely need to install the Ecdat package to load the
# data we use via:
#install.packages("Ecdat")
library(Ecdat)

library(gridExtra) # Required for using grid.arrange()
```

21.1 Re-creating the Tennessee STAR Boxplot

This handout provides the solution to the in-class plotting exercise where we recreated a plot of the Tennessee Star dataset. In this handout we start with a simple boxplot and work our way up to the final visualization. When then look at some extensions.

In our target plot, we want to visualize reading test scores (`treadssk`) by the type of classroom randomly assigned (`classsk`) along with variables showing whether the student was eligible for free or reduced price lunch in kindergarten (`freelunk`) and race (`race`). The plot we are trying to mimic looks like this:

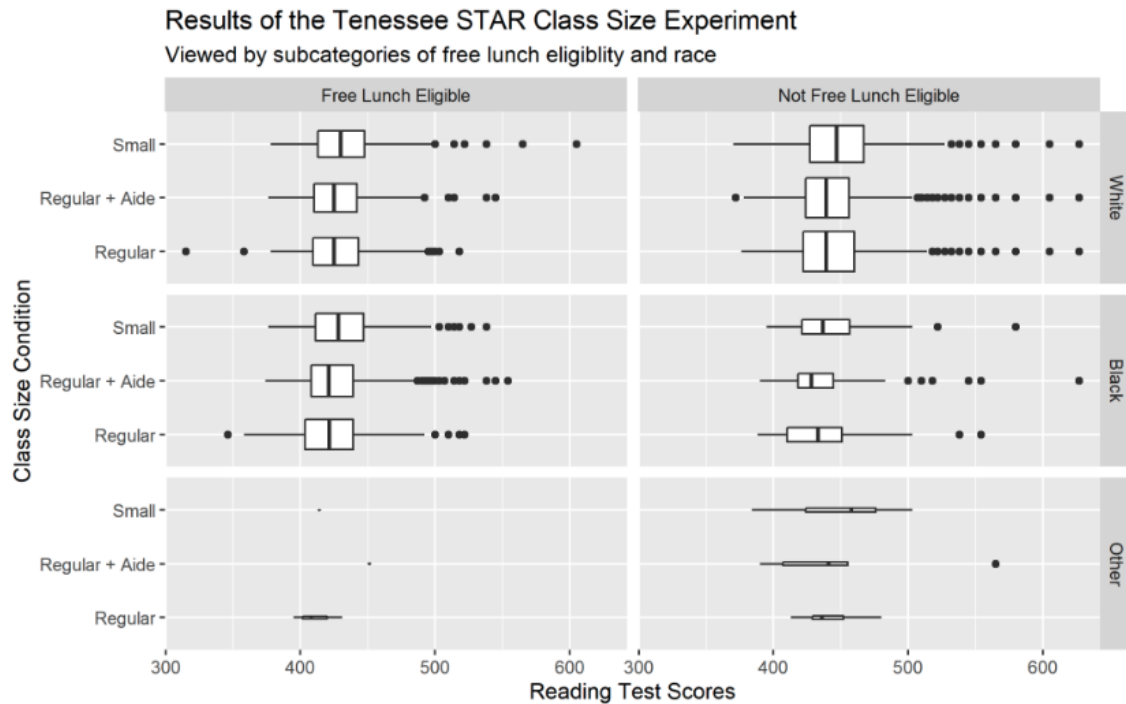
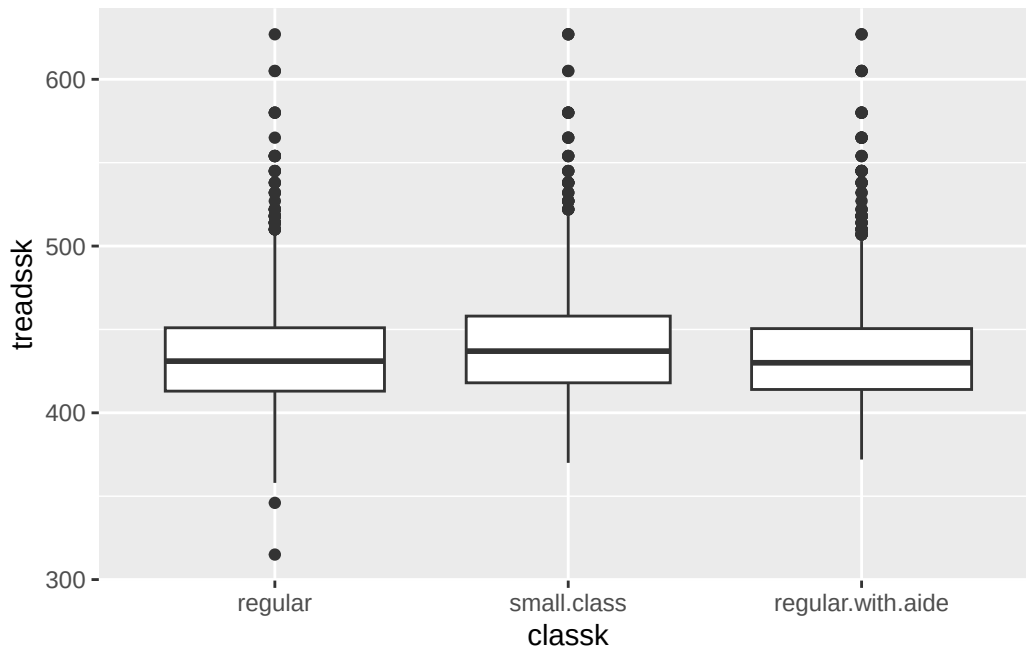


Figure 21.1: The target plot we are trying to recreate

It's useful to start small, and then build up iteratively to more complex plots. Start with a simple `geom_boxplot` of test scores, and then gradually make it more complex to try and emulate the original. Some manipulations are easier done in the `ggplot` itself, whereas other things might be easier if you manipulate the data *before* you give it to `ggplot`.

Let's start by a single boxplot that separate students only by class sizes:

```
star <- Ecdat::Star
star %>%
  ggplot(aes(x = classsk, y = treadssk)) +
  geom_boxplot(varwidth = TRUE)
```

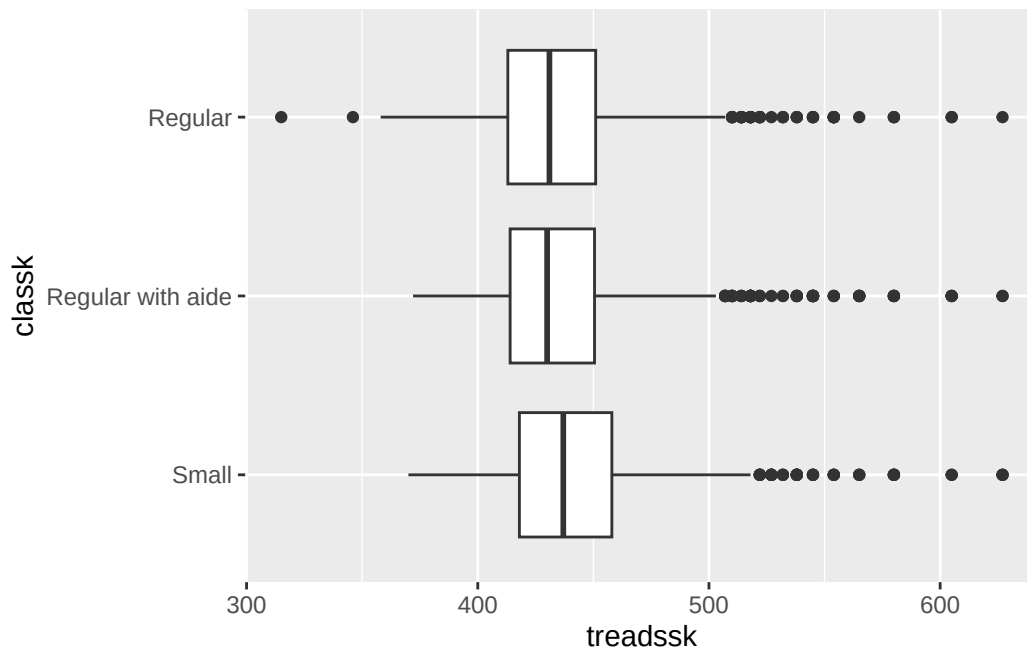


If we want to change the labels of each group, we can convert our `classk` to a factor. The order of the levels will determine the order of the boxes in the plot. Also, the boxes are vertical instead of horizontal, as in the example. We can fix this with `coord_flip()`. Here we go:

```
# some pre-processing to make the class sizes a proper factor
star <- mutate(star, classk = factor(classk,
                                     levels = c("small.class",
                                                 "regular.with.aide",
                                                 "regular"),
                                     labels = c("Small",
                                                "Regular with aide",
                                                "Regular"))))

# build simple boxplot
star %>%
  ggplot(aes(x = classk, y = treadssk)) +
  # When varwidth = TRUE,
  # the width of each boxplot is proportional to the
  # sample size of that category. This means that categories with
  # more data points will have wider boxplots, while categories with
  # fewer data points** will have narrower boxplots.
  # If varwidth = FALSE (default), all boxplots have the
  # same width regardless of sample size.
  geom_boxplot(varwidth = TRUE) +
```

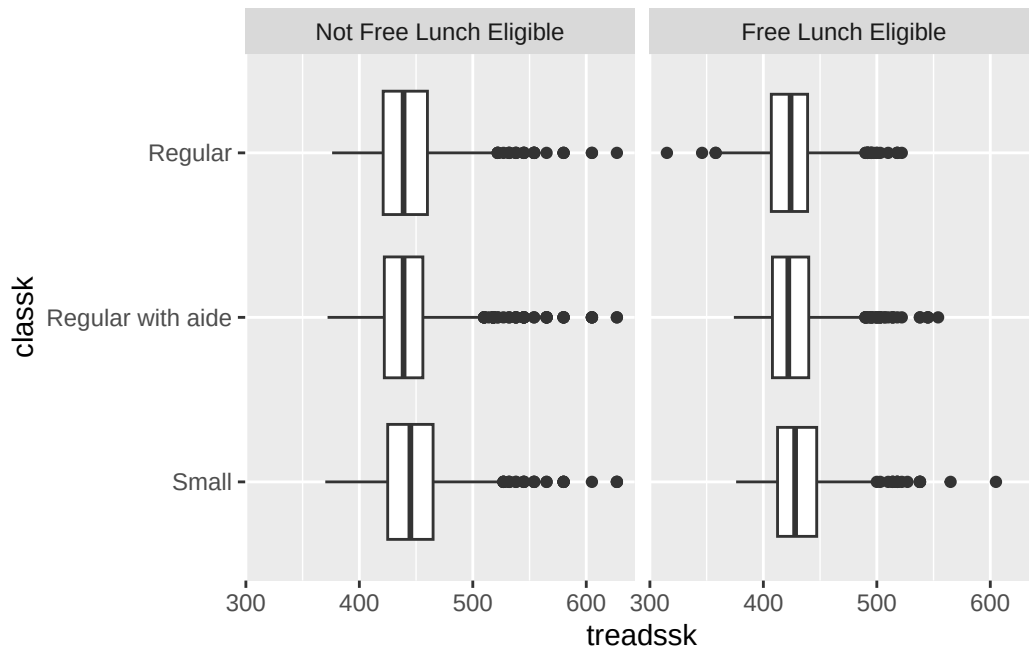
```
coord_flip() # Flip axes for better readability
```



Now let's add in just free lunch as an additional categorical variable.

```
# again, pre-process to make sure that free lunch is properly labeled factor
star <- mutate(star, freelunk = recode(freelunk,
                                       no = "Not Free Lunch Eligible",
                                       yes = "Free Lunch Eligible"))
```

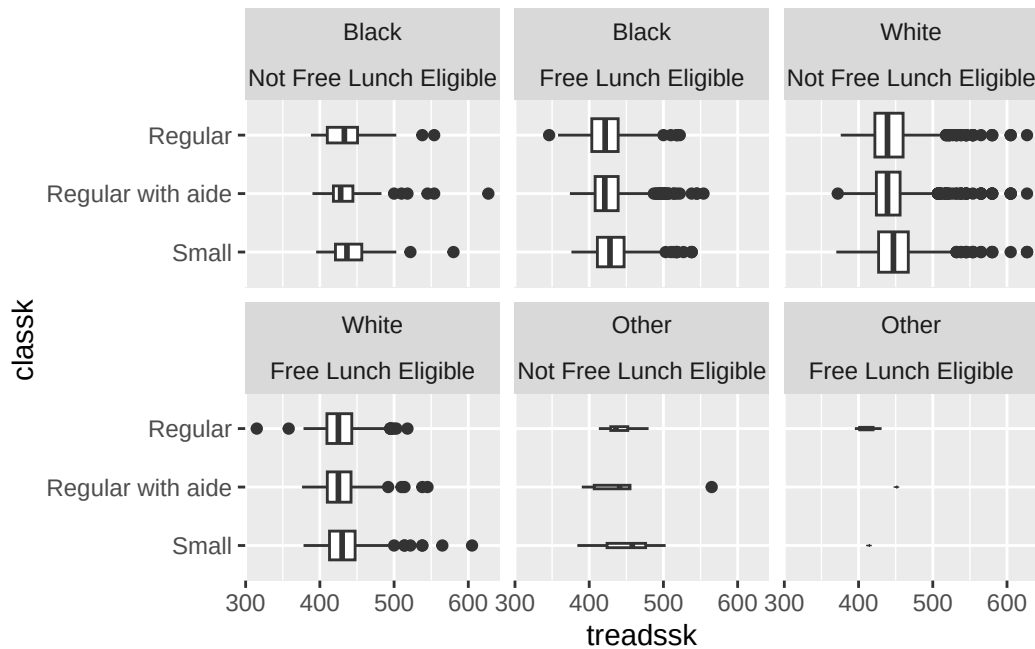
```
star %>%
  ggplot(aes(x = classk, y = treadssk)) +
  facet_wrap( ~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip()
```



Add in race as another dimension of variability:

```
# making sure race is also a properly labeled factor
star <- mutate(star, race = factor(race,
                                   levels = c("black", "white",
                                                "other"),
                                   labels = c("Black", "White",
                                                "Other"))))

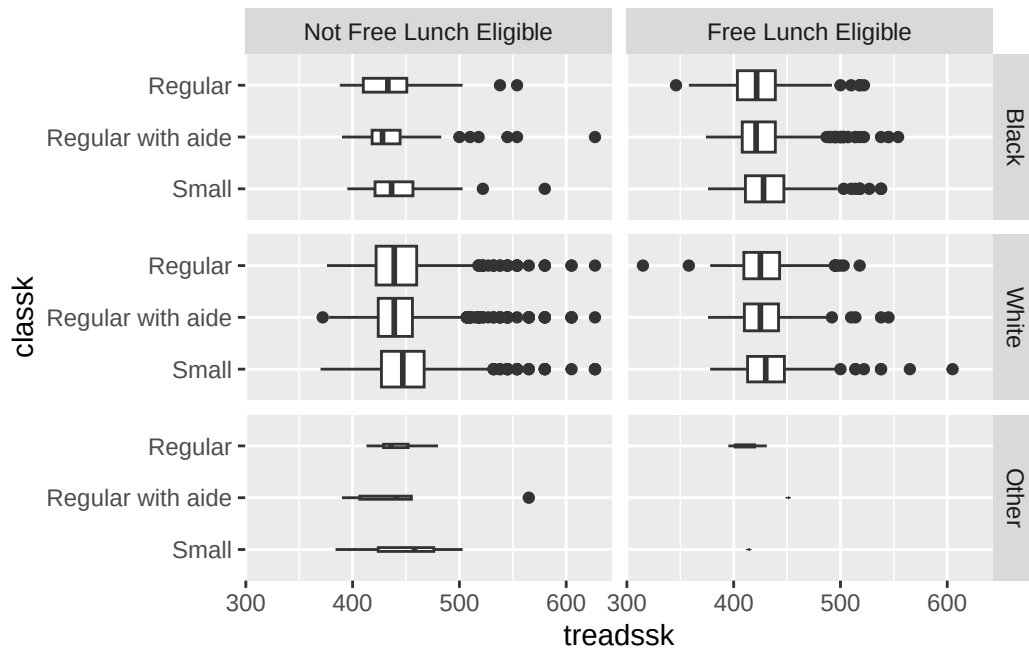
star %>%
  ggplot(aes(x = classk, y = treadssk)) +
  facet_wrap( race~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip()
```



It looks almost there right now. But `facet_wrap` seems to create six completely separate panels. Let's try `facet_grid` instead.

Side note: if our goal is simply to create a “good” plot instead of replicating an existing one, is `facet_wrap` or `facet_grid` the better choice here? Why?.

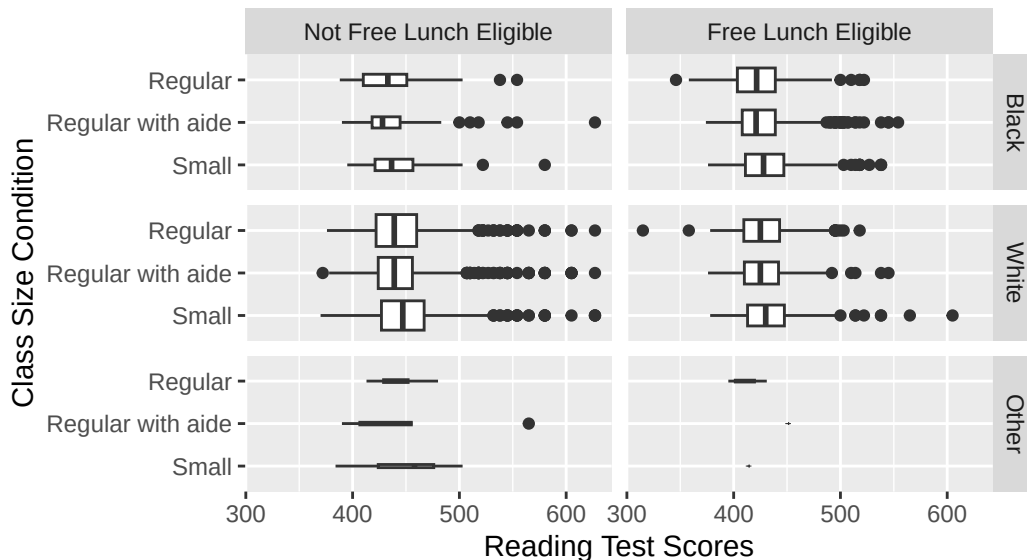
```
star %>%
  ggplot(aes(x = classk, y = treadssk)) +
  facet_grid( race~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip()
```



Last but not least, let's add in titles and axes:

```
star %>%
  ggplot(aes(x = classk, y = treadssk)) +
  facet_grid( race~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip() +
  labs(title = "Results of the Tennessee STAR Class Size Experiment",
       subtitle = "Viewed by subcategories of free lunch eligibility and race",
       y = "Reading Test Scores",
       x = "Class Size Condition")
```

Results of the Tennessee STAR Class Size Experiment Viewed by subcategories of free lunch eligibility and race



In this iterative process, we managed to build a clean (i.e., not much chart junk) and accessible (e.g., relatively low cognitive load to process, friendly to people with color vision deficiencies) plot to represent the data. Viewers are able to see that (visually, as we haven't done any statistical tests yet) class size appears to impact reading test scores differently for students with different races and family incomes.

21.1.1 (Optional) Recreating the boxplot without manipulating the original dataset

We can label things on the fly rather than making the factors in our dataset different, like so:

```
# Loading the Tennessee Star dataset in a new data frame
star_new <- Ecdat::Star

# The ggplot code below shows us how we can modify the facet labels
# without manipulating the dataset. We pass the new dataset star_new
# to ggplot, and use the labeller function inside facet_grid to rename
# facet labels. In the .cols argument, we specify the data label we want
# to see in the plot, and set it equal to the value the label corresponds to.

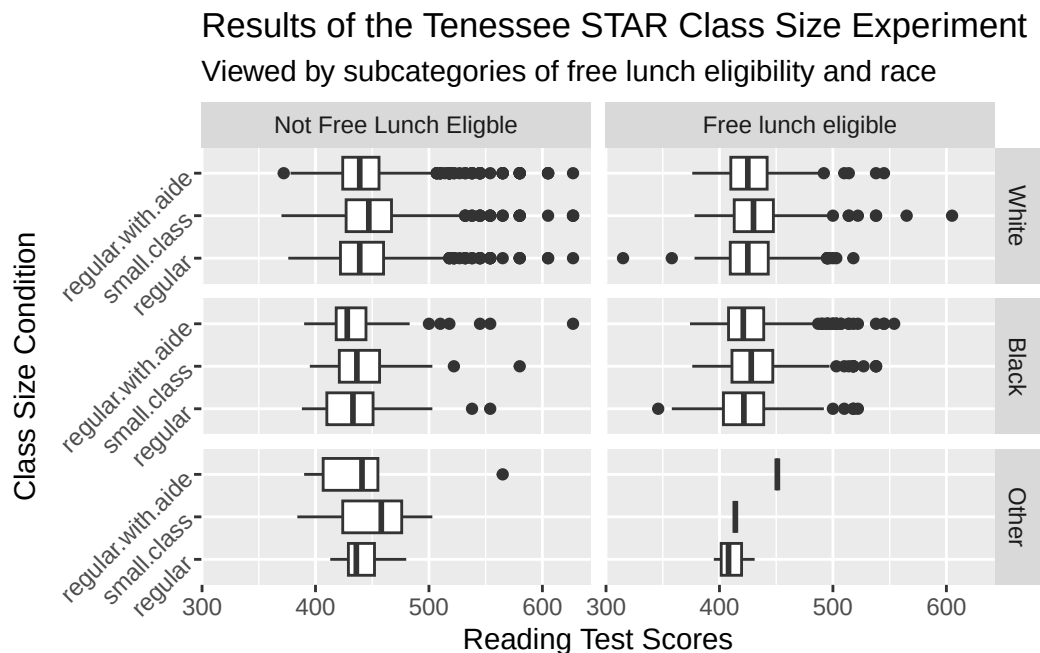
ggplot(data=star_new, aes(x=treadssk, y=classk)) +
  geom_boxplot() +
  facet_grid(race~freelunk,
```

```

# the facet columns correspond to the freelunk variable
# here, we say where you see "yes", label it "Free lunch eligible"
# Similarly, we label the remaining facet columns and rows
labeller = labeller(.cols = c("yes" = "Free lunch eligible",
                             "no" = "Not Free Lunch Eligible"),
                   .rows = c("white" = "White",
                             "black" = "Black",
                             "other" = "Other")) +

labs(x = "Reading Test Scores",
     y = "Class Size Condition",
     title = "Results of the Tennessee STAR Class Size Experiment",
     subtitle = "Viewed by subcategories of free lunch eligibility and race") + # we use t
# to be tilted at a 45-degree angle for a cleaner look
theme(axis.text.y = element_text(angle = 45))

```



Which you do is a matter of coding taste, for the most part.

21.2 Bundling plots with grid.arrange()

If we want to place two completely separate plots side by side without faceting, we can use the `grid.arrange()` function. `grid.arrange()` allows you to arrange multiple ggplot2 plots in a grid layout. This is useful when you want to display multiple visualizations side by side in a single output.

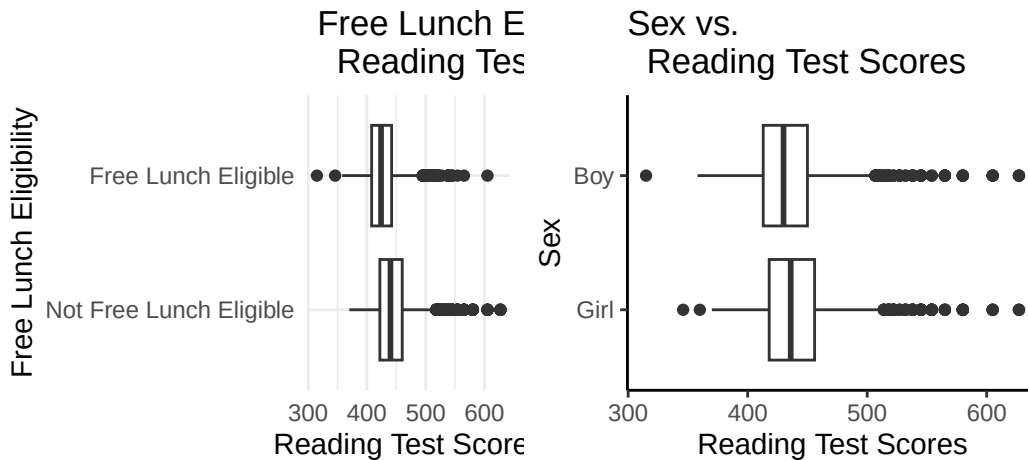
How is `grid.arrange()` different from faceting?

- `grid.arrange()` is used to arrange **completely separate ggplot objects** together in a grid
- Each plot can have **different aesthetics, geoms, themes, or datasets**
- Faceting (`facet_wrap()` or `facet_grid()`) is used **within a single ggplot object** to split data into multiple subplots based on a categorical variable
- Faceting is ideal when **all plots share the same x/y variables**, while `grid.arrange()` is better for combining **independent plots** that may have different structures

The code chunk below demonstrates use of this function.

```
# Let's go back to the Tennessee Star dataset,  
# and create two boxplots, one between reading test scores  
# and free lunch eligibility, and the second between  
# reading test scores and sex.  
  
# Assign the first plot to a variable called plot_1  
plot_1 <- star %>%  
  ggplot(aes(x=treadssk, y=freelunk)) +  
  geom_boxplot() +  
  theme_minimal() +  
  labs( #Adding some labels here for x-axis, y-axis and the plot title  
        title = " Free Lunch Eligibility vs.\n  Reading Test Scores",  
        x = "Reading Test Scores",  
        y = "Free Lunch Eligibility"  
      )  
  
# Assign the first plot to a different variable called plot_2  
plot_2 <- star %>%  
  ggplot(aes(x=treadssk, y=sex)) +  
  geom_boxplot() +  
  theme_classic() +  
  labs( #Adding some labels here for x-axis, y-axis and the plot title  
        title = "Sex vs.\n  Reading Test Scores",  
        x = "Reading Test Scores",  
        y = "Sex"  
      ) + # Below, we use the scale_y_discrete function to rename labels on y-axis  
# to capitalize the first letter in the label  
  scale_y_discrete(labels = c("boy" = "Boy", "girl" = "Girl"))  
  
# Using grid.arrange(), we arrange plot_1 and plot_2 in a single row
```

```
# with two columns
grid.arrange(plot_1, plot_2, ncol = 2, nrow = 1)
```



21.3 Improvements on the Tennessee STAR Boxplot

21.3.1 What if we want to add a different color for the small class sizes to highlight the distributions in that category?

You can set colors quite easily. You may find the [documentation page](#) for `scale_manual` helpful. For a list of color names, this [link](#) will be handy. Here is a demonstration:

```
star %>%
  ggplot(aes(x = classk, y = treadssk, col=classk)) +
  facet_grid( race~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip() +
  labs(title = "Results of the Tennessee STAR Class Size Experiment",
       subtitle = "Viewed by subcategories of free lunch eligibility and race",
       y = "Reading Test Scores",
       x = "Class Size Condition") +
  # The scale_color_manual function below manually sets colors for the classk variable.
  # The first category (small class size) is highlighted in orange,
  # while the other class size categories are set to gray.
  # This helps emphasize the small class size condition in the visualization.
  scale_color_manual(values=c("orange", "gray", "gray"))
```

Results of the Tennessee STAR Class Size Experiment Viewed by subcategories of free lunch eligibility and race

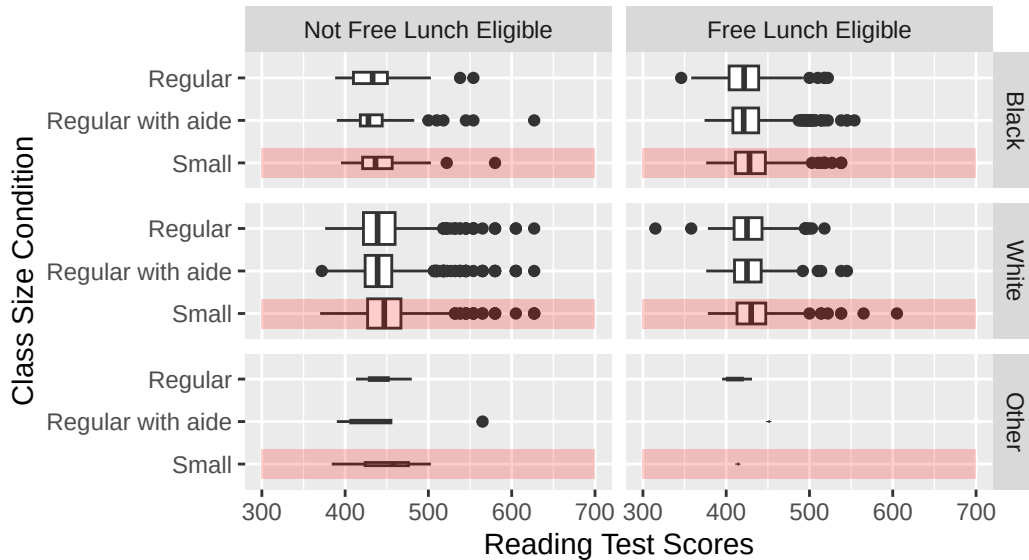


21.3.2 What if we want to annotate all the boxplots corresponding to the the small class size category?

You can also annotate your plot. You may find the [documentation page](#) for `annotate` helpful.

```
star %>%
  ggplot(aes(x = classk, y = treadssk)) +
  facet_grid(race~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip() +
  labs(title = "Results of the Tennessee STAR Class Size Experiment",
       subtitle = "Viewed by subcategories of free lunch eligibility and race",
       y = "Reading Test Scores",
       x = "Class Size Condition") +
  annotate("rect", xmin = 0.65, xmax = 1.35, ymin = 300, ymax = 700,
         alpha = .2, fill="red")
```

Results of the Tennessee STAR Class Size Experiment Viewed by subcategories of free lunch eligibility and race



Explanation of the `annotate()` function

The `annotate()` function is used to **add custom shapes or text** to a ggplot. Here, “rect” (rectangle) is used to **highlight** a specific region in the plot.

- **`xmin = 0.65`, `xmax = 1.35`**: `classk` is a **factor**, so ggplot assigns it numeric positions (e.g., 1 = small class, 2 = regular with aide, 3 = regular). The values 0.65 and 1.35 **define the “height” of the rectangle**, but because of `coord_flip()`, this actually affects the **y-axis** instead of the x-axis.
- **`ymin = 300`, `ymax = 700`**: Defines the **horizontal** range of the rectangle (test scores from 300 to 700), since `coord_flip()` swaps the x- and y-axes.
- **If `coord_flip()` was not used** and we instead used `x = treadssk`, `y = classk` in the ggplot aesthetics, we would need to swap `xmin` with `ymin` and `xmax` with `ymax`: `xmin = 300`, `xmax = 700` (test scores would be on the x-axis) and `ymin = 0.65`, `ymax = 1.35` (class size categories would be on the y-axis). This is because, without `coord_flip()`, the **x-axis stays horizontal and the y-axis stays vertical**.
- **`alpha = 0.2`**: This controls transparency (0 = fully transparent, 1 = fully opaque). A value of 0.2 makes the rectangle **semi-transparent**, so the data underneath is still visible.
- **`fill = “red”`**: The rectangle is filled with red, drawing attention to the selected region.

21.4 What about adding text to the annotated Tennessee Star Boxplot?

We next demonstrate adding some text to the annotated plot above. In particular we add the useful eye catching “Here!” in red.

```
p <- star %>%
  ggplot(aes(x = classsk, y = treadssk)) +
  facet_grid( race~freelunk) +
  geom_boxplot(varwidth = TRUE) +
  coord_flip() +
  labs(title = "Results of the Tennessee STAR Class Size Experiment",
        subtitle = "Viewed by subcategories of free lunch eligibility and race",
        y = "Reading Test Scores",
        x = "Class Size Condition") +
  annotate("rect", xmin = 0.65, xmax = 1.35, ymin = 300, ymax = 700,
         alpha = .2, fill="red")

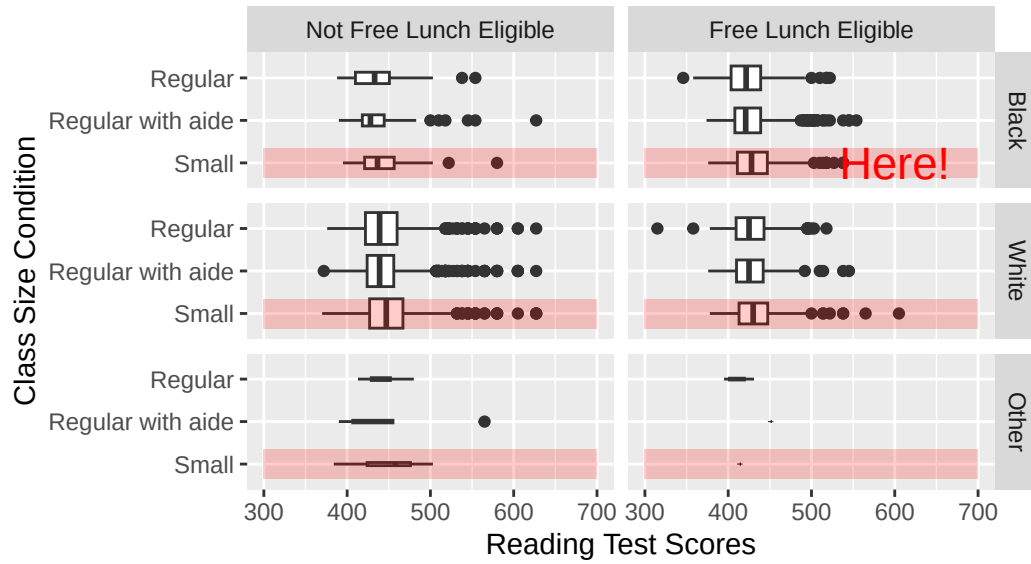
# ----- Creating a Data Frame for Text Annotation -----

text <- data.frame(
  classsk = 1, # To position the text at class size condition = 1 (small class)
  treadssk = 600, # To position the text at treadssk = 600 (as seen in the figure)
  # Below, we specify the free lunch eligibility category wherein to place the text
  freelunk = factor("Free Lunch Eligible",
                    levels = c("Not Free Lunch Eligible", "Free Lunch Eligible")),
  # Below, we specify the race category wherein to place the text
  race = factor("Black", levels=c("Black", "White", "Other")))

# ----- Adding Text Annotation to the Plot -----
# - `geom_text()` adds a text label to the plot using data from the `text` data frame.
# - `data = text` tells ggplot where to position the text.
# - `label = "Here!"` is the actual text that will appear on the plot.
# - `col = "red"` sets the text color to red.
# - `size = 6` controls the font size.
# - Since `coord_flip()` is used, `classsk = 1` (normally an x value) now determines
# the vertical position, and `treadssk = 600` (normally a y value) determines
# the horizontal position.

p + geom_text(data = text, label = "Here!", col="red", size=6)
```

Results of the Tennessee STAR Class Size Experiment Viewed by subcategories of free lunch eligibility and race



Voila!

22 Plotting Two Datasets at Once

It's easy (though not always advisable) to plot two data sets at once with `ggplot`. First, we load `tidyverse` and our HSB data. We then create a school-level aggregate data set of just the mean SES values.

```
library(tidyverse)
library(haven)

# clear memory
rm(list = ls())

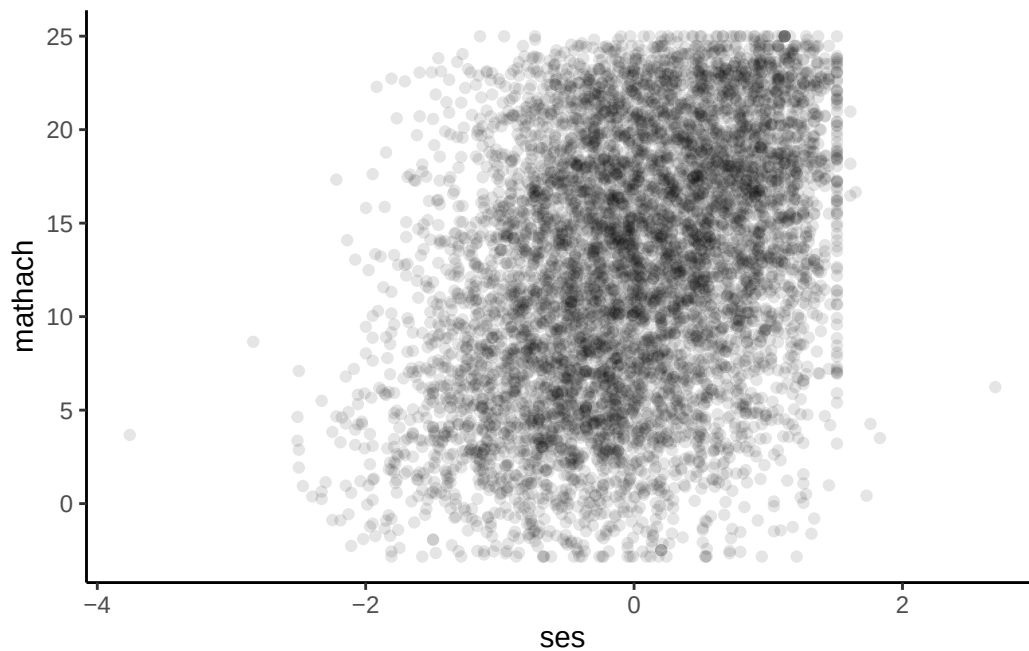
theme_set(theme_classic())

# load HSB data
hsb <- read_dta("data/hsb.dta") |>
  select(mathach, ses, schoolid)

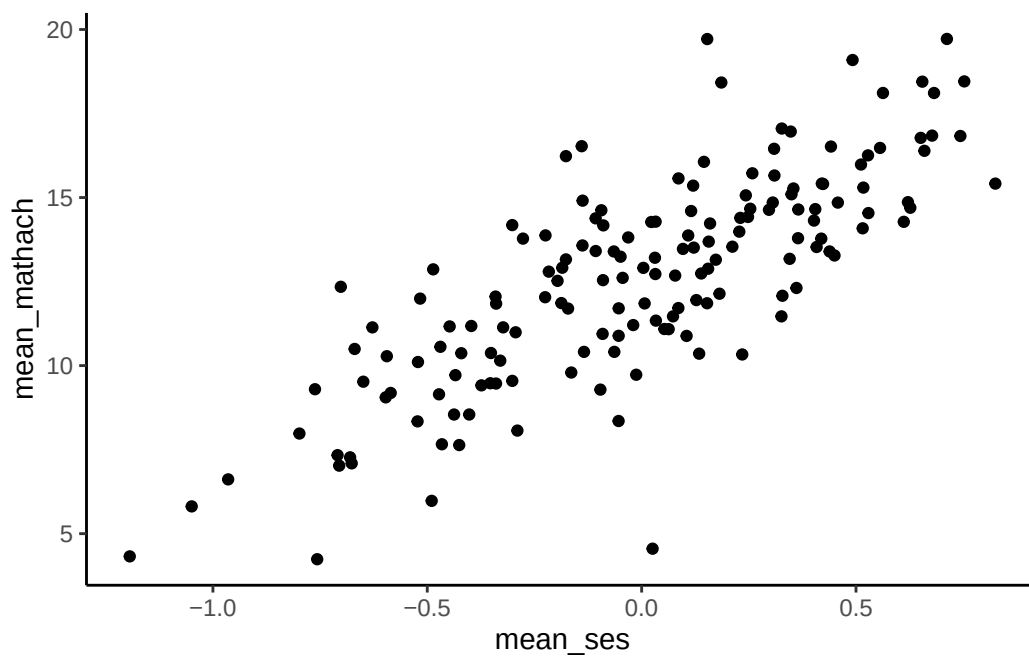
sch <- hsb |>
  group_by(schoolid) |>
  summarise(mean_ses = mean(ses),
            mean_mathach = mean(mathach))
```

Let's say we wanted to plot *both* the individual students *and* the school means. This is easy enough to do separately:

```
ggplot(hsb, aes(x = ses, y = mathach)) +
  geom_point(alpha = 0.1)
```



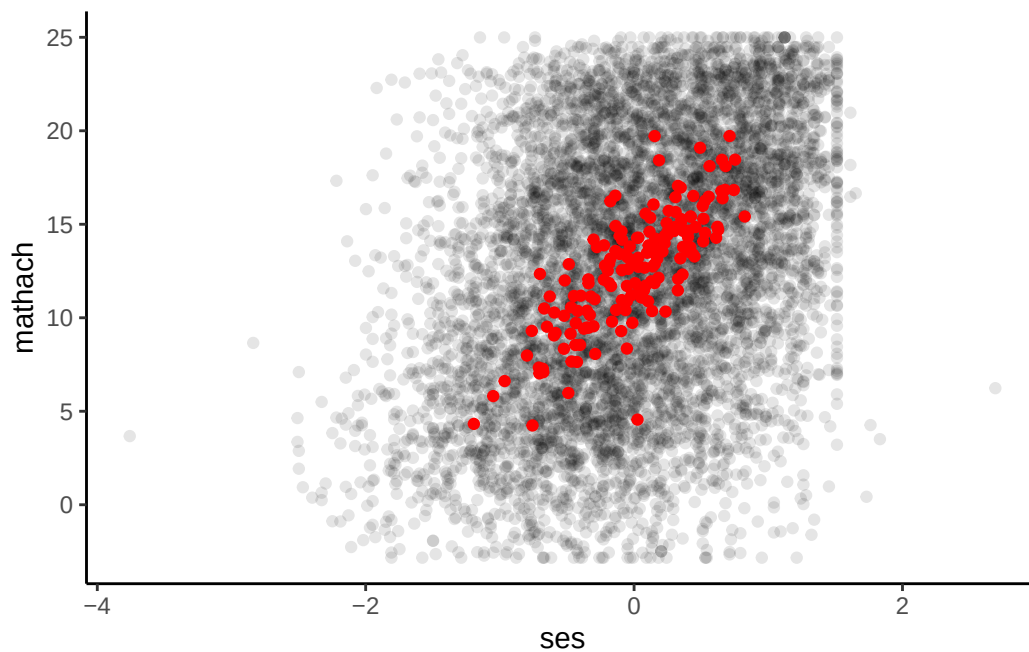
```
ggplot(sch, aes(x = mean_ses, y = mean_mathach)) +  
  geom_point()
```



We can superimpose both plots as follows. Essentially, the first argument in `ggplot` provides the data, and by default, this is passed to all subsequent layers of the plot. We can override

this behavior by specifying a different data set (and aesthetic mappings, if desired) *within an individual layer* of `ggplot`, such as `geom_point`.

```
ggplot(hsb, aes(x = ses, y = mathach)) +  
  geom_point(alpha = 0.1) +  
  geom_point(data = sch, aes(x = mean_ses, y = mean_mathach), color = "red")
```



23 Plotting aggregate data with ggPlot

This handout covers how to plot univariate distributions of data (i.e. histograms and bar charts) if your data have already been aggregated (e.g. you have the counts for each value of a variable but not the individual observations). In particular, this handout demonstrates how to create histograms with ggplot when the data have already been aggregated or tallied.

23.0.1 Preface

Before we get started, let's load the packages we'll need.

```
library(tidyverse)
library(ggplot2)
```

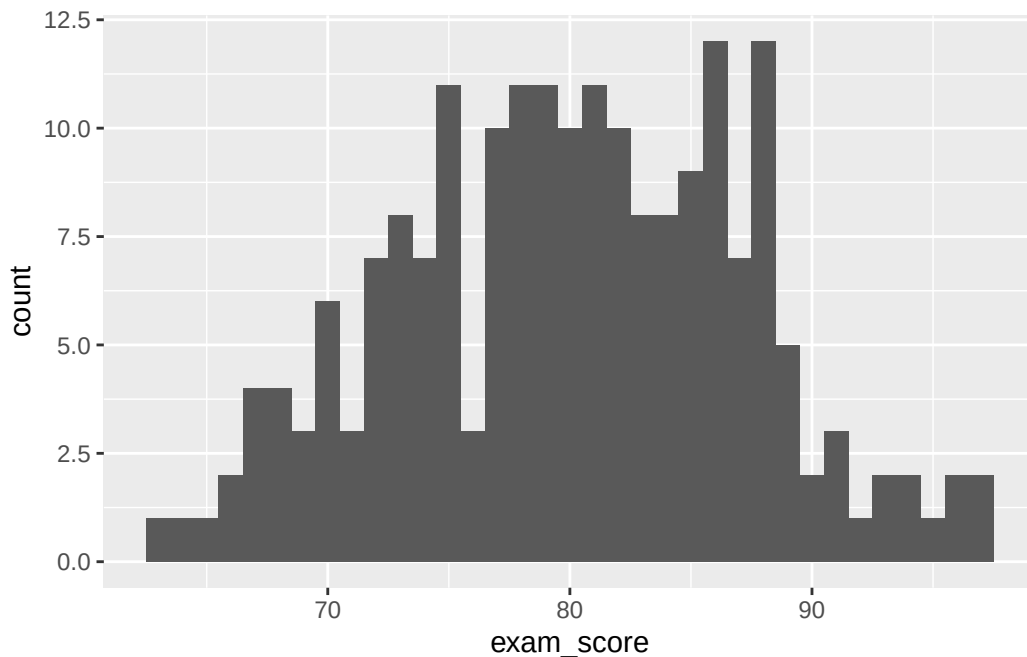
24 The Situation

Typically, we create histograms from individual data points. Suppose we have data on students. The data include a student ID number, a course identifier, the grade the student got in the course, and their final exam score (out of 100 points). Here's some example data.

```
num_students <- 200
df <- data.frame(sid=rep(1:num_students, 2),
                 course=c(rep('math', num_students), rep('ela', num_students)),
                 grade=sample(c('A', 'B', 'C', 'D', 'F'),
                             2*num_students,
                             replace=TRUE),
                 exam_score=sample(0:100, 2*num_students, replace=TRUE,
                                  prob=dnorm(0:100, mean=80, sd=7)))
```

Let's make a histogram of the exam score data for math. This should be familiar.

```
ggplot(df %>% filter(course=='math'), aes(x=exam_score)) +
  geom_histogram(binwidth=1)
```



There's a lot we could do to make this a publication-ready graphic - label the axes, add a title, etc. - but our focus here is on the core plotting, so we'll skip it.

But what if the data were already aggregated by exam score? Let me show you what I mean in code.

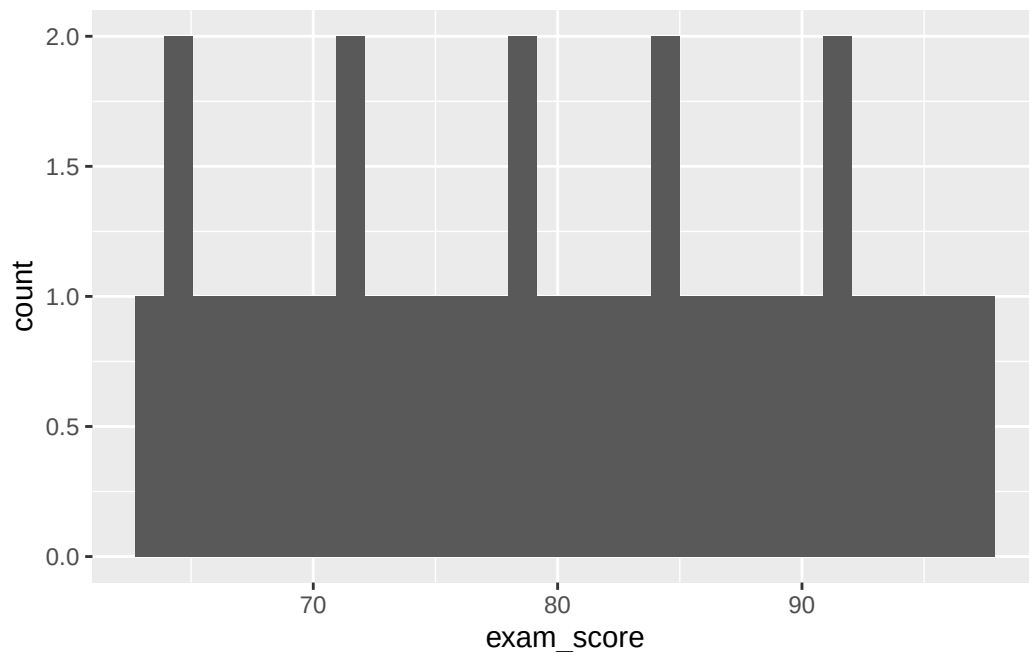
```
exam_agg <- df %>%  
  filter(course=='math') %>%  
  group_by(exam_score) %>%  
  summarize(n=n())  
exam_agg
```

```
# A tibble: 35 x 2  
  exam_score    n  
    <int> <int>  
1         63     1  
2         64     1  
3         65     1  
4         66     2  
5         67     4  
6         68     4  
7         69     3  
8         70     6  
9         71     3  
10        72     7  
# i 25 more rows
```

The dataset `exam_agg` contains only two variables: the exam score and the number of students who received that score. Sometimes, this is the way we receive data. How can we make a histogram out of this? Let's try it the old way:

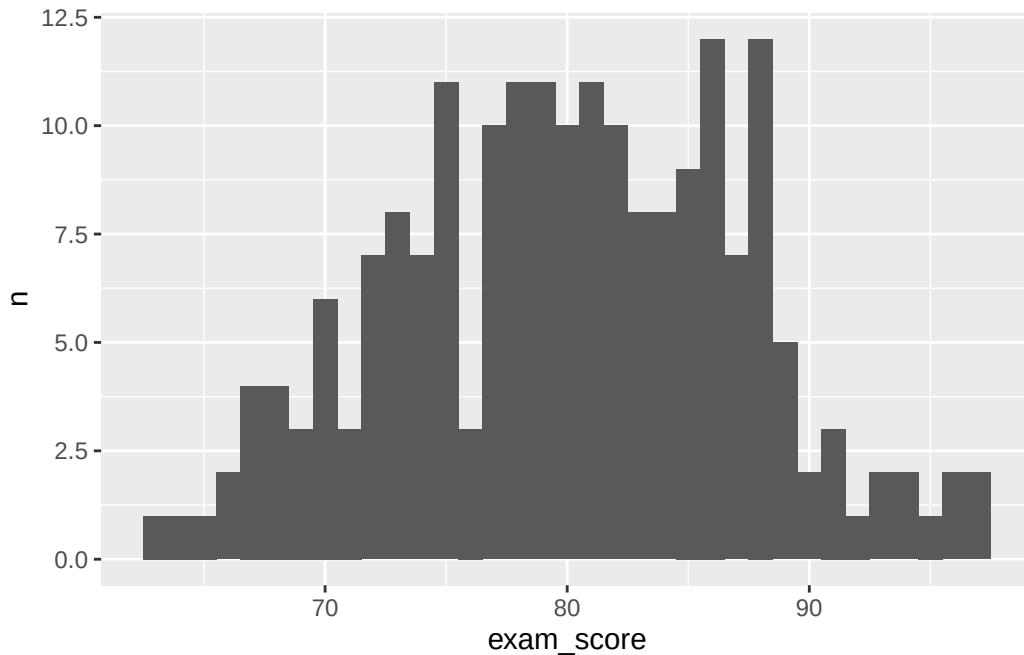
```
ggplot(exam_agg, aes(x=exam_score)) +  
  geom_histogram()
```

``stat_bin()` using `bins = 30`. Pick better value with `binwidth`.`



Whoa! That doesn't look right. The problem is that `geom_histogram()` counts how many times each score appears in the data, but after we aggregated, each score only appears once. We need to tell R that the counts are stored in the `n` column of the dataset. Here's how we do it:

```
ggplot(exam_agg, aes(x=exam_score, y=n)) +  
  geom_bar(stat='identity', width=1)
```

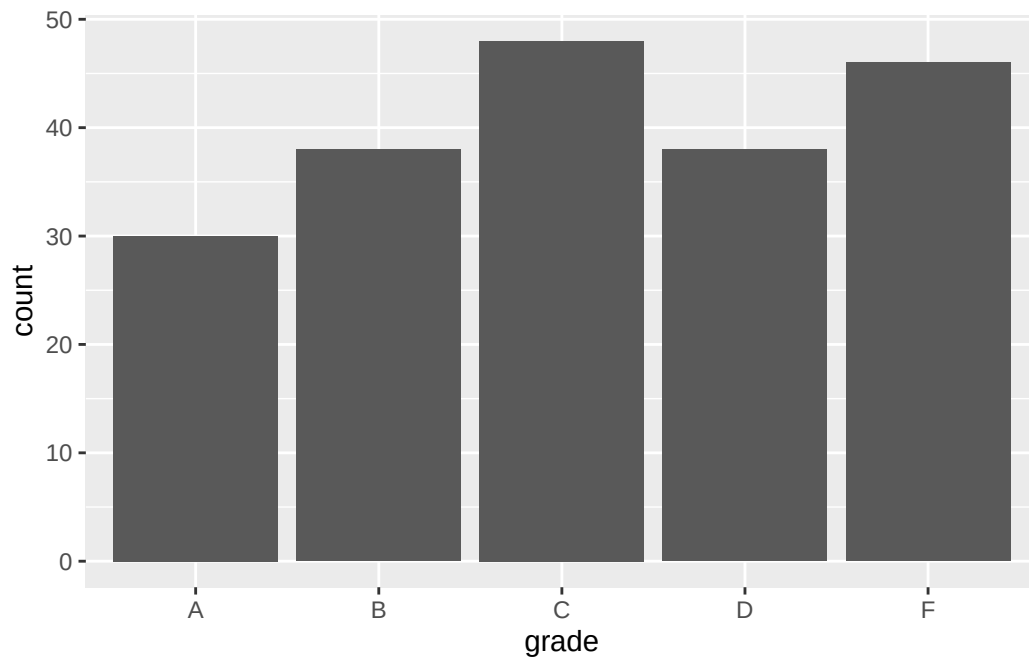


We've switched from `geom_histogram()` to `geom_bar()`. This allows us to specify a separate variable for the y-axis. Inside the `geom_bar()` call, we need to specify that `stat='identity'`. This tells ggplot that it should just use the raw value of the y-variable we gave it. If we don't make this explicit, R doesn't know what to do with the second variable.

24.1 Categorical data and bar charts

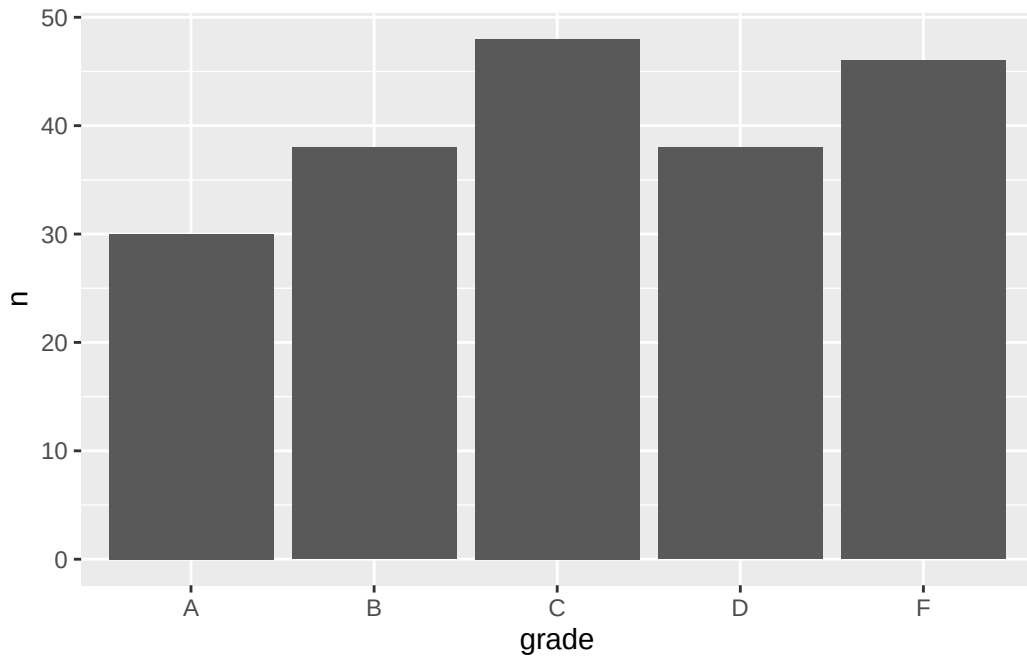
We can do the same thing with a categorical variable, course grades. First we'll plot using the raw data, and then we'll switch to the aggregated version. For the aggregated case, we'll again rely on the `stat` argument of the `geom_bar()` function.

```
# Plot from raw data
# For a categorical variable, we use geom_bar
df %>% filter(course=='ela') %>%
  ggplot(aes(x=grade)) +
  geom_bar()
```



```
# Now let's aggregate the data
grade_agg <- df %>% filter(course=='ela') %>%
  group_by(grade) %>%
  summarize(n=n())

# Now reproduce the bar plot from the aggregated data
ggplot(grade_agg, aes(x=grade, y=n)) + geom_bar(stat='identity')
```



Looks exactly the same! (Except for the y-axis label.)

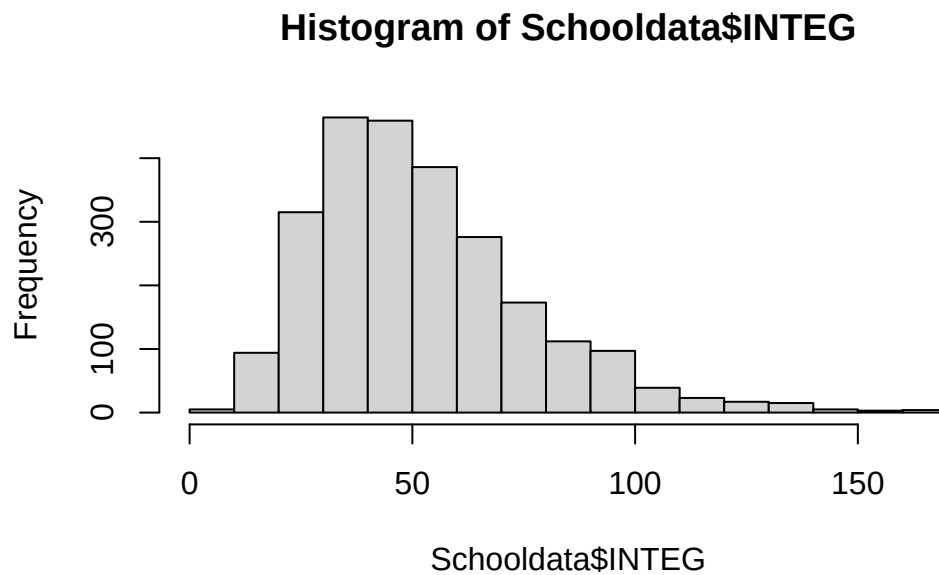
24.2 A Closing Thought

Sometimes the data we get have already been aggregated. In this case, we might still want to visualize distributions of the variables that have been grouped. The above guide introduces the `stat` argument of `geom_bar()` as a way to get this done.

25 Jitter and heatmap examples

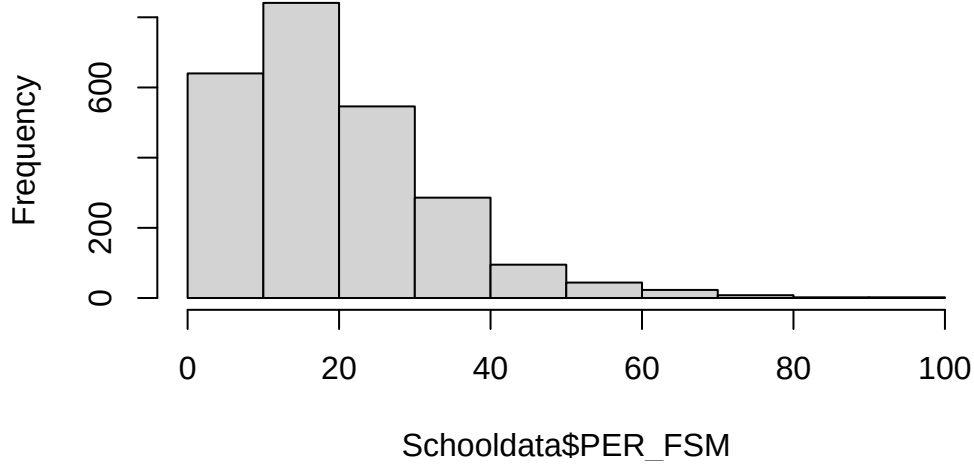
26 Jitter examples

```
# Histogram of ouput variable (INTEG)
hist(Schooldata$INTEG)
```

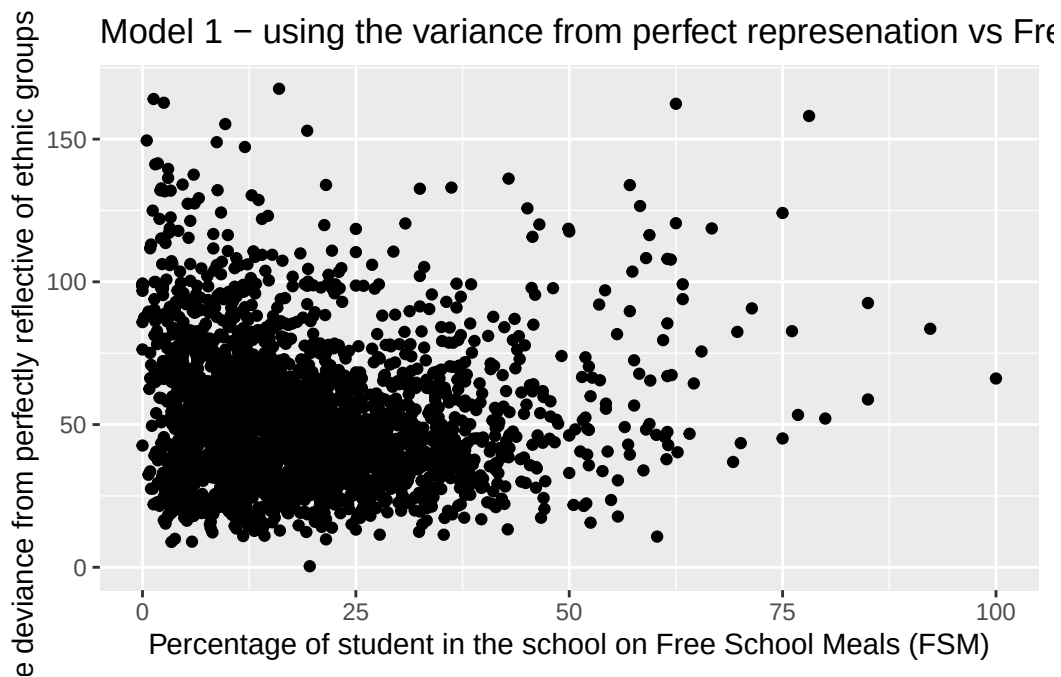


```
# Histogram of predictor variable (PER_FSM)
hist(Schooldata$PER_FSM)
```

Histogram of Schooldata\$PER_FSM



```
# Scatterplot of ouput variable (INTEG) on predictor (FSM)
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +
  geom_point() +
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals",
        x = "Percentage of student in the school on Free School Meals (FSM)",
        y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```



```
fit1 <- lm(INTEG ~ PER_FSM, data=Schooldata)
summary(fit1)
```

Call:

```
lm(formula = INTEG ~ PER_FSM, data = Schooldata)
```

Residuals:

Min	1Q	Median	3Q	Max
-51.901	-17.485	-3.918	12.352	115.029

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	54.10637	0.86768	62.358	<2e-16 ***
PER_FSM	-0.09395	0.03680	-2.553	0.0107 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

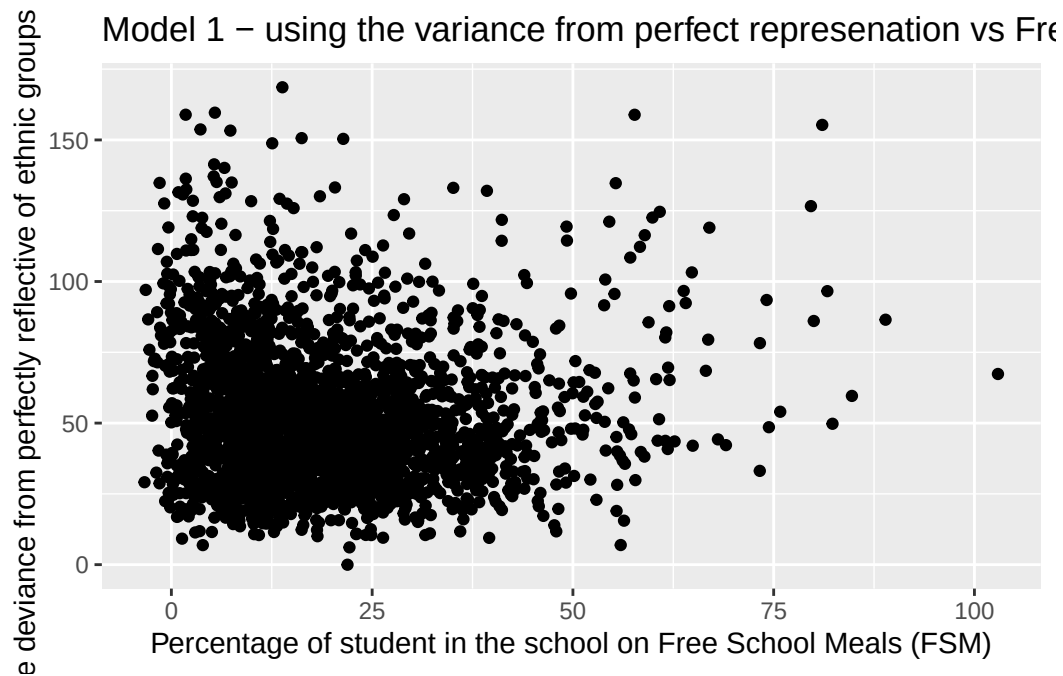
Residual standard error: 24.4 on 2485 degrees of freedom

Multiple R-squared: 0.002616, Adjusted R-squared: 0.002215

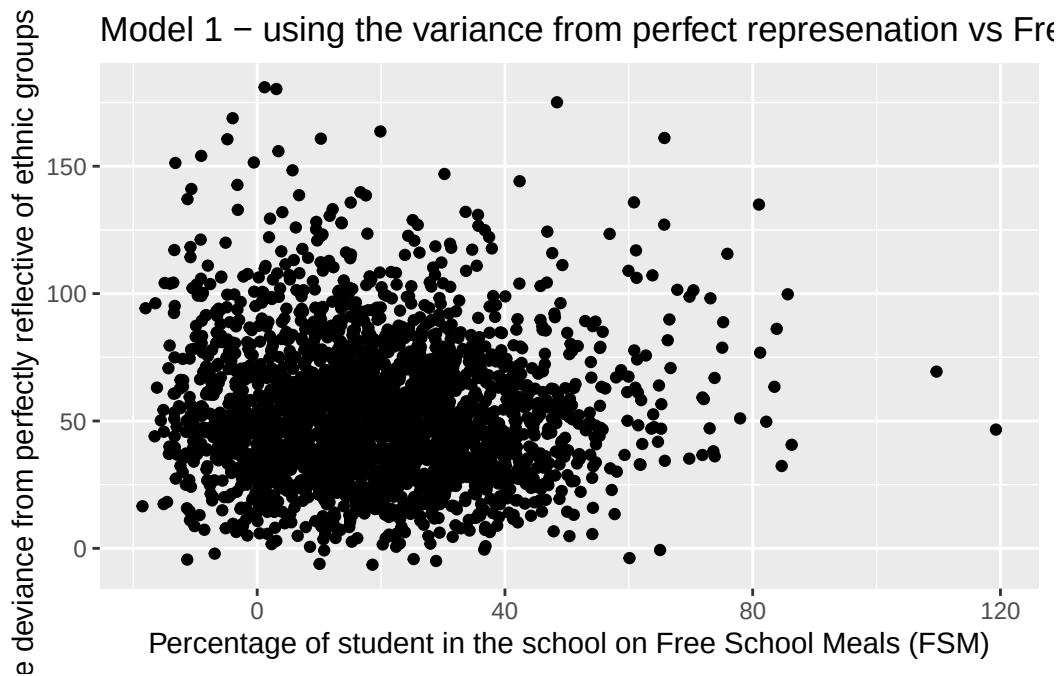
F-statistic: 6.518 on 1 and 2485 DF, p-value: 0.01074

We can try jittering more (jittering here is not a good choice, by the way). The width and height control how much jitter to do both left-right and up-down.

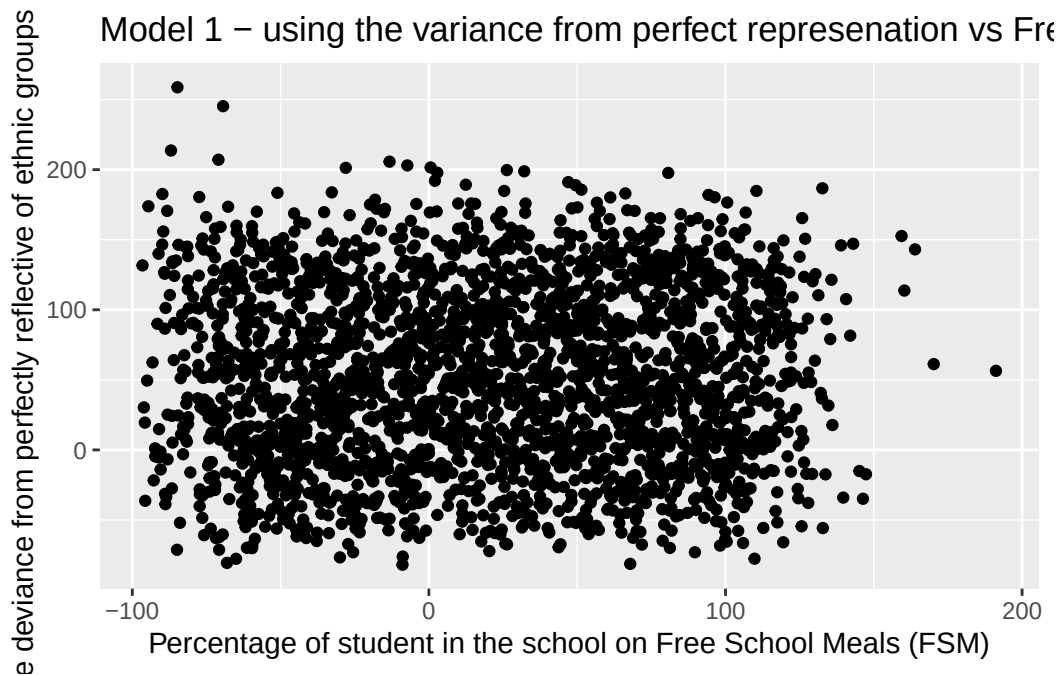
```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +
  geom_jitter(width = 5, height=5) +
  labs(title= "Model 1 - using the variance from perfect represenation vs Free School Meals"
       x = "Percentage of student in the school on Free School Meals (FSM)",
       y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```



```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +
  geom_jitter(width = 20, height=20) +
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals"
       x = "Percentage of student in the school on Free School Meals (FSM)",
       y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```



```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +
  geom_jitter(width = 100, height=100) +
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals"
       x = "Percentage of student in the school on Free School Meals (FSM)",
       y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```



You can also generate jittered data with `jitter()`:

```
a = 1:10
```

```
a
```

```
[1] 1 2 3 4 5 6 7 8 9 10
```

```
jitter( a )
```

```
[1] 1.163285 2.091552 3.197858 3.918542 4.877540 6.146654 6.867742
[8] 8.175742 9.104601 10.144863
```

```
jitter( a, factor = 0.5 )
```

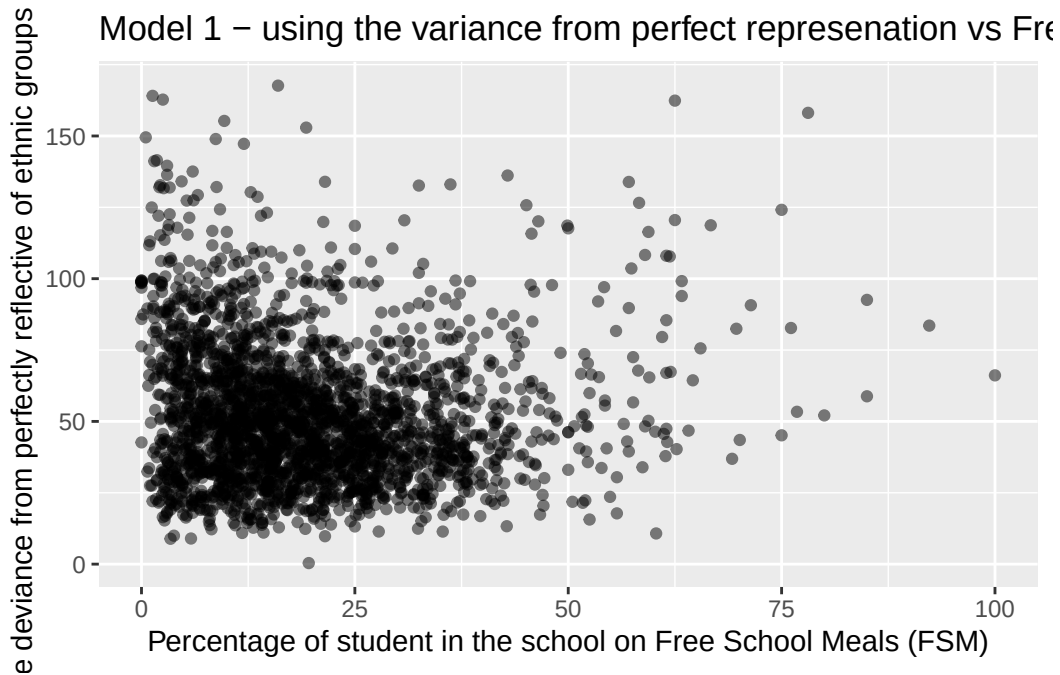
```
[1] 1.033350 1.952547 3.049305 4.093829 4.968599 5.986190 7.055124
[8] 7.993313 9.002655 10.008129
```

```
jitter( a, factor = 0.0005 )
```

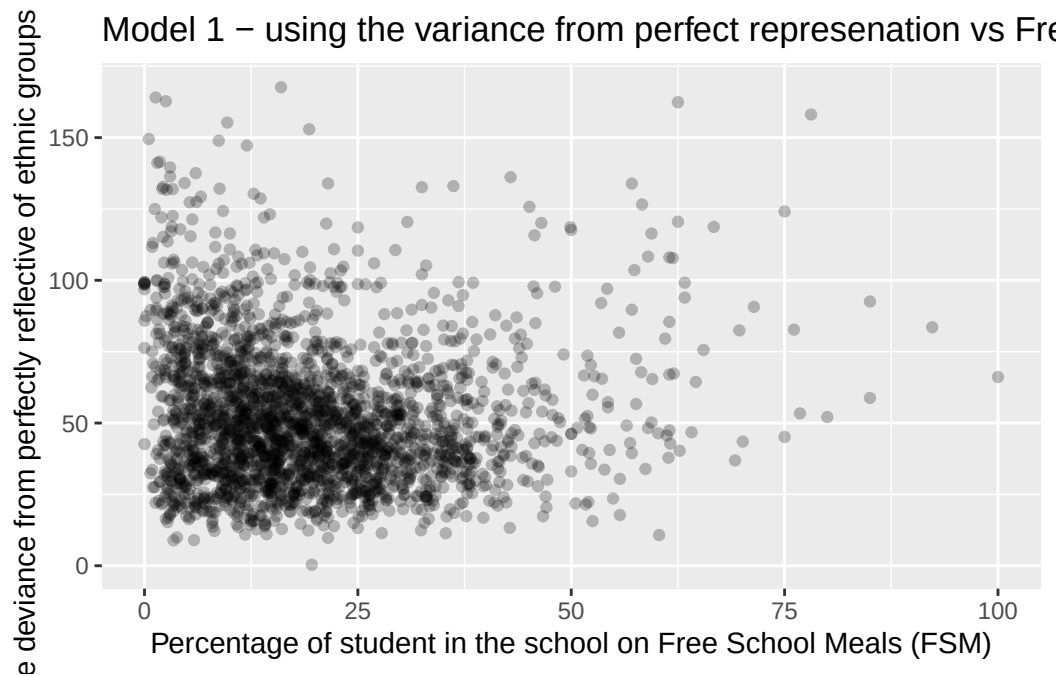
```
[1] 1.000084 1.999921 3.000000 4.000062 5.000080 6.000058 7.000090
[8] 8.000027 9.000074 10.000019
```

```
## Try alpha
```

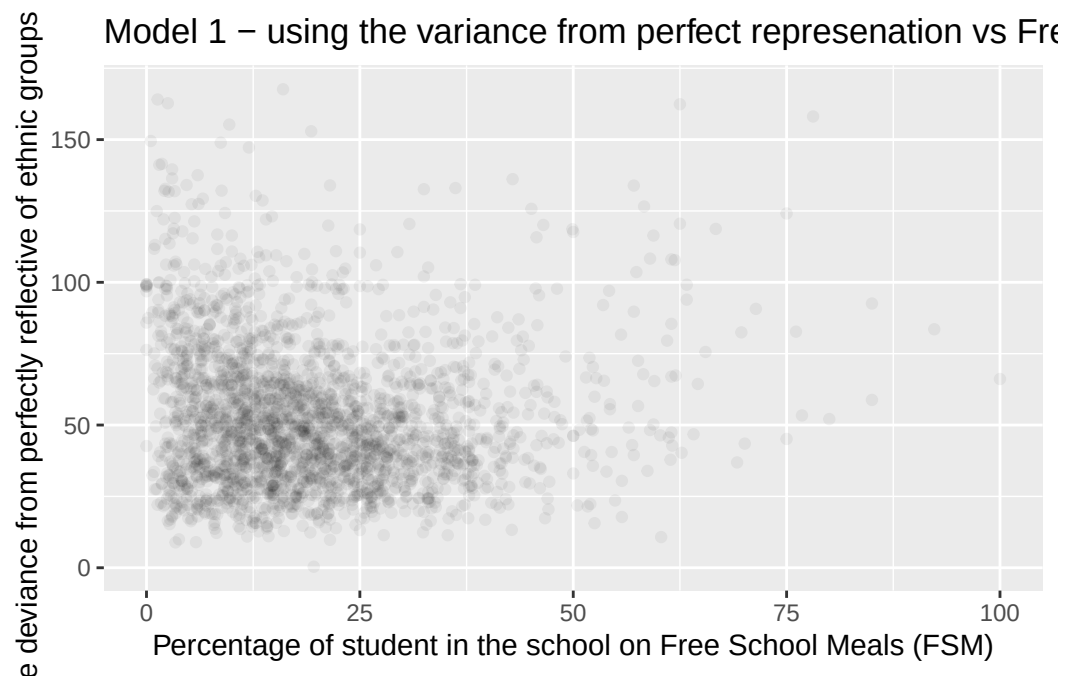
```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +  
  geom_point(alpha = 0.5) +  
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals",  
        x = "Percentage of student in the school on Free School Meals (FSM)",  
        y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```



```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +  
  geom_point(alpha = 0.25) +  
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals",  
        x = "Percentage of student in the school on Free School Meals (FSM)",  
        y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```

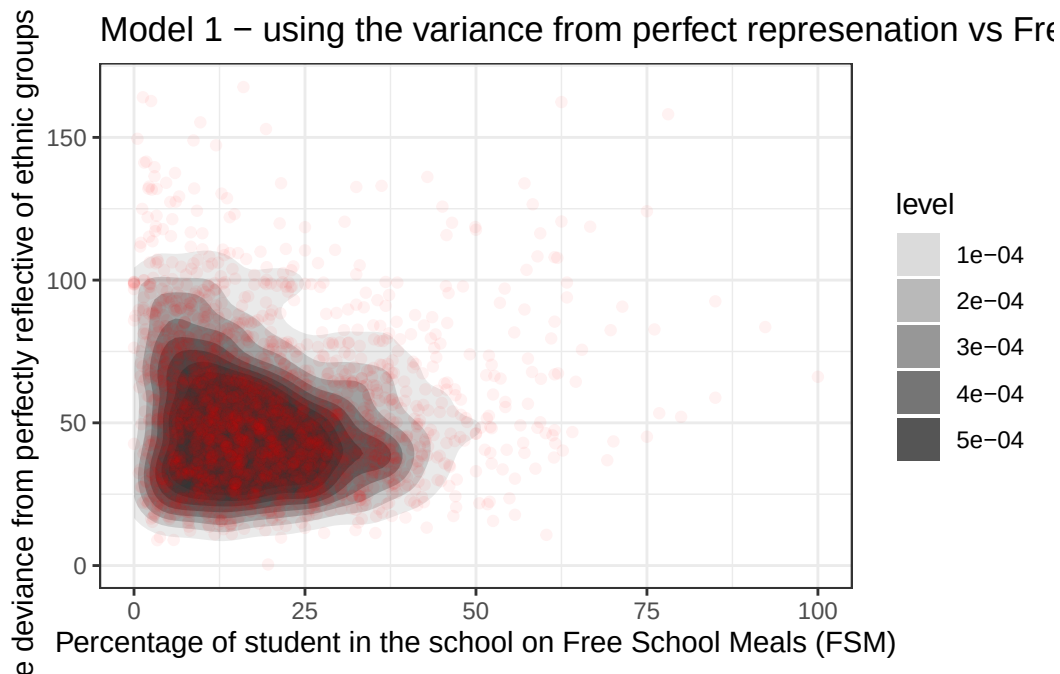
```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +
  geom_point(alpha = 0.05) +
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals",
        x = "Percentage of student in the school on Free School Meals (FSM)",
        y = "Percentage deviance from perfectly reflective of ethnic groups in local area")
```



27 Making a heat map

```
ggplot(Schooldata, aes(x = PER_FSM, y = INTEG)) +  
  stat_density2d(aes(alpha=..level..), geom="polygon") +  
  labs(title= "Model 1 - using the variance from perfect representation vs Free School Meals",  
        x = "Percentage of student in the school on Free School Meals (FSM)",  
        y = "Percentage deviance from perfectly reflective of ethnic groups in local area") +  
  geom_point(colour="red", alpha=0.05) +  
  theme_bw()
```

Warning: The dot-dot notation (`..level..`) was deprecated in ggplot2 3.4.0.
i Please use `after\_stat(level)` instead.



28 Using pivot methods to make line graphs

28.1 Overview

In this short example I demonstrate how to aggregate data and make a line graph of the results. In particular, the setting is we have a bunch of data on church donations, along with the age of the donor.

We are going to look at what proportion of the donations are above a specific size as a function of age. We are also going to look at whether the distribution of donations from those belonging to a strict evangelical doctrine are different.

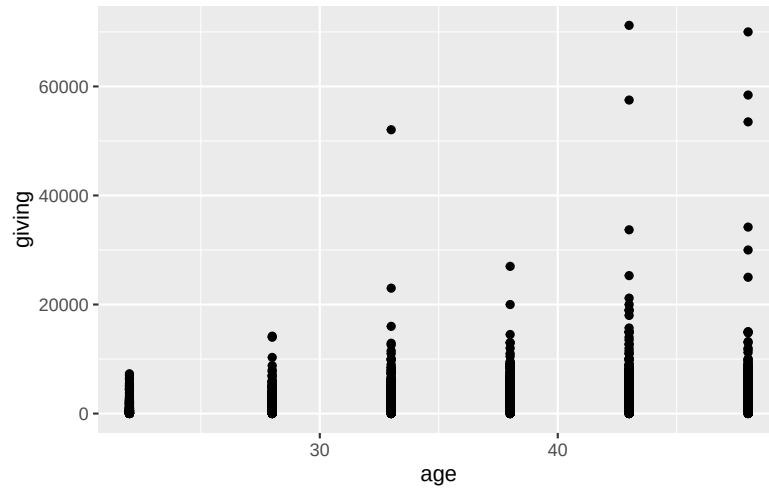
28.2 Our data

We load the data and look at the donations. It is hard to see trends given the heavy skew of donation size.

```
library( tidyverse )

church = read.table( "data/CHURCH.txt" )
names(church) = c( "church", "memberID", "giving", "age", "strict" )

ggplot( church, aes( age, giving ) ) +
  geom_point()
```



28.3 Aggregating

We group by age category and church type, and calculate the proportion of donations above \$5000 and above \$1000.

```
schurch = church %>% group_by( age, strict ) %>%
  summarise( prop_large = mean( giving > 5000 ),
             prop_medium = mean( giving > 1000 ) )
```

``summarise()`` has grouped output by 'age'. You can override using the ``.groups`` argument.

```
schurch
```

```
# A tibble: 12 x 4
# Groups:   age [6]
   age strict prop_large prop_medium
  <int> <int>      <dbl>      <dbl>
1    22     0    0.0141    0.141
2    22     1    0.0526    0.4
3    28     0    0.0287    0.217
4    28     1    0.0667    0.55
5    33     0    0.0253    0.333
6    33     1    0.0848    0.635
7    38     0    0.0353    0.357
8    38     1    0.161    0.672
9    43     0    0.0812    0.452
```

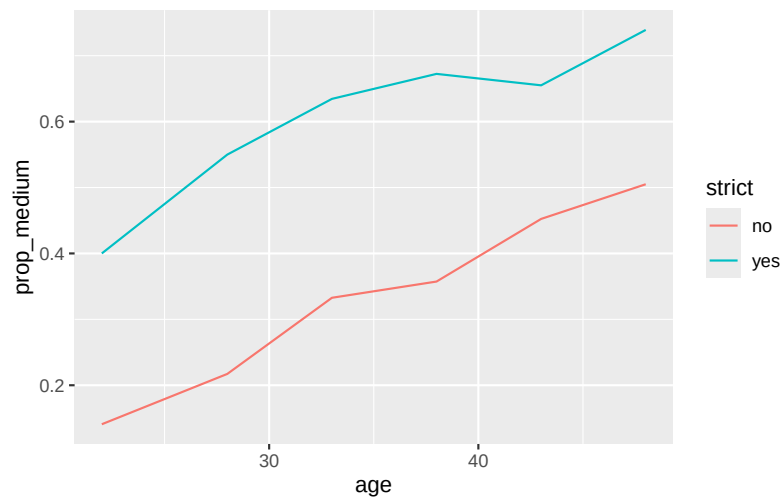
10	43	1	0.131	0.655
11	48	0	0.0532	0.505
12	48	1	0.186	0.739

28.4 Graphing (take 1)

We can make a graph of the proportion of medium donations vs age relatively easily:

```
schurch$strict = factor( schurch$strict, levels = c(0,1),
                          labels=c("no","yes" ) )

ggplot( schurch, aes( age, prop_medium, group=strict, col=strict ) ) +
  geom_line()
```



28.5 Graphing (take 2)

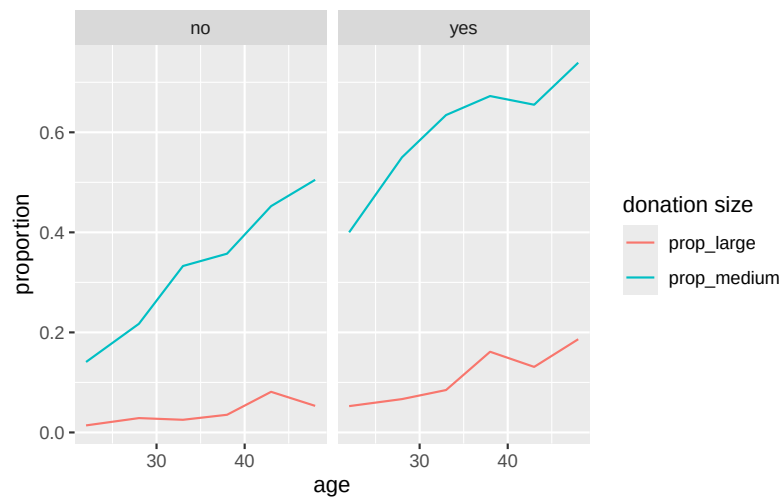
To get a line for both medium and large donations, we need to reshape our data. In particular, we want to have two groups of observations, one for each type:

```
schurchL <- schurch %>% pivot_longer( cols=c(prop_large, prop_medium),
                                       names_to = "size", values_to="proportion" )
schurchL
```

```
# A tibble: 24 x 4
# Groups:   age [6]
   age strict size      proportion
  <int> <fct> <chr>      <dbl>
1    22 no   prop_large  0.0141
2    22 no   prop_medium 0.141
3    22 yes  prop_large  0.0526
4    22 yes  prop_medium 0.4
5    28 no   prop_large  0.0287
6    28 no   prop_medium 0.217
7    28 yes  prop_large  0.0667
8    28 yes  prop_medium 0.55
9    33 no   prop_large  0.0253
10   33 no   prop_medium 0.333
# i 14 more rows
```

See how now each row is a specific age, donation size and proportion. This sets us up nicely for ggplot since each variable for plotting is ready to go:

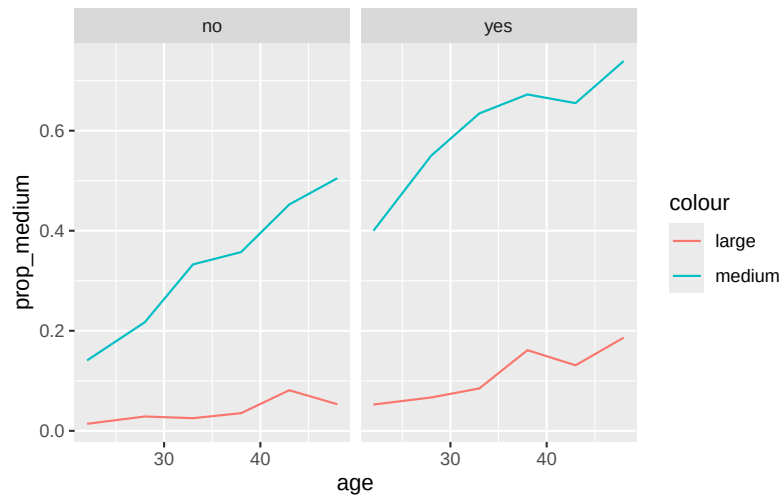
```
ggplot( schurchL, aes( age, proportion, col=size ) ) +
  facet_wrap( ~ strict ) +
  geom_line() +
  labs( col="donation size" )
```



28.6 Graphing (another road)

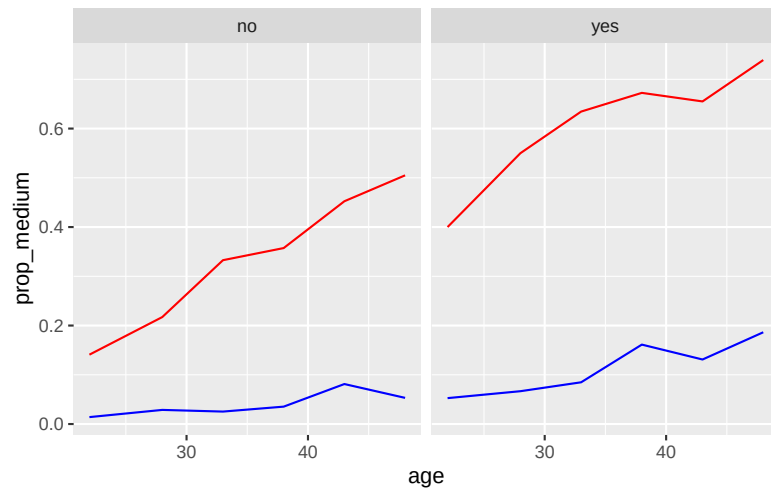
We can also plot individual lines without doing a `pivot_longer` by specifying different outcomes in the `ggplot` command:

```
ggplot( schurch, aes( age ) ) +  
  facet_wrap( ~strict ) +  
  geom_line( aes( y = prop_medium, col="medium" ) ) +  
  geom_line( aes( y = prop_large, col="large" ) )
```



Notice we are putting color to specific values inside each `geom_line`: this tells `ggplot` to make a legend of those different values. If we put these *outside* the `aes()` we can specify the color but we will not get an automatic legend:

```
ggplot( schurch, aes( age ) ) +  
  facet_wrap( ~strict ) +  
  geom_line( aes( y = prop_medium ), col="red" ) +  
  geom_line( aes( y = prop_large ), col="blue" )
```

29 Prediction and Plotting

To make nice plots from funky, curved models, the easiest way is to use the model to predict values for different covariate combinations, store all of this in a dataframe, and then use `ggplot` to plot your results using `geom_line` which will give you nice smoothed curves through the points.

This document hilites some useful methods for doing this. See R for DS 23.3. We need the `tidyverse` and the `modelr` package:

```
library( tidyverse )
library( modelr )
```

29.1 Some important functions

Some important functions are: `add_predictions()` and `add_residuals()`

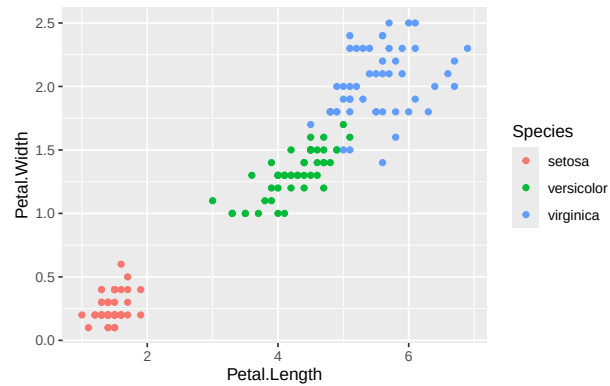
Also see `gather_pred()` to compare models.

Here is an example using the iris dataset

```
head(iris)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

```
ggplot( iris, aes( x = Petal.Length, y = Petal.Width, col=Species ) ) +
  geom_point()
```



Let's fit a simple linear model ignoring species

```
M0 = lm( Petal.Length ~ Petal.Width, data=iris )
summary( M0 )
```

Call:

```
lm(formula = Petal.Length ~ Petal.Width, data = iris)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.33542	-0.30347	-0.02955	0.25776	1.39453

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.08356	0.07297	14.85	<2e-16 ***
Petal.Width	2.22994	0.05140	43.39	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4782 on 148 degrees of freedom

Multiple R-squared: 0.9271, Adjusted R-squared: 0.9266

F-statistic: 1882 on 1 and 148 DF, p-value: < 2.2e-16

Let's fit a linear model with a species interaction:

```
M1 = lm( Petal.Length ~ Petal.Width * Species, data=iris )
summary( M1 )
```

```

Call:
lm(formula = Petal.Length ~ Petal.Width * Species, data = iris)

Residuals:
    Min       1Q   Median       3Q      Max
-0.84099 -0.19343 -0.03686  0.16314  1.17065

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.3276     0.1309  10.139 < 2e-16 ***
Petal.Width       0.5465     0.4900   1.115  0.2666
Speciesversicolor  0.4537     0.3737   1.214  0.2267
Speciesvirginica   2.9131     0.4060   7.175 3.53e-11 ***
Petal.Width:Speciesversicolor  1.3228     0.5552   2.382  0.0185 *
Petal.Width:Speciesvirginica   0.1008     0.5248   0.192  0.8480
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3615 on 144 degrees of freedom
Multiple R-squared:  0.9595,    Adjusted R-squared:  0.9581
F-statistic: 681.9 on 5 and 144 DF,  p-value: < 2.2e-16

```

Now let's add some predictions to our original data:

```

my.iris = iris
my.iris = add_predictions(iris, M1 )
head( my.iris )

```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species	pred
1	5.1	3.5	1.4	0.2	setosa	1.436861
2	4.9	3.0	1.4	0.2	setosa	1.436861
3	4.7	3.2	1.3	0.2	setosa	1.436861
4	4.6	3.1	1.5	0.2	setosa	1.436861
5	5.0	3.6	1.4	0.2	setosa	1.436861
6	5.4	3.9	1.7	0.4	setosa	1.546160

29.2 More on the model matrix

See `model_matrix`, a slightly better version of `model.matrix` in that it gives you a tibble rather than a matrix, so it is easier to deal with.

```

model_matrix( Petal.Length ~ Petal.Width * Species, data=iris )

# A tibble: 150 x 6
  `(Intercept)` Petal.Width Speciesversicolor Speciesvirginica
      <dbl>         <dbl>          <dbl>          <dbl>
1             1         0.2             0             0
2             1         0.2             0             0
3             1         0.2             0             0
4             1         0.2             0             0
5             1         0.2             0             0
6             1         0.4             0             0
7             1         0.3             0             0
8             1         0.2             0             0
9             1         0.2             0             0
10            1         0.1             0             0
# i 140 more rows
# i 2 more variables: `Petal.Width:Speciesversicolor` <dbl>,
#   `Petal.Width:Speciesvirginica` <dbl>

```

29.3 data_grid()

This function, in the `modelr` package, given a dataframe, will generate a simple dataframe with one row for each possible combination of variables you want:

```

library( modelr )
my.df = data.frame( A=c(1,2,1,2), B=c(10,20,20,20), C=c(10,20,30,40) )
data_grid( my.df, A, B )

```

```

# A tibble: 4 x 2
      A      B
  <dbl> <dbl>
1     1    10
2     1    20
3     2    10
4     2    20

```

You can also get “typical values” (usually the median or most commonly occurring) for things not on your grid using a modeling argument:

```

mod <- lm(Petal.Length ~ Petal.Width * Species, data=iris)
data_grid(iris, .model = mod)

```

```

# A tibble: 3 x 2
  Petal.Width Species
      <dbl> <chr>
1         1.3 setosa
2         1.3 versicolor
3         1.3 virginica

data_grid(iris, Petal.Width = seq_range(Petal.Width, 9), .model = mod)

# A tibble: 27 x 2
  Petal.Width Species
      <dbl> <chr>
1         0.1 setosa
2         0.1 versicolor
3         0.1 virginica
4         0.4 setosa
5         0.4 versicolor
6         0.4 virginica
7         0.7 setosa
8         0.7 versicolor
9         0.7 virginica
10        1 setosa
# i 17 more rows

```

Similar is `seq_range()` that gives you a nice sequence of numbers over a range of values

```

X = rnorm( 10000 )
seq_range( X, 5 )

[1] -3.48639099 -1.71313412  0.06012274  1.83337961  3.60663647

# drop 5% of data on tails
seq_range( X, 5, trim=0.05 )

[1] -1.92615236 -0.94714269  0.03186697  1.01087664  1.98988631

# use pretty numbers
seq_range( X, 5, trim=0.05, pretty=TRUE )

[1] -2 -1  0  1  2

```

You can combine with `data_grid` to get a nice data frame of points to use for your “canonical points” for prediction or whatnot:

```
my.df = data.frame( A=rnorm(1000), B=rexp(1000), C=rnorm(1000,mean=5) )
data_grid( my.df,
            A=seq_range(A,n=3),
            B=seq_range(B,n=3) )
```

```
# A tibble: 9 x 2
      A      B
  <dbl> <dbl>
1 -3.15  0.000755
2 -3.15  3.84
3 -3.15  7.67
4  0.0156 0.000755
5  0.0156 3.84
6  0.0156 7.67
7  3.19  0.000755
8  3.19  3.84
9  3.19  7.67
```

30 Plotting

The above is useful for plotting:

```
grid = data_grid(iris, Petal.Width = seq_range(Petal.Width, 9), .model = mod)
grid2 = add_predictions(grid, M1, "M1" )
grid2 = add_predictions(grid2, M0, "M0" )
head( grid2 )
```

```
# A tibble: 6 x 4
  Petal.Width Species      M1      M0
      <dbl> <chr>    <dbl> <dbl>
1      0.1 setosa      1.38  1.31
2      0.1 versicolor 1.97  1.31
3      0.1 virginica  4.31  1.31
4      0.4 setosa      1.55  1.98
5      0.4 versicolor 2.53  1.98
6      0.4 virginica  4.50  1.98
```

```
# more simply done with this
grid2 = spread_predictions( grid, M0, M1 )
head( grid2 )
```

```
# A tibble: 6 x 4
  Petal.Width Species      M0      M1
      <dbl> <chr>    <dbl> <dbl>
1      0.1 setosa      1.31  1.38
2      0.1 versicolor 1.31  1.97
3      0.1 virginica  1.31  4.31
4      0.4 setosa      1.98  1.55
5      0.4 versicolor 1.98  2.53
6      0.4 virginica  1.98  4.50
```

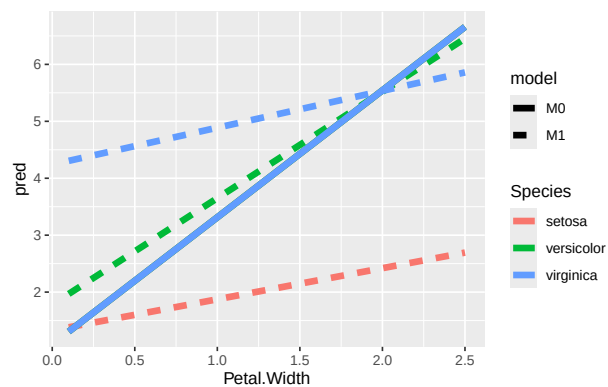
```
# but for plotting, we want long form
grid2 = gather_predictions( grid, M0, M1 )
head( grid2 )
```



```
# A tibble: 6 x 4
  model Petal.Width Species    pred
  <chr>      <dbl> <chr>      <dbl>
1 M0          0.1 setosa        1.31
2 M0          0.1 versicolor  1.31
3 M0          0.1 virginica   1.31
4 M0          0.4 setosa        1.98
5 M0          0.4 versicolor  1.98
6 M0          0.4 virginica   1.98
```

Ok, now let's plot:

```
ggplot( grid2, aes(x= Petal.Width, y= pred, col=Species, lty=model ) ) +
  geom_line(lwd=2)
```

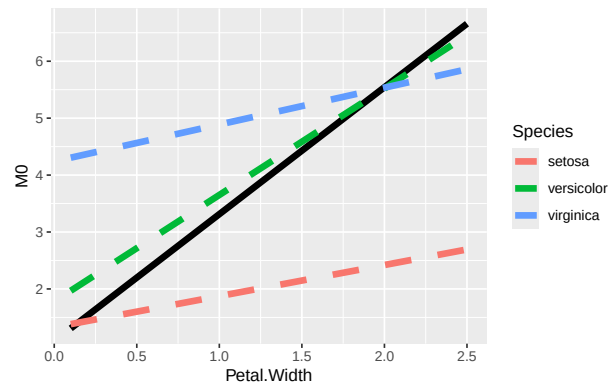


Note that our plot is not perfect: the three lines for the three M0 species predictions are all the same and thus we see only blue (the green and red get blotted out). We could get fancy:

```
grid2 = spread_predictions( grid, M0, M1 )
head( grid2 )
```

```
# A tibble: 6 x 4
  Petal.Width Species    M0    M1
      <dbl> <chr>      <dbl> <dbl>
1      0.1 setosa        1.31  1.38
2      0.1 versicolor  1.31  1.97
3      0.1 virginica   1.31  4.31
4      0.4 setosa        1.98  1.55
5      0.4 versicolor  1.98  2.53
6      0.4 virginica   1.98  4.50
```

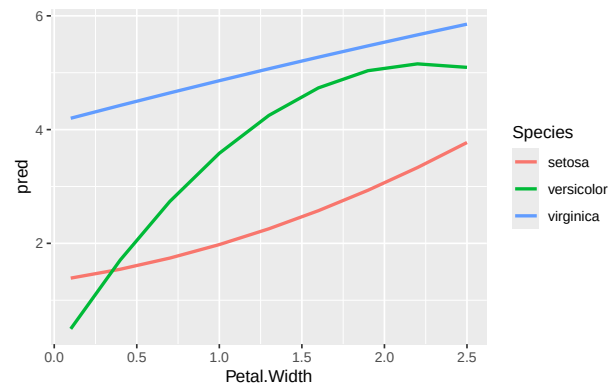
```
ggplot( grid2, aes(x= Petal.Width ) ) +
  geom_line( aes( y=M0 ), col="black", lwd=2 ) +
  geom_line( aes( y=M1, col=Species ), lwd=2, lty=2 )
```



30.1 Bonus: Interactions and curved lines

The above generalizes. Let's add a quadratic model:

```
M2 = lm( Petal.Length ~ (Petal.Width + I(Petal.Width^2)) * Species, data=iris )
grid2 = add_predictions( grid, M2 )
ggplot( grid2, aes(x= Petal.Width, y= pred, col=Species ) ) +
  geom_line( lwd=1)
```



31 Illustration of prediction with a model

Reading: R for DS (1st edition), 23.3 (23.1 and 23.2 are fairly optional)

31.1 Fitting the model on our data

We are going to show how to fit a model and then make plots of the model fit. This is a useful skill, especially when the model gets more and more complex.

First load the data and drop a bunch of variables we are not going to be using. Note how the pipe flows from load data to drop X to drop all the **empathy** variables.

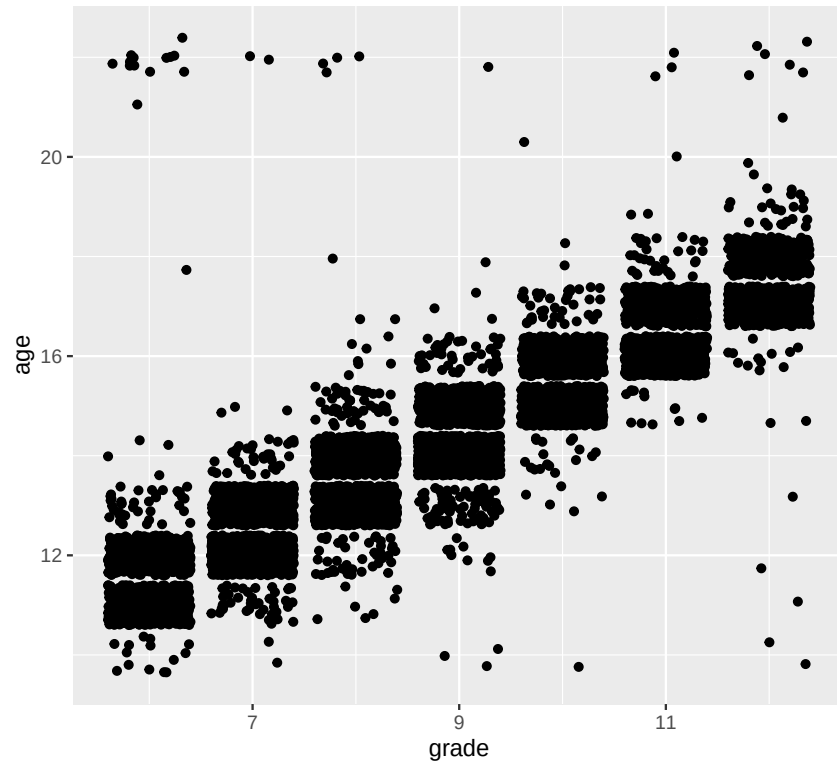
```
emp = read.csv( "data/empathy_data.csv", as.is=TRUE ) %>%  
  select( -X, -starts_with("empathy" ) )  
head( emp )
```

	grade	age	female	race	schid
1	7	12	1	Other	1
2	7	13	1	Black	1
3	7	12	1	Asian	1
4	7	13	1	Black	1
5	7	12	0	Black	1
6	7	12	0	White	1

For illustration, we are going to investigate the incredibly uninteresting relationship between age and grade. Here is our raw data

```
ggplot( data=emp, aes( x=grade, y=age ) ) +  
  geom_jitter()
```

Warning: Removed 410 rows containing missing values or values outside the scale range (``geom_point()``).



Let's fit a simple model:

```
M0 <- lm( age ~ grade * female, data=emp )
```

31.2 Prediction and Model Visualization

Now we can use our model to predict. We are going to use some stuff from the `modelr` package which helps with this kind of activity.

First load `modelr` and then make all combos of grade and gender:

```
library( modelr )
grid = emp %>% data_grid( grade, female )
grid
```

```
# A tibble: 24 x 2
  grade female
  <int> <int>
1     6     0
2     6     1
```

```

3      6      NA
4      7      0
5      7      1
6      7      NA
7      8      0
8      8      1
9      8      NA
10     9      0
# i 14 more rows

```

This is a new, fake, dataset. We can use it to predict what we would expect to see as an outcome for each case (row) in this set. We get one prediction for each row in our new data.

```
predict( M0, newdata=grid )
```

```

      1      2      3      4      5      6      7      8
11.58937 11.39423      NA 12.56097 12.38665      NA 13.53257 13.37908
      9     10     11     12     13     14     15     16
      NA 14.50417 14.37151      NA 15.47577 15.36394      NA 16.44736
     17     18     19     20     21     22     23     24
16.35636      NA 17.41896 17.34879      NA      NA      NA      NA

```

We can be a bit more clever, using more magic from the tidyverse. The following code adds a new column to our grid dataframe with predictions for each entry.

```
grid <- grid %>% add_predictions( M0 )
head( grid )
```

```

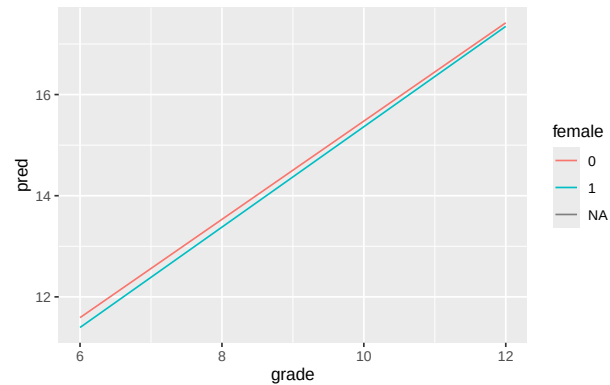
# A tibble: 6 x 3
  grade female pred
  <int>   <int> <dbl>
1     6     0 11.6
2     6     1 11.4
3     6    NA  NA
4     7     0 12.6
5     7     1 12.4
6     7    NA  NA

```

Our mini-dataset of predictions makes it super easy to plot trends (we first make female a factor so it doesn't give a continuous color axis)

```
grid$female = as.factor( grid$female )
ggplot( grid, aes( x=grade, y=pred, col=female, group=female ) ) + geom_line()
```

Warning: Removed 10 rows containing missing values or values outside the scale range (`geom\_line()`).



31.3 Residuals

We can also add residuals to our original data:

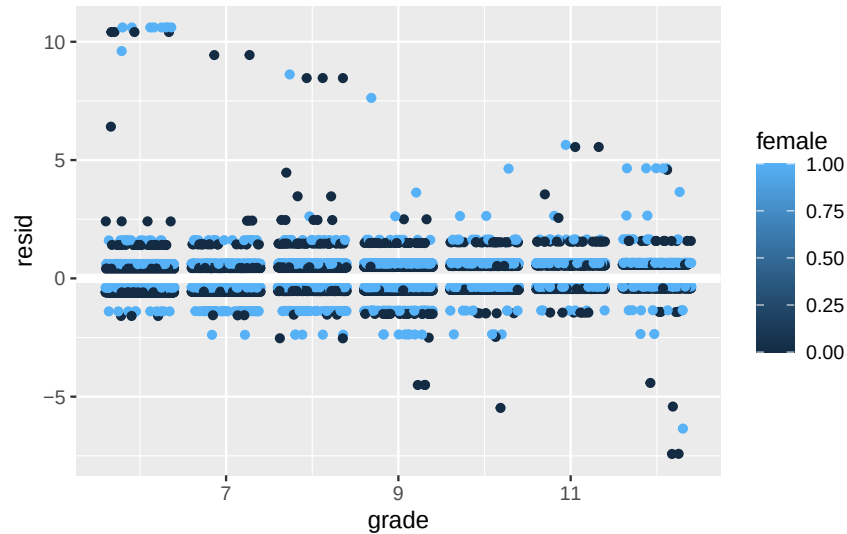
```
emp <- add_residuals( emp, M0 )
head( emp )
```

	grade	age	female	race	schid	resid
1	7	12	1	Other	1	-0.3866544
2	7	13	1	Black	1	0.6133456
3	7	12	1	Asian	1	-0.3866544
4	7	13	1	Black	1	0.6133456
5	7	12	0	Black	1	-0.5609714
6	7	12	0	White	1	-0.5609714

This allows for diagnostic plots

```
ggplot( emp, aes( x=grade, y=resid, col=female ) ) + geom_jitter() +
  geom_ref_line(h=0)
```

Warning: Removed 438 rows containing missing values or values outside the scale range (`geom\_point()`).



31.3.1 Commentary

We could have also made our dataset by hand. For example, we could do something like this:

```
another_grid = data.frame( grade=seq( 1,5, length.out=10 ), female=0 )
```

This allows us to put in values that we didn't necessarily have in our original dataset. The `data_grid` is a very convenient way to avoid doing this.

31.3.2 Bonus weirdness

If you have multiple models you want to compare, you might try these commands

```
M0 <- lm( age ~ grade, data=emp )
M1 <- lm( age ~ grade * female, data=emp )
M2 <- lm( age ~ grade * female + I(grade^2), data=emp )
grid <- data_grid( emp, age, female )
grid <- spread_predictions( emp, M0, M1, M2 )
head( grid )
```

	grade	age	female	race	schid	resid	M0	M1	M2
1	7	12	1	Other	1	-0.3866544	12.47108	12.38665	12.39010
2	7	13	1	Black	1	0.6133456	12.47108	12.38665	12.39010
3	7	12	1	Asian	1	-0.3866544	12.47108	12.38665	12.39010
4	7	13	1	Black	1	0.6133456	12.47108	12.38665	12.39010
5	7	12	0	Black	1	-0.5609714	12.47108	12.56097	12.56186

```
6      7  12      0 White      1 -0.5609714 12.47108 12.56097 12.56186
```

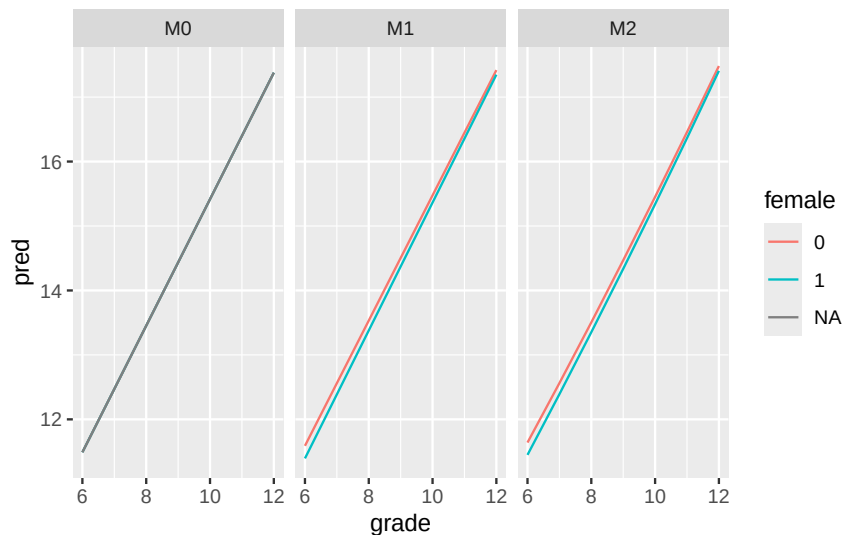
For plotting, use `gather_predictions` instead:

```
grid <- data_grid( emp, grade, female )
grid <- gather_predictions( grid, M0, M1, M2 )
head(grid)
```

```
# A tibble: 6 x 4
  model grade female  pred
  <chr> <int>   <int> <dbl>
1 M0      6      0  11.5
2 M0      6      1  11.5
3 M0      6     NA  11.5
4 M0      7      0  12.5
5 M0      7      1  12.5
6 M0      7     NA  12.5
```

```
grid$female = as.factor( grid$female )
ggplot( grid, aes( x=grade, y=pred, group=female, col=female ) ) +
  facet_wrap( ~ model ) +
  geom_line()
```

Warning: Removed 19 rows containing missing values or values outside the scale range (``geom_line()``).



Part IV

On Coding

32 Basic Data Manipulation with tidyverse

Let's use the following simple table and run through various tidyverse commands. This is not a complete reference! See the text of the textbook for further details.

```
library( tidyverse )
table1

# A tibble: 6 x 4
  country      year  cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 1999     745   19987071
2 Afghanistan 2000    2666   20595360
3 Brazil       1999   37737   172006362
4 Brazil       2000   80488   174504898
5 China        1999  212258  1272915272
6 China        2000  213766  1280428583
```

32.1 filter() (Grab the rows you want, 5.1)

```
filter( table1, year > 1999 )

# A tibble: 3 x 4
  country      year  cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 2000    2666   20595360
2 Brazil       2000   80488   174504898
3 China        2000  213766  1280428583
```

Remember, if you want to save the results of your command, you need to put it in a new variable, like so:

```
my.table <- filter( table1, year > 1999 )
my.table
```

```
# A tibble: 3 x 4
  country    year cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 2000   2666   20595360
2 Brazil      2000  80488   174504898
3 China       2000 213766  1280428583
```

When you do something like this, you should see it appear in your workplace.

32.2 arrange() (Sort your rows the way you want, 5.2)

```
arrange( table1, cases )
```

```
# A tibble: 6 x 4
  country    year cases population
  <chr>      <dbl> <dbl>      <dbl>
1 Afghanistan 1999    745   19987071
2 Afghanistan 2000   2666   20595360
3 Brazil      1999  37737   172006362
4 Brazil      2000  80488   174504898
5 China       1999 212258  1272915272
6 China       2000 213766  1280428583
```

```
arrange( table1, desc( country ), population )
```

```
# A tibble: 6 x 4
  country    year cases population
  <chr>      <dbl> <dbl>      <dbl>
1 China     1999 212258  1272915272
2 China     2000 213766  1280428583
3 Brazil    1999  37737   172006362
4 Brazil    2000  80488   174504898
5 Afghanistan 1999    745   19987071
6 Afghanistan 2000   2666   20595360
```

32.3 select() (Grab the columns you want, 5.3)

```
select( table1, country, population )
```

```
# A tibble: 6 x 2
  country      population
  <chr>         <dbl>
1 Afghanistan  19987071
2 Afghanistan  20595360
3 Brazil       172006362
4 Brazil       174504898
5 China        1272915272
6 China        1280428583
```

32.4 mutate() (Make new variables out of your old ones, 5.4)

```
table1 <- mutate( table1, case.per.1000 = 1000 * cases / population )
table1
```

```
# A tibble: 6 x 5
  country      year  cases population case.per.1000
  <chr>      <dbl> <dbl>      <dbl>         <dbl>
1 Afghanistan  1999     745   19987071     0.0373
2 Afghanistan  2000    2666   20595360     0.129
3 Brazil       1999   37737  172006362     0.219
4 Brazil       2000   80488  174504898     0.461
5 China        1999  212258  1272915272     0.167
6 China        2000  213766  1280428583     0.167
```

(We will use this new variable later, so I am saving it in our table)

32.5 group_by() and summarize (Summarize your data by subgroup, 5.6)

```
tbl <- group_by( table1, country )
summarize( tbl, av.pop = mean( population ), av.cases = mean( cases ) )
```

```
# A tibble: 3 x 3
  country      av.pop av.cases
  <chr>      <dbl> <dbl>
1 Afghanistan  20291216.  1706.
2 Brazil       173255630  59112.
3 China        1276671928.  213012
```

Same thing, with the pipe!

```
table1 %>% group_by( country ) %>%  
  summarize( av.pop = mean( population ), av.cases = mean( cases ) )
```

```
# A tibble: 3 x 3  
  country      av.pop av.cases  
  <chr>      <dbl> <dbl>  
1 Afghanistan 20291216.  1706.  
2 Brazil      173255630  59112.  
3 China       1276671928. 213012
```

32.6 Special: grouped mutates (Making new variables within subgroups, 5.6)

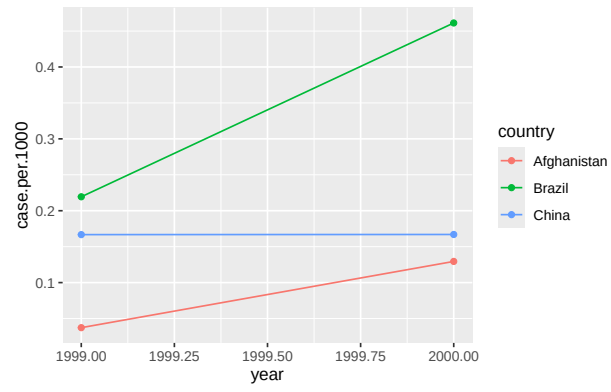
This combo is for doing things like group mean centering your data:

```
table1 %>% group_by( year ) %>% mutate( case.per.1000.cent = case.per.1000 - mean( case.per.1000 ) )
```

```
# A tibble: 6 x 6  
# Groups:   year [2]  
  country      year cases population case.per.1000 case.per.1000.cent  
  <chr>      <dbl> <dbl>      <dbl>      <dbl>      <dbl>  
1 Afghanistan 1999    745  19987071    0.0373    -0.104  
2 Afghanistan 2000   2666  20595360    0.129     -0.123  
3 Brazil      1999  37737  172006362    0.219     0.0783  
4 Brazil      2000  80488  174504898    0.461     0.209  
5 China       1999 212258  1272915272    0.167     0.0256  
6 China       2000 213766  1280428583    0.167    -0.0856
```

And a plot

```
ggplot( table1, aes( x=year, y=case.per.1000, col=country ) ) + geom_line() + geom_point()
```



33 A Deep Dive on `group_by()`

The `group_by()` functionality in the Tidyverse is a powerhouse function, with many applications for summarizing and cleaning data. This handout walks through several common use cases for the `groupby_()` function.

34 Importing the ‘dplyr’ package

The `group_by()` function technically belongs to the `dplyr` package, but we can also load it by importing the `tidyverse` shortcut.

```
library(dplyr)
library(tidyverse)
```

For this walkthrough, we will be using a dataset of information on NFL players. We will use the `group_by()` function to analyze information across NFL teams. Our dataset consists of 16 columns and 1,822 players (one player per row).

```
nfl
```

```
# A tibble: 1,822 x 16
  `Player Name`      Team Number Position Salary Height Weight Birthdate Age
  <chr>             <chr>   <dbl> <chr>      <dbl> <chr>   <dbl> <chr>   <dbl>
1 Abdullah, Husain  KC        39 S        715000 "6'0\"~ 204 7/27/85 29
2 Abdul-Quddus, Isa NO         42 S        465000 "6'1\"~ 205 8/3/89 25
3 Aboushi, Oday    NYJ        75 OL        589750 "6'5\"~ 308 6/5/91 24
4 Abraham, John    ARI        55 DE       2300000 "6'4\"~ 263 5/6/78 37
5 Acho, Emmanuel  PHI        53 LB        525920 "6'2\"~ 240 11/10/90 24
6 Acho, Sam        ARI        94 LB        628607 "6'3\"~ 257 9/6/88 26
7 Adams, Joe       CAR        15 WR        641888 "5'11\"~ 185 11/22/89 25
8 Adams, Kris      NYG        89 WR        510000 "6'3\"~ 194 9/4/87 27
9 Adams, Kyle      CHI        86 TE        465000 "6'4\"~ 255 1/19/88 27
10 Adams, Michael  TB         27 CB          NA "5'8\"~ 181 6/17/85 30
# i 1,812 more rows
# i 7 more variables: Experience <dbl>, College <chr>, City <chr>, State <chr>,
#   Race <chr>, birth_date <date>, birth_month <dbl>
```

34.1 What does `group_by()` do?

When you call `group_by()` on a dataframe, you are asking R to “organize” your data according to the grouping variable. For example, let’s group our data on NFL players by team. If you look carefully at the output below, you will see that our data has 1,822 rows and 16 columns, just like it did before. However, it now also has 32 groups (one for each NFL team).

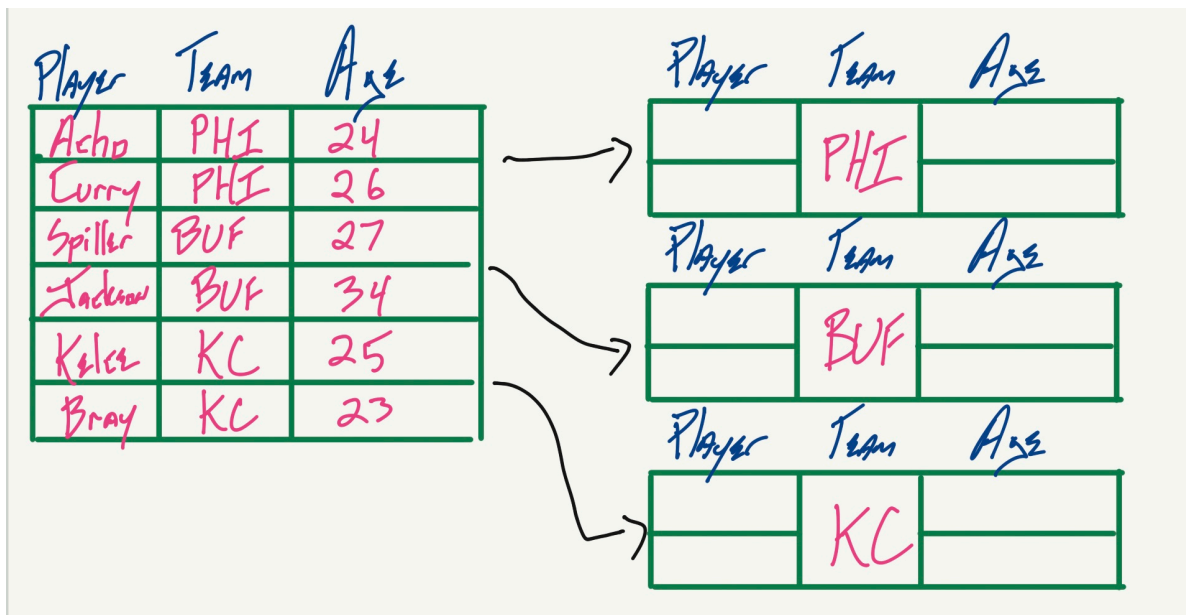

```
nfl %>% group_by(Team)
```

```
# A tibble: 1,822 x 16
```

```
# Groups: Team [32]
```

```
  `Player Name`    Team Number Position Salary Height Weight Birthdate   Age
  <chr>            <chr>   <dbl> <chr>      <dbl> <chr>   <dbl> <chr>    <dbl>
1 Abdullah, Husain KC        39 S        715000 "6'0\"~ 204 7/27/85    29
2 Abdul-Quddus, Isa NO        42 S        465000 "6'1\"~ 205 8/3/89     25
3 Aboushi, Oday    NYJ        75 OL        589750 "6'5\"~ 308 6/5/91     24
4 Abraham, John    ARI        55 DE       2300000 "6'4\"~ 263 5/6/78     37
5 Acho, Emmanuel  PHI        53 LB        525920 "6'2\"~ 240 11/10/90   24
6 Acho, Sam        ARI        94 LB        628607 "6'3\"~ 257 9/6/88     26
7 Adams, Joe       CAR        15 WR        641888 "5'11\"~ 185 11/22/89   25
8 Adams, Kris      NYG        89 WR        510000 "6'3\"~ 194 9/4/87     27
9 Adams, Kyle      CHI        86 TE        465000 "6'4\"~ 255 1/19/88    27
10 Adams, Michael  TB         27 CB          NA "5'8\"~ 181 6/17/85    30
# i 1,812 more rows
# i 7 more variables: Experience <dbl>, College <chr>, City <chr>, State <chr>,
# Race <chr>, birth_date <date>, birth_month <dbl>
```

Conceptually, here is a visualization of what R does when you `group_by(Team)`. Behind the scenes, your data is partitioned according to values of your grouping variables. This becomes extremely useful if we want to calculate group-specific variables or create group-specific summary variables.



34.2 Introducing `summarize()`

The `summarize()` function is extremely useful for creating summary statistics on aggregated data, such as calculating the number of observations, the mean of a particular variable, etc. First, let's look at what happens when we use `summarize()` on its own. In the example below, we use `summarize()` on the *ungrouped* `nfl` data to find the overall mean salary of players (\$2,142,272).

```
nfl %>% summarize(mean_salary = mean(Salary, na.rm = TRUE))
```

```
# A tibble: 1 x 1
  mean_salary
    <dbl>
1    2142272.
```

We can also use `summarize()` to get a variety of summary features using different functions to suit our purposes. We can also name these summary variables whatever we want (“mean_salary”, “n_obs”, “number_of_observations”, etc.). The code below calculates the average salary, the standard deviation of salaries, average age of players, and the total number of observations.

Below, you might also notice the `n()` function. This “short and sweet” function tells us the number of observations that we are summarizing across. In this case, `n()` returns 1,822—the same number of rows in our NFL data frame!

```
nfl %>%
  summarize(mean_salary = mean(Salary, na.rm = TRUE), # gets the overall average salary
            sd_salary = sd(Salary, na.rm = TRUE), # gets the standard deviation of salaries
            mean_age = mean(Age, na.rm = TRUE), # we can also look at other variables
            n_obs = n() # gets the number of observations
  )
```

```
# A tibble: 1 x 4
  mean_salary sd_salary mean_age n_obs
    <dbl>      <dbl>    <dbl> <int>
1    2142272.  2776358.    27.8  1822
```

34.3 Combining `group_by()` with `summarize()`

Now, let's add in `group_by()` to find the average (mean) and maximum salary *within each team*. Let's also get a count of the number of players on each team. Our output below has 32 rows—one for each value of the `Team` grouping variable.

Looking at the output, we can see that the Cleveland Browns (CLE) have an average salary of \$1,924,681 and that the Baltimore Ravens (BAL) have an average salary of \$2,145,966. Note that we set `mean(..., na.rm = TRUE)`, so we are ignoring any players with missing salary data when calculating our team averages.

```
nfl %>%
  group_by(Team) %>%
  summarize(num_players = n(),
            mean_sal = mean(Salary, na.rm = TRUE),
            max_sal = max(Salary, na.rm = TRUE))
```

A tibble: 32 x 4

	Team	num_players	mean_sal	max_sal
	<chr>	<int>	<dbl>	<dbl>
1	ARI	58	2110725.	15750000
2	ATL	55	2278403.	18958333
3	BAL	57	2145966.	20100000
4	BUF	55	1992520.	16000000
5	CAR	61	2228745	12666667
6	CHI	54	2369191.	14000000
7	CIN	59	1955534.	11175000
8	CLE	55	1924681.	10062500
9	DAL	56	2491651.	17071429
10	DEN	60	2145368.	19200000

i 22 more rows

34.3.1 Getting fancy: Grouping with more than one variable

However, we can also group by more than one variable! For example, we can extend our code to get the average salary within each player position on each team. Let's extend the code above to find the average and maximum salary within each team and position on that team. In this case, the `n()` summary variable tells us how many observations are within each grouping cell (i.e. the number of players in each position on each team).

What do you notice that is similar or different from the output above with a single grouping variable?

```

nfl %>%
  group_by(Team, Position) %>%
  summarize(num_players = n(),
            min_sal = min(Salary, na.rm = TRUE),
            mean_sal = mean(Salary, na.rm = TRUE),
            max_sal = max(Salary, na.rm = TRUE))

# A tibble: 471 x 6
# Groups:   Team [32]
   Team Position num_players min_sal mean_sal max_sal
  <chr> <chr>         <int>   <dbl>   <dbl>   <dbl>
1 ARI    C             1 3250000 3250000  3250000
2 ARI    CB             8  498367 1615209.  4607375
3 ARI    DE             5  555000 3152000 11000000
4 ARI    DL             2  405000 4869166.  9333333
5 ARI    K              1 1500000 1500000  1500000
6 ARI    LB             8  495000 1317682.  3166667
7 ARI    LS             1 1090667 1090667  1090667
8 ARI    NT             1 2067000 2067000  2067000
9 ARI    OL             9  539856 2250741.  6000000
10 ARI   P              1 1150000 1150000  1150000
# i 461 more rows

```

34.4 Using mutate() with group_by()

On its own, the `mutate()` function allows us to create and modify variables in a dataframe (see Chapter 32 for more information about `mutate()`). For example, we can use `mutate` to convert players' weight from pounds to kilograms.

In this basic example, note that we have not grouped the `nfl` dataframe.

```

nfl %>%
  select(`Player Name`, Weight) %>%
  mutate(weight_kg = Weight / 2.2) %>%
  head()

# A tibble: 6 x 3
  `Player Name`      Weight weight_kg
  <chr>           <dbl>   <dbl>
1 Abdullah, Husain    204     92.7
2 Abdul-Quddus, Isa  205     93.2

```

3	Aboushi, Oday	308	140
4	Abraham, John	263	120.
5	Acho, Emmanuel	240	109.
6	Acho, Sam	257	117.

However, when we combine `group_by()` and `mutate()` we can create new variables for individual players that reflect group-level attributes. For example, let's look at how we could find how much younger each player is than the oldest player on their team.

In this case, the grouping variable essentially allows us to create a new column for each individual in the NFL using summary variables within each NFL team.

```
# let's create a new dataframe to hold our new data
nfl_1 <- nfl %>%
  select(`Player Name`, Team, Age) %>%
  group_by(Team) %>%
  mutate(years_younger = max(Age, na.rm = TRUE) - Age,
         team_max_age = max(Age, na.rm = TRUE) # we don't need this, but you can include it
        ) %>%
  arrange(Team) # re-order our data to help us understand how grouping is occurring at the t

nfl_1 %>% head()
```

```
# A tibble: 6 x 5
# Groups:   Team [1]
  `Player Name` Team    Age years_younger team_max_age
  <chr>         <chr> <dbl>         <dbl>         <dbl>
1 Abraham, John  ARI     37             2             39
2 Acho, Sam      ARI     26            13             39
3 Alexander, Lorenzo ARI     32             7             39
4 Arenas, Javier ARI     27            12             39
5 Bell, Jeremiah ARI     37             2             39
6 Bethel, Justin ARI     25            14             39
```

If you look at the output above, you will notice that our new `nfl_1` dataframe is still grouped by `Team`. We might not want this, especially if we want to create other variables based on other grouping factors later. We can remove the groups from the `nfl_1` dataset like this:

```
nfl_1 <- nfl_1 %>% ungroup()
```

34.5 Learning Check: What is the difference between pairing `group_by()` with `summarize()` vs `group_by()` with `mutate()`?

Take a moment to review the output of the code above. How many rows do we have in our resulting dataframes after calling `group_by(Team)` with `summarize()` vs `group_by(Team)` with `mutate()`?

You might notice that there are far fewer rows in the output from commands using `summarize()`. This is because the `summarize()` function collapses data down to return a single value per group level (ex. one row for each of the 32 teams in the NFL). By contrast, using `group_by()` with `mutate()` preserves all rows in the data while allowing the user to synthesize and make use of group-level information.

34.6 Practical Application: Centering Data

Centering data is a common practice in social science and education research. For example, we might often be less interested in a student's absolute test score and more interested in how they performed relative to an average value.

For example, we established above that the average salary in the entire NFL is \$2,142,272. We can use centering to calculate the amount each player earns above or below this overall average salary using the code below. This is called “grand mean” centering (i.e. we are comparing each player's salary to the overall average).

Note that the code below tabulates the average salary `mean(Salary)` without first calling `group_by()`.

```
nfl <- nfl %>%
  group_by(Team) %>%
  mutate(grand_cent_sal = (Salary - mean(Salary, na.rm = TRUE))) %>%
  ungroup() # ungroup our data once we have what we need

nfl %>% select(Team, Salary, grand_cent_sal)
```

```
# A tibble: 1,822 x 3
  Team   Salary grand_cent_sal
  <chr>   <dbl>         <dbl>
1 KC      715000     -2159274.
2 NO      465000     -1468431.
3 NYJ     589750     -1284454.
4 ARI    2300000       189275.
5 PHI     525920     -1932906.
```

```

6 ARI      628607      -1482118.
7 CAR      641888      -1586857
8 NYG      510000      -1578188.
9 CHI      465000      -1904191.
10 TB              NA              NA
# i 1,812 more rows

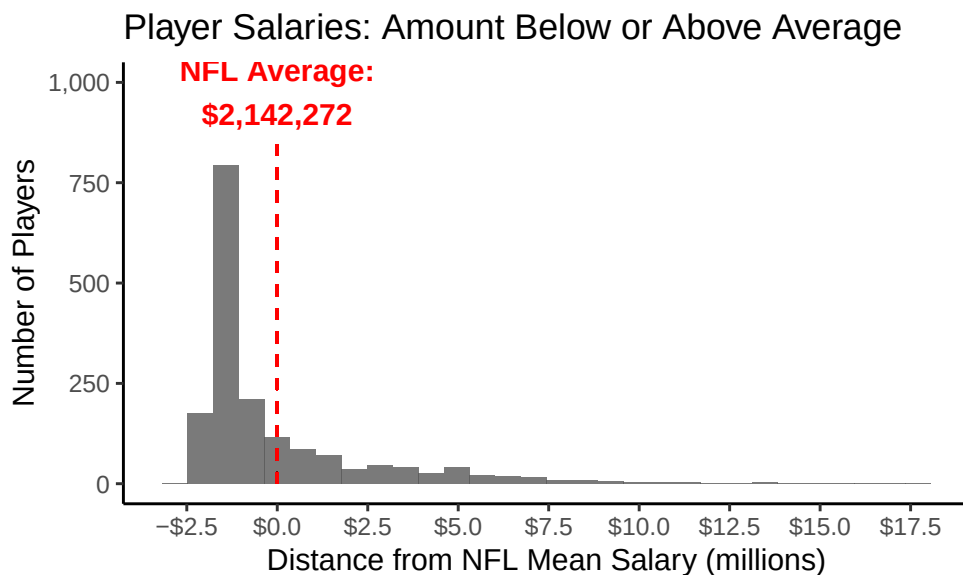
```

We can visualize the resulting `grand_cent_sal` data series using `ggplot()`.

```

ggplot(nfl, aes(x = (grand_cent_sal) / 1000000)) +
  geom_histogram(alpha = 0.8) +
  geom_segment(x = 0, xend = 0, y = 0, yend = 875, linetype = 'dashed', color = 'red', ) +
  annotate(geom = 'text', x = 0, y = 975, label = str_wrap("NFL Average: $2,142,272", 12),
    color = 'red', fontface = 'bold') +
  scale_x_continuous(breaks = seq(-5, 18.5, 2.5), labels = scales::dollar_format(accuracy = 0),
  scale_y_continuous(breaks = seq(0, 1000, 250), limits = c(0, 1000), labels = scales::comma),
  labs(title = "Player Salaries: Amount Below or Above Average",
    x = "Distance from NFL Mean Salary (millions)",
    y = "Number of Players")

```



34.6.1 Group Mean Centering

Now let's assume that it actually might be more informative to compare players' salaries to those of other players *on the same team*. In this case, we want to group our data by team before calculating our centered salary variable. This is called group mean centering.

```

nfl <- nfl %>%
  group_by(Team) %>%
  mutate(group_cent_sal = (Salary - mean(Salary, na.rm = TRUE)),
         team_average_salary = mean(Salary, na.rm = TRUE))

nfl %>% select(Team, Salary, group_cent_sal, team_average_salary) %>% head()

# A tibble: 6 x 4
# Groups:   Team [5]
  Team    Salary group_cent_sal team_average_salary
  <chr>    <dbl>         <dbl>             <dbl>
1 KC      715000      -2159274.         2874274.
2 NO      465000      -1468431.         1933431.
3 NYJ     589750      -1284454.         1874204.
4 ARI    2300000         189275.         2110725.
5 PHI     525920      -1932906.         2458826.
6 ARI     628607      -1482118.         2110725.

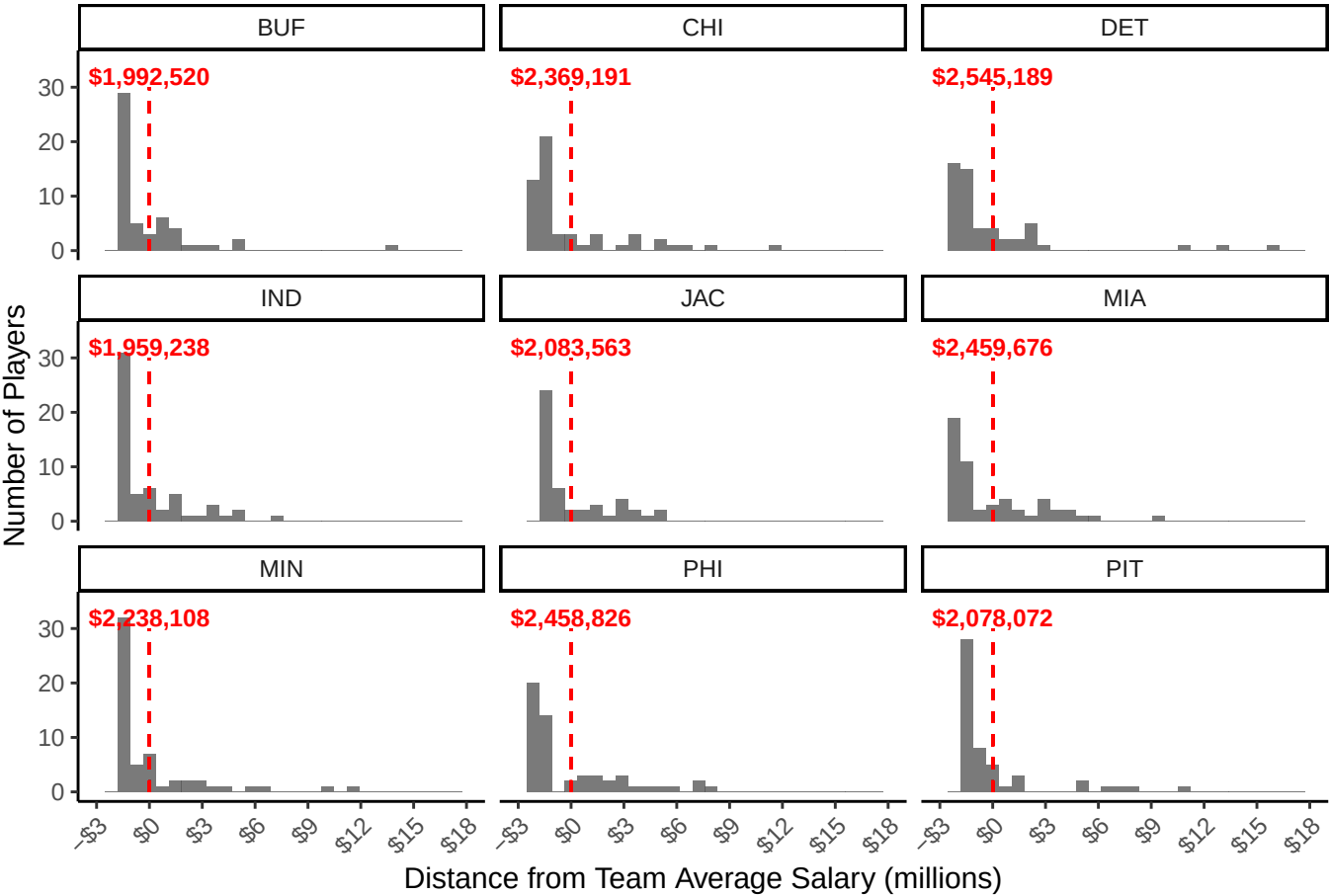
# choose some teams to plot
teams <- sample(unique(nfl$Team), 9)

# get the average salary for each team
team_average_sal_df <- nfl %>%
  filter(Team %in% teams) %>%
  group_by(Team) %>%
  summarize(team_avg_sal = mean(Salary, na.rm = T))

ggplot(nfl %>% filter(Team %in% teams),
       aes(x = (group_cent_sal) / 1000000)) +
  geom_histogram(alpha = 0.8) +
  facet_wrap(~ Team) +
  geom_segment(x = 0, xend = 0, y = 0, yend = 30, linetype = 'dashed', color = 'red', ) +
  geom_text(data = team_average_sal_df, aes(x = 0, y = 32, label = scales::dollar(team_avg_sal)),
           color = 'red', size = 3,
           fontface = 'bold', inherit.aes = F) +
  scale_x_continuous(breaks = seq(-6, 18.5, 3), limits = c(-3, 18),
                    labels = scales::dollar_format(accuracy = 1)) +
  scale_y_continuous(breaks = seq(0, 30, 10), limits = c(0, 35)) +
  labs(title = "Player Salaries: Amount Below or Above Team Average",
       x = "Distance from Team Average Salary (millions)",
       y = "Number of Players") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```


Player Salaries: Amount Below or Above Team Average



35 Intro to Regression

In S022 we assume a general understanding of linear regression, but sometimes one might like to review those concepts! In the following we first give some readings about regression, and then give a code walkthrough of how to do it in R.

The suggested readings (which can be accessed via the Library Reserves Tool on the Canvas site) are:

- **Gelman, A. & Hill, J. Chapter 4. In Data analysis using regression and multilevel/hierarchical models. Cambridge University Press, pp 53-74.** This chapter talks a bit more about regression, in particular transforming and centering variables, both being important tools for making your regression models more interpretable.
- **Kaplan, D.T. (2012). Chapter 7. In Statistical modeling: A fresh approach. (n.p.): Project Mosaic, pp. 129-151.** This reading reviews what regression is, and how to think about regression coefficients. An excellent review that points out that we can view regression as a summary of trends in our data. (i.e., use regression without believing in the model being correct). This is more in line with how data scientists think about models.
- **Gerber, A. S. & Green, D. (2012). Chapter 3. In Field experiments: Design, analysis, and interpretation. New York, NY: W. W. Norton & Company, pp. 51-86.** This chapter is a nice overview of what a standard error is, and how we should think about statistical uncertainty. It will deepen your understanding of the S040 material.

35.1 Simple Regression

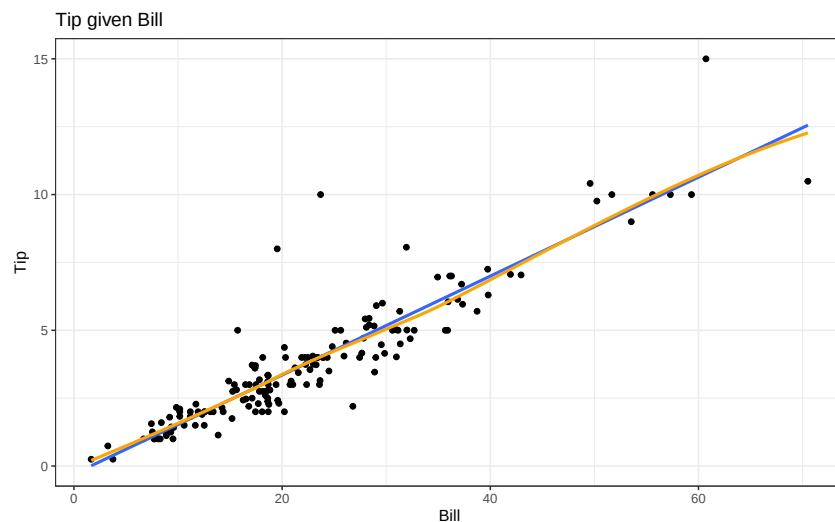
We now give a walk-through on how to fit simple linear regression models in R. Linear regression is the main way researchers tend to examine the relationships between multiple variables. This document runs through some code without too much discussion, with the assumption that you are already familiar with interpretation of such models. If you want some help with the interpretation, see the suggested readings above!

For our walk-through, we are going to use an example dataset, `RestaurantTips`, that records tip amounts for a series of bills. Let's first regress `Tip` on `Bill`. Before doing regression, we should plot the data to make sure using simple linear regression is reasonable. For kicks, we

add in an automatic regression line as well by taking advantage of ggplot's `geom_smooth()` method:

```
# load the data into memory
data(RestaurantTips)

# plot Tip on Bill
ggplot( RestaurantTips, aes(x = Bill, y = Tip) ) +
  geom_point() +
  geom_smooth( method="lm", se=FALSE ) +
  geom_smooth( method="loess", se=FALSE, col="orange" ) +
  labs(title = "Tip given Bill")
```



That looks pretty darn linear! There are a few unusually large tips, but no extreme outliers, and variability appears to be constant at all levels of Bill , so we proceed:

```
# fit the linear model
mod <- lm(Tip ~ Bill, data = RestaurantTips)
summary(mod)
```

Call:

```
lm(formula = Tip ~ Bill, data = RestaurantTips)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.391	-0.489	-0.111	0.284	5.974

```

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.29227    0.16616   -1.76    0.081 .
Bill         0.18221    0.00645   28.25   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.98 on 155 degrees of freedom
Multiple R-squared:  0.837, Adjusted R-squared:  0.836
F-statistic: 798 on 1 and 155 DF,  p-value: <2e-16

```

The first line tells R to fit the regression. The thing on the left of the `~` is our outcome, the things on the right are our covariates or predictors. R then saves the results of all that work under the name `mod` (short for model - you can call it anything you want). Once we fit the model, we used `summary()` command to print the output to the screen.

Results relevant to the intercept are in the **(Intercept)** row and results relevant to the slope are in the **Bill** row (**Bill** is the explanatory variable). The **Estimate** column gives the estimated coefficients, the **Std. Error** column gives the standard error for these estimates, the **t value** is simply estimate/SE, and the p-value is the result of a hypothesis test testing whether that coefficient is significantly different from 0.

We also see the RMSE as **Residual standard error** and R^2 as **Multiple R-squared**. The last line of the regression output gives details relevant to an ANOVA table for testing our model against no model. It has the F-statistic, degrees of freedom, and p-value.

You can pull the coefficients of your model out with the `coef()` command:

```

coef(mod)

(Intercept)      Bill
      -0.292      0.182

coef(mod)[1] # intercept

(Intercept)
      -0.292

coef(mod)[2] # slope

Bill
0.182

```

```
coef(mod)["Bill"] # alternate way.
```

```
Bill  
0.182
```

Alternatively, you can use the `tidy()` function from `broom` to turn the regression results into a tidy data frame, which makes it easier to work with:

```
tidy(mod)
```

```
# A tibble: 2 x 5
```

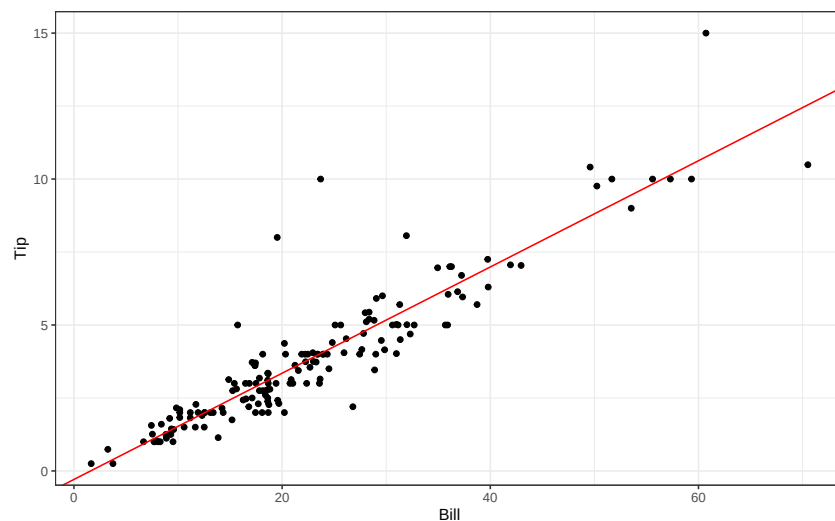
term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	-0.292	0.166	-1.76	8.06e- 2
2 Bill	0.182	0.00645	28.2	5.24e-63

```
tidy(mod)[[2,2]] # slope
```

```
[1] 0.182
```

We can plot our regression line on top of the scatterplot manually using the `geom_abline()` layer in `ggplot`:

```
ggplot( RestaurantTips, aes( Bill, Tip ) ) +  
  geom_point() +  
  geom_abline( intercept = -0.292, slope = 0.182, col="red" )
```



35.2 Multiple Regression

We now include the additional explanatory variables of number in party (**Guests**) and whether or not they pay with a credit card (**Credit**):

```
tip.mod <- lm(Tip ~ Bill + Guests + Credit, data=RestaurantTips )
summary(tip.mod)
```

Call:

```
lm(formula = Tip ~ Bill + Guests + Credit, data = RestaurantTips)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.384	-0.478	-0.108	0.272	5.984

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.25468	0.20273	-1.26	0.21
Bill	0.18302	0.00846	21.64	<2e-16 ***
Guests	-0.03319	0.10282	-0.32	0.75
Credit	0.04217	0.18282	0.23	0.82

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.985 on 153 degrees of freedom

Multiple R-squared: 0.838, Adjusted R-squared: 0.834

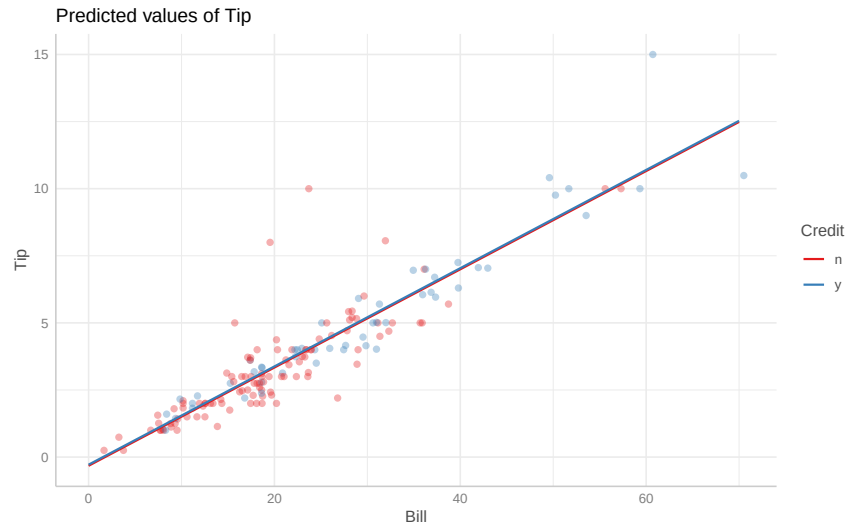
F-statistic: 263 on 3 and 153 DF, p-value: <2e-16

This output should look very similar to the output for one variable, except now there is a row corresponding to each explanatory variable. Our two-category (y, n) **Credit** variable was automatically converted to a 0-1 dummy variable (with “y” being 1 and “n” our baseline).

You can make plots and tables of your fit models. For one easy kind of regression graph, try **ggeffects**:

```
# graph model 2, with Bill on X, Credit as color, and Guests held constant at the mean
ggeffect(tip.mod, terms = c("Bill", "Credit")) |>
  plot(show_data = TRUE, show_ci = FALSE)
```

Data points may overlap. Use the ``jitter`` argument to add some amount of random variation to the location of data points and avoid overplotting.



For making tables, `?@sec-make-regression-tables`.

35.3 Categorical Variables (and Factors)

You can include any explanatory categorical variable in a multiple regression model, and R will automatically create corresponding 0/1 variables. For example, if you were to include gender coded as male/female, R would create a variable `GenderMale` that is 1 for males and 0 for females.

35.3.1 Numbers Coding Categories.

If you have multiple levels of a category, but your levels are coded with numbers you have to be a bit careful because R can treat this as a quantitative (continuous) variable by mistake in some cases. You will know it did this if you only see the single variable on one line of your output. For categorical variables with k categories, you should see $k - 1$ lines.

To make a variable categorical, even if the levels are numbers, convert the variable to a factor with `as.factor` or `factor`:

```
# load the US states data
data( USStates )

# convert Region to a factor
USStates <- USStates |>
  mutate(Region = factor(Region))
```

35.3.2 Setting new baselines.

We can reorder the levels if desired (the first is our baseline).

```
levels( USStates$Region )

[1] "MW" "NE" "S"  "W"

USStates$Region = relevel(USStates$Region, "S" )
levels( USStates$Region )

[1] "S"  "MW" "NE" "W"
```

Now any regression will use the south as baseline.

35.3.3 Testing for significance of a categorical variable.

When deciding whether to keep a categorical variable, we need to test how important all the dummy variables for that category are to the model all at once. We do this with ANOVA. Here we examine whether region is useful for predicting the percent vote for Clinton in 2016:

```
mlm = lm( ClintonVote ~ Region, data=USStates)
anova( mlm )
```

Analysis of Variance Table

```
Response: ClintonVote
      Df Sum Sq Mean Sq F value Pr(>F)
Region   3   1643     548   6.99 0.00057 ***
Residuals 46   3603       78
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

It is quite important.

We can also compare for region beyond some other variable:

```
mlm2 = lm( ClintonVote ~ HouseholdIncome + HouseholdIncome + HighSchool +
           EighthGradeMath, data=USStates)

mlm3 = lm( ClintonVote ~ HouseholdIncome + HouseholdIncome + HighSchool +
           EighthGradeMath + Region, data=USStates)
anova( mlm2, mlm3 )
```


Analysis of Variance Table

Model 1: ClintonVote ~ HouseholdIncome + HouseholdIncome + HighSchool + EighthGradeMath

Model 2: ClintonVote ~ HouseholdIncome + HouseholdIncome + HighSchool + EighthGradeMath + Region

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	46	3287				
2	43	2649	3	638	3.45	0.025 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Region is still important, beyond including some further controls. Interpreting this mess of a regression is not part of this document; this document shows you how to run regressions but it doesn't discuss whether you should or not.

35.3.4 Missing levels in a factor

R often treats categorical variables as factors. This is often useful, but sometimes annoying. A factor has different **levels** which are the different values it can be. For example:

```
data(FishGills3)
levels(FishGills3$Calcium)

[1] ""      "High"  "Low"   "Medium"

table(FishGills3$Calcium)
```

	High	Low	Medium
0	30	30	30

Note the weird nameless level; it also has no actual observations in it. Nevertheless, if you make a boxplot, you will get an empty plot in addition to the other three. This error was likely due to some past data entry issue. You can drop the unused level:

```
FishGills3$Calcium = droplevels(FishGills3$Calcium)
```

You can also turn a categorical variable into a numeric one like so:

```
summary( FishGills3$Calcium )
```

High	Low	Medium
30	30	30

```
asnum = as.numeric( FishGills3$Calcium )
asnum
```

```
[1] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3
[39] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
[77] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

Regression on only a categorical variable is fine:

```
mylm = lm( GillRate ~ Calcium, data=FishGills3 )
mylm
```

Call:

```
lm(formula = GillRate ~ Calcium, data = FishGills3)
```

Coefficients:

(Intercept)	CalciumLow	CalciumMedium
58.2	10.3	0.5

R has made you a bunch of dummy variables automatically. Here “high” is the baseline, selected automatically. We can also force it so there is no baseline by removing the intercept, in which case the coefficients are the means of each group.

```
myymm = lm( GillRate ~ 0 + Calcium, data=FishGills3 )
myymm
```

Call:

```
lm(formula = GillRate ~ 0 + Calcium, data = FishGills3)
```

Coefficients:

CalciumHigh	CalciumLow	CalciumMedium
58.2	68.5	58.7

35.4 Some extensions (optional)

35.4.1 Confidence Intervals

To get confidence intervals around each parameter in your model, try this:

```
confint(tip.mod)
```

```
                2.5 % 97.5 %  
(Intercept) -0.655  0.146  
Bill         0.166  0.200  
Guests      -0.236  0.170  
Credity     -0.319  0.403
```

You can also create them easily using `tidy` and `mutate`:

```
tip.mod |>  
  tidy() |>  
  mutate(upper = estimate + 1.96*std.error,  
         lower = estimate - 1.96*std.error)
```

A tibble: 4 x 7

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>	upper <dbl>	lower <dbl>
1	(Intercept)	-0.255	0.203	-1.26	2.11e- 1	0.143	-0.652
2	Bill	0.183	0.00846	21.6	2.07e-48	0.200	0.166
3	Guests	-0.0332	0.103	-0.323	7.47e- 1	0.168	-0.235
4	Credity	0.0422	0.183	0.231	8.18e- 1	0.400	-0.316

35.4.2 Prediction

Suppose a server at this bistro is about to deliver a \$20 bill, and wants to predict their tip. They can get a predicted value and 95% (this is the default level, change with `level`) prediction interval with

```
new.dat = data.frame( Bill = c(20) )  
predict(mod,new.dat, interval = "prediction")
```

```
    fit lwr upr  
1 3.35 1.41 5.29
```

They should expect a tip somewhere between \$1.41 and \$5.30.

If we know a bit more we can use our more complex model called `tip.mod` from above:

```
new.dat = data.frame( Bill = c(20), Guests=c(1), Credit=c("n") )
predict(tip.mod,new.dat,interval = "prediction")
```

```
      fit   lwr   upr
1 3.37  1.41  5.34
```

This is the predicted tip for one guest paying with cash for a \$20 tip. It is wider than our original interval because our model is a bit more unstable (it turns out guest number and credit card aren't that relevant or helpful).

Compare the prediction interval to the confidence interval

```
new.dat = data.frame( Bill = c(20), Guests=c(1), Credit=c("n") )
predict(tip.mod, new.dat, interval = "confidence")
```

```
      fit   lwr   upr
1 3.37  3.09  3.65
```

This predicts the mean tip for all single guests who pay a \$20 bill with cash. Our interval is smaller because we are generating a confidence interval for where the mean is, and are ignoring that individuals will vary around that mean. Confidence intervals are different from prediction intervals.

35.4.3 Removing Outliers

If you can identify which rows the outliers are on, you can do this by hand (say the rows are 5, 10, 12).

```
new.data = old.data[ -c(5,10,12), ]
lm( Y ~ X, data=new.data )
```

Some technical details: The `c(5,10,12)` is a list of 3 numbers. The `c()` is the concatenation function that takes things makes lists out of them. The “-list” notation means give me my old data, but without rows 5, 10, and 12. Note the comma after the list. This is because we identify elements in a dataframe with row, column notation. So `old.data[1,3]` would be row 1, column 3.

If you notice your points all have X bigger than some value, say 20.5, you could use filtering to keep everything less than some value:

```
new.data = filter( old.data, X <= 20.5 )
```

35.4.4 Missing data

If you have missing data, `lm` will automatically drop those cases because it doesn't know what else to do. It will tell you this, however, with the `summary` command.

```
data(AllCountries)
dev.lm = lm( BirthRate ~ Rural + Health + ElderlyPop, data=AllCountries )
summary( dev.lm )
```

Call:

```
lm(formula = BirthRate ~ Rural + Health + ElderlyPop, data = AllCountries)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.592	-3.728	-0.791	3.909	16.218

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	26.5763	1.6795	15.82	< 2e-16 ***
Rural	0.0985	0.0224	4.40	1.9e-05 ***
Health	-0.0995	0.0930	-1.07	0.29
ElderlyPop	-1.0249	0.0881	-11.64	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.83 on 174 degrees of freedom

(39 observations deleted due to missingness)

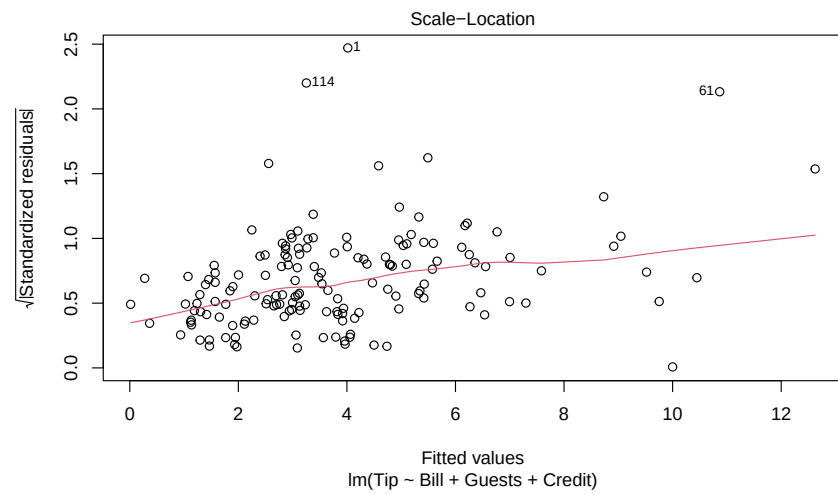
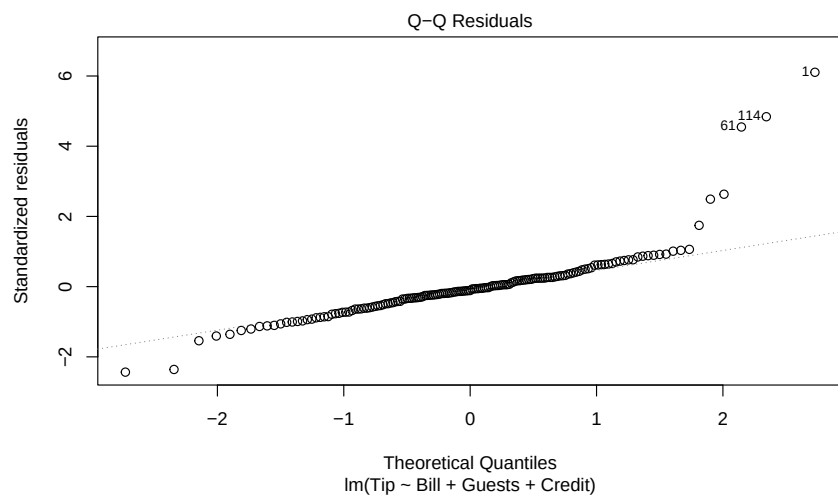
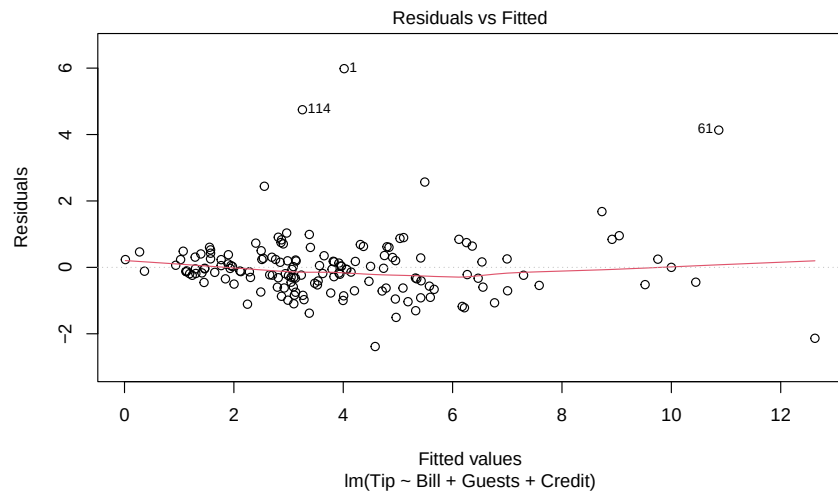
Multiple R-squared: 0.663, Adjusted R-squared: 0.657

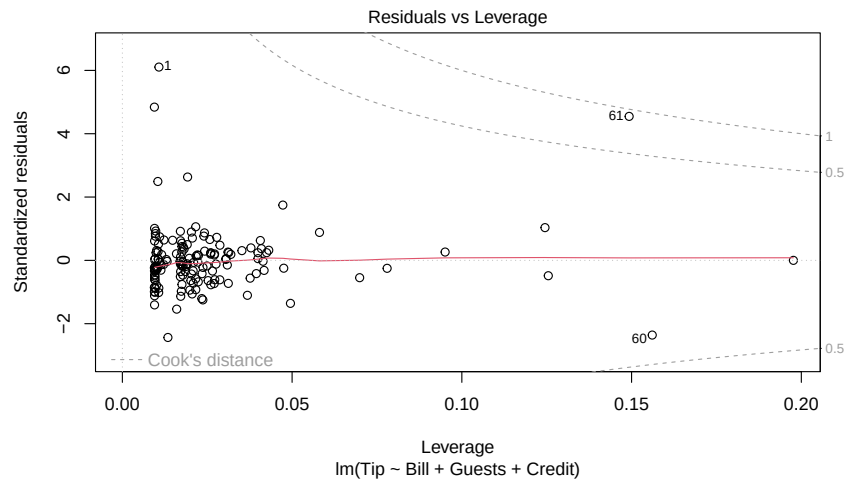
F-statistic: 114 on 3 and 174 DF, p-value: <2e-16

35.4.5 Residual plots and model fit

If we throw out model into the `plot` function, we get some nice regression diagnostics.

```
plot(tip.mod)
```

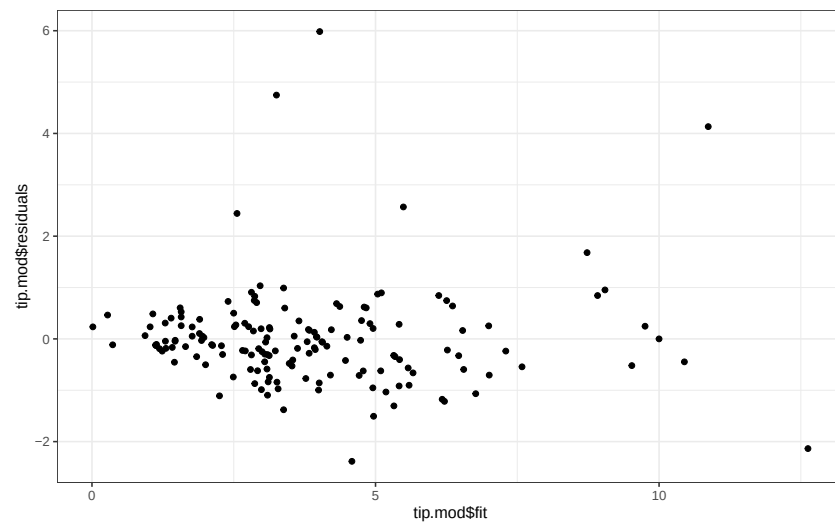




To generate classic model fit diagnostics with more control, we need to calculate residuals, make a residual versus fitted values plot, and make a histogram of the residuals. We can make some quick and dirty plots with `qplot` (standing for “quick plot”) like so:

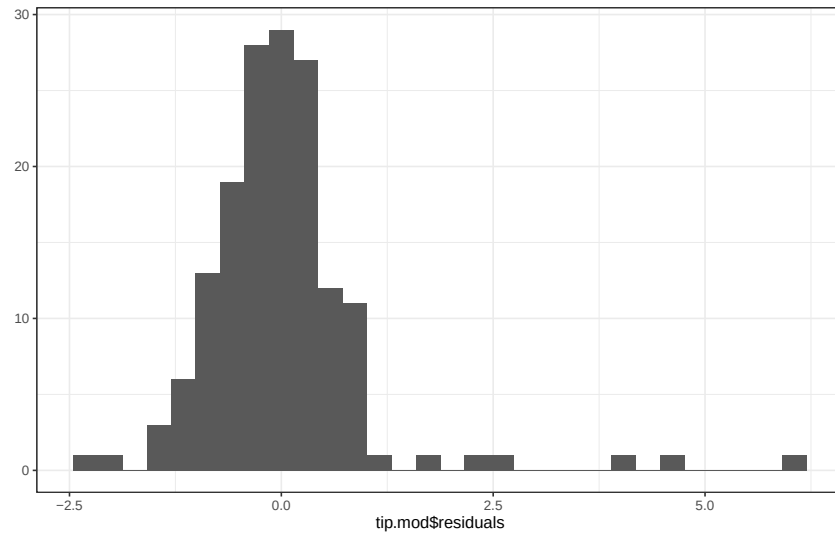
```
qplot(tip.mod$fit, tip.mod$residuals )
```

Warning: ``qplot()`` was deprecated in `ggplot2 3.4.0`.



and

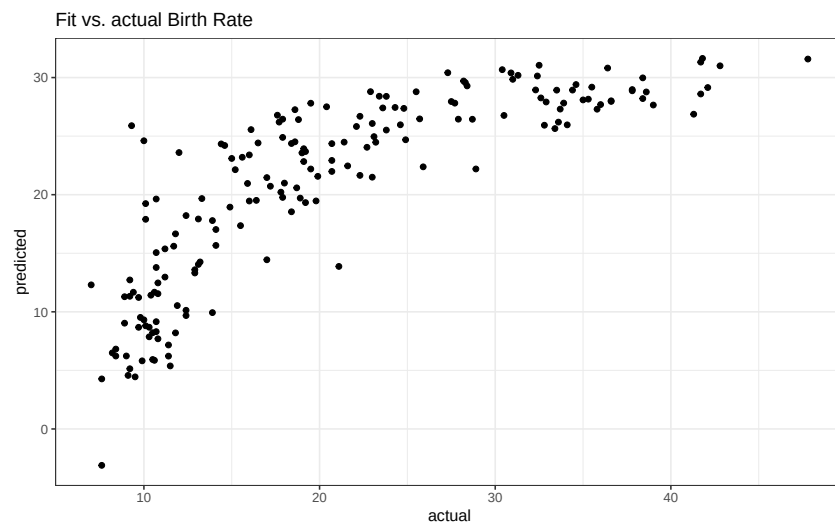
```
qplot(tip.mod$residuals, bins=30)
```



We see no real pattern other than some extreme outliers. The residual histogram suggests we are not really normally distributed, so we should treat our SEs and p -values with caution. These plots are the canonical “model-checking” plots you might use.

Another is the “fitted outcomes vs. actual outcomes” plot of:

```
predicted = predict( dev.lm )
actual = dev.lm$model$BirthRate
qplot( actual, predicted, main="Fit vs. actual Birth Rate" )
```



Note the `dev.lm` variable has a `model` variable inside it. This is a data frame of the **used** data for the model (i.e., if cases were dropped due to missingness, they will not be in the model).

We then grab the birth rates from this, and make a scatterplot. If we tried to skip this, and use the original data, we would get an error because our original data set has some observations that were dropped.

Note we can't just add our predictions to `AllCountries` since we would get an error due to this dropped data issue:

```
AllCountries$predicted = predict( dev.lm )
```

```
Error in `$$<-.data.frame`(`*tmp*`, predicted, value = c(`1` = 31.630301617421, :  
  replacement has 179 rows, data has 217
```

We can, however, predict like this:

```
AllCountries$predicted = predict( dev.lm, newdata=AllCountries )
```

The `newdata` tells `predict` to generate a prediction for each row in `AllCountries` rather than each row in the left over data after `lm` dropped cases with missing values.

36 Doing things over and over again

In this handout, we will look at several ways of doing things over and over again in R. This comes up all the time. Three main ones we have seen are the following:

- Fit a loess curve to my data for each of a series of smoothing parameters.
- Repeatedly bootstrap my dataset and analyze it.
- Run a simulation where we generate data and analyze it over and over.
- Scrape a series of web pages

Coding wise, there are a few ways of repeating yourself. We are going to walk through them and compare these tools to each other.

37 The replicate() command

This simple command repeats a line of code a given number of times. If the line of code gives you a number back each time it is run, then you will end up with a list of numbers.

To illustrate, say we want to look at rolling dice. Here is some code that provides a function that will roll some number of 6-sided dice:

```
roll_dice = function( ndice ) {  
  rolls = sample( 1:6, ndice, replace=TRUE )  
  sum( rolls )  
}
```

```
# Roll a single die  
roll_dice( 1 )
```

```
[1] 2
```

```
# Roll 3 dice and add them up.  
roll_dice( 3 )
```

```
[1] 14
```

If we want to get the sum of three dice over and over, we can replicate:

```
rolls = replicate( 10, roll_dice( 3 ) )  
rolls
```

```
[1] 8 12 12 12 12 6 9 13 11 11
```

Note how the `rolls` variable is a nice numeric vector, easy to work with. It is easy to do calculations with it, like take the average:

```
mean( rolls )
```

```
[1] 10.6
```

Here we use this to see how often we roll above a 15:

```
rolls = replicate( 10000, roll_dice( 3 ) )  
mean( rolls > 15 )
```

```
[1] 0.0452
```

38 The rerun() command

The `rerun()` command, from the tidyverse, is almost exactly like `replicate`, but instead of giving a numeric vector of numbers back, it gives an R list. For brevity, I rerun twice in the following:

```
rolls = rerun( 2, roll_dice(3) )
```

```
Warning: `rerun()` was deprecated in purrr 1.0.0.
```

```
i Please use `map()` instead.
```

```
# Previously
```

```
rerun(2, roll_dice(3))
```

```
# Now
```

```
map(1:2, ~ roll_dice(3))
```

```
rolls
```

```
[[1]]
```

```
[1] 10
```

```
[[2]]
```

```
[1] 10
```

Each element of the list is itself a list of numbers! Why would someone make such an annoying command like that? This is useful if the function we are rerunning gives us back multiple things. For example:

```
roll_dice_extended = function( ndice ) {  
  rolls = sample( 1:6, ndice, replace=TRUE )  
  c( mean = mean( rolls ), median = median( rolls ), max = max( rolls ) )  
}  
roll_dice_extended()
```

```
mean median    max  
  3.5    3.0    6.0
```

Now when we rerun we get this:

```
rolls = rerun( 2, roll_dice_extended(3) )

Warning: `rerun()` was deprecated in purrr 1.0.0.
i Please use `map()` instead.
# Previously
rerun(2, roll_dice_extended(3))

# Now
map(1:2, ~ roll_dice_extended(3))
```

```
rolls
```

```
[[1]]
      mean  median    max
2.666667 2.000000 5.000000
```

```
[[2]]
      mean median    max
      4      4      5
```

But this is hard to work with. All our numbers are nested and weird. But there is a solution, which is the “rerun + bind_rows combo”. Once we have a list of our answers, we can “stack” them with `bind_rows`:

```
bind_rows( rolls )

# A tibble: 2 x 3
  mean median    max
  <dbl> <dbl> <dbl>
1  2.67      2      5
2   4       4      5
```

We can also give each row a name:

```
bind_rows( rolls, .id="runID" )

# A tibble: 2 x 4
  runID mean median    max
  <chr> <dbl> <dbl> <dbl>
1  1    2.67      2      5
2  2     4       4      5
```

Side Note: I recommend making your function give back a simple little dataframe of all your stuff. It is less prone to having weird errors. The `bind_rows()` method does really well with data.frames or tibbles. Here is an updated version of the above:

```
roll_dice_extended = function( ndice ) {  
  rolls = sample( 1:6, ndice, replace=TRUE )  
  data.frame( mean = mean( rolls ), median = median( rolls ), max = max( rolls ) )  
}  
roll_dice_extended()
```

```
      mean median max  
1 3.833333      4   6
```

```
rolls = rerun( 6, roll_dice_extended(3) )
```

Warning: `rerun()` was deprecated in purrr 1.0.0.

i Please use `map()` instead.

Previously

```
rerun(6, roll_dice_extended(3))
```

Now

```
map(1:6, ~ roll_dice_extended(3))
```

```
rolls = bind_rows( rolls, .id="runID" )
```

```
rolls
```

```
  runID    mean median max  
1     1 3.666667      4   6  
2     2 3.333333      4   5  
3     3 3.000000      3   4  
4     4 3.666667      5   5  
5     5 4.000000      5   6  
6     6 6.000000      6   6
```

38.1 Warning: `replicate()` doesn't do well with fancy functions

The `replicate()` command doesn't act nice in the following:

```
rolls = replicate( 2, roll_dice_extended(3) )  
rolls
```

```
, , 1
```

```
      mean median max  
1 4      4      6
```

```
, , 2
```

```
      mean      median max  
1 3.333333 3      5
```

That doesn't look like a fun thing to work with. (It is a 3 dimensional array of output, in case you are wondering.) Use `rerun + bind_rows`; it is easier to control and understand.

38.2 Take-away

If your function that you want to repeat returns a single number with each call, use `replicate()`. If it returns more than one thing, use the `rerun() + bind_rows()` combination.

39 map(), another way of repeating yourself

The above runs the exact same code over and over. Sometimes you want to run the same function on a collection of different things (e.g., fit a loess line for each of a series of bandwidths). This is done with the `map()` command which takes a list of things, and then calls a function for each thing on that list.

To illustrate, we will see how the largest number rolled changes as a function of the number of dice rolled. To start, let's roll 1 die, then 2 dice, then 3 dice, and each time calculate the average, median, and max:

```
dice = 1:3
result = map( dice, roll_dice_extended )
result
```

```
[[1]]
      mean median max
1      6      6    6
```

```
[[2]]
      mean median max
1  3.5     3.5    5
```

```
[[3]]
      mean median max
1 3.666667      4    6
```

Note unlike `rerun` or `replicate` we are not **calling** our `roll_dice_extended` function, we are just giving the name of it. You can tell since we do not have the `()` after `roll_dice_extended`, we just pass the name of the function we want to call. We are asking `map` to call `roll_dice_extended` over and over. For `rerun` or `replicate`, by comparison, we provide a stand-alone complete line of code that would run by itself. For `map` we just give the name of a function to run.

We can make our output nicer with the same `bind_rows` trick from `rerun()`:

```
bind_rows( result )
```

	mean	median	max
1	6.000000	6.0	6
2	3.500000	3.5	5
3	3.666667	4.0	6

Even better is using the `map_df()` method, which works nicely **provided your function returns a dataframe**:

```
result = map_df( dice, roll_dice_extended )
result$ndice = dice
result
```

	mean	median	max	ndice
1	1.000000	1	1	1
2	5.000000	5	6	2
3	3.666667	4	5	3

40 Repeating yourself when you are repeating yourself

Our little simulation is a bit sad in that we only have a single trial for each of our number of dice scenarios. What we really want is to know the distribution of the maximum roll for each number of dice. We do this by repeating ourself and then repeating this repeating ourself!

Study this:

```
one_trial = function( ndice ) {  
  rolls = rerun( 1000, roll_dice_extended( ndice ) )  
  rolls = bind_rows( rolls )  
  tbl = table( rolls$max )  
  tbl  
}  
one_trial( 1 )
```

Warning: `rerun()` was deprecated in purrr 1.0.0.

i Please use `map()` instead.

Previously

```
rerun(1000, roll_dice_extended(ndice))
```

Now

```
map(1:1000, ~ roll_dice_extended(ndice))
```

```
  1  2  3  4  5  6  
152 149 176 147 186 190
```

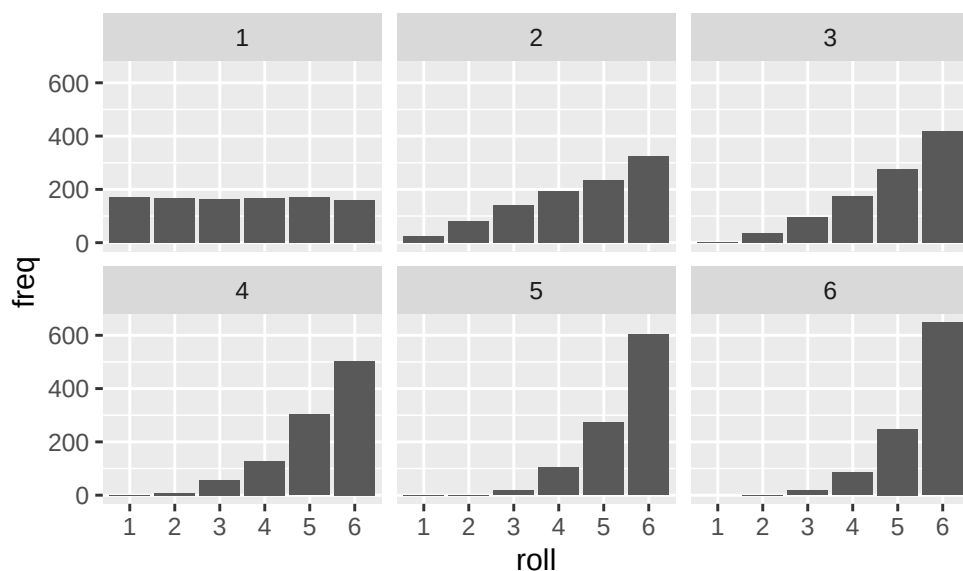
Then this:

```
result = map( 1:6, one_trial )  
result = bind_rows( result )  
result$ndice = 1:6  
result
```

```
# A tibble: 6 x 7
  `1`      `2`      `3`      `4`      `5`      `6`      ndice
  <table[1d]> <table[1d]> <table[1d]> <table[1d]> <table[1d]> <table[1d]> <int>
1 170      167      165      167      170      161         1
2  26       81     142     193     235     323         2
3   2       34      94     174     277     419         3
4   1        8      57     128     302     504         4
5   1        1      18     105     273     602         5
6 NA        1      17      85     248     649         6
```

We can then make a nice plot:

```
resL = pivot_longer( result, `1`:`6`,
                     names_to="roll", values_to="freq" ) %>%
  filter( !is.na( freq ) )
resL$freq = as.numeric( resL$freq )
ggplot( resL, aes( roll, freq ) ) +
  facet_wrap( ~ndice ) +
  geom_col( )
```



Note. The `as.numeric` line is because the `table` command makes table objects which are really just lists of numbers, but it seems cleaner to tell R to knock it off and just let it be a number. Not doing this still works fine, it just gives a warning in `ggplot()`

41 Advanced stuff

In the following we look at a few cool functions that help with mapping.

41.1 relocate()

Not really mapping specific, but still a nice way to move a variable to the start of a data frame.

```
relocate( resL, freq )
```

```
# A tibble: 35 x 3
  freq ndice roll
  <dbl> <int> <chr>
1   170     1 1
2   167     1 2
3   165     1 3
4   167     1 4
5   170     1 5
6   161     1 6
7    26     2 1
8    81     2 2
9   142     2 3
10  193     2 4
# i 25 more rows
```

41.2 pull()

This will grab a column from a data frame in a list of pipe commands, which can make it easier to plug into some other tools.

Here we pull the roll column and then hand it to the `table()` command to count the number of instances of each roll.

```
resL %>% pull( roll ) %>%
  table()
```

```
.
1 2 3 4 5 6
5 6 6 6 6 6
```

41.3 set_names(list, names)

On the fly name a list before handing to map!

```
dice = 2:6
dice %>% set_names( paste0( dice, " dice" ) ) %>%
  map_df( roll_dice_extended, .id = "scenario" )
```

	scenario	mean	median	max
1	2 dice	3.000000	3.0	5
2	3 dice	3.333333	3.0	4
3	4 dice	2.500000	1.5	6
4	5 dice	3.400000	3.0	4
5	6 dice	3.833333	4.0	6

41.4 unpack() and pack()

These will translate a data frame column into individual columns. You can end up with a data frame column if you use map to make a new column in your data:

```
scenarios = tibble( n_dice = 1:6 )
scenarios = scenarios %>%
  mutate( result = map_df( n_dice, roll_dice_extended ) )
scenarios
```

```
# A tibble: 6 x 2
  n_dice result$mean $median $max
  <int>      <dbl>   <dbl> <int>
1     1         4       4       4
2     2        3.5     3.5       4
3     3        1.67     2       2
4     4        4.25     5       6
5     5        3.8      4       5
6     6        3.17     4       4
```

Notice the weird \$ in the printout? This is because the three columns are inside the dataframe of `result`. You can unpack it like so:

```
unpack( scenarios, result )
```

```
# A tibble: 6 x 4
  n_dice mean median  max
  <int> <dbl> <dbl> <int>
1     1  4     4     4
2     2  3.5  3.5     4
3     3  1.67  2     2
4     4  4.25  5     6
5     5  3.8   4     5
6     6  3.17  4     4
```

to get your nice, normal dataframe.

42 Exploring map and friends (tidyverse)

43 Overview

This document pulls out elements from R for DS Chapter 21, which is on ways of doing iteration. And then it jumps to the `map()` function and list columns extensively used in Chapter 25. It is all within the tidyverse:

```
library( tidyverse )
```

44 The map() function (R for DS 21.5)

The core idea of `map` is to take a list and call a function on each thing on that list. This list could be a list of lists (i.e. a dataframe), which can be very important, because then you can call a function on each column in your dataframe.

Generally what `map` gives back is another list, with each element being what we got back from the function from each call. If your first list has named items, your result will have those names as well. This can be quite useful.

To illustrate we make a silly list with a list inside it, and then call `map()`:

```
lst = list( 1, B=3, inner_list = c(10, 20, 30) )
lst

[[1]]
[1] 1

$B
[1] 3

$inner_list
[1] 10 20 30

map( lst, mean )

[[1]]
[1] 1

$B
[1] 3

$inner_list
[1] 20
```

See? We got the mean out of each element in our list.

By adding `_dbl` or similar, we can explicitly tell `map()` to turn this list into a vector of some type:

```
map_dbl( lst, mean )
```

	B	inner_list
1	3	20

We also have `map_int`, `map_chr`, `map_lgl`, and `map_df`. Be warned, if you call one of these and the things you get back are the wrong type, you get an error:

```
map_df( lst, mean )
```

```
Error in `dplyr::bind_rows()`:  
! Argument 1 must be a data frame or a named atomic vector.
```

44.1 Making data frames

The `map_df()` is especially important. But you need something that gives back a dataframe for it to work! This dataframe can be a 1 row dataframe, however. For example:

```
my.func = function( numbers ) {  
  data.frame( mean=mean(numbers), sd=sd(numbers) )  
}  
map_df( lst, my.func )
```

	mean	sd
1	1	NA
2	3	NA
3	20	10

44.2 The `unlist()` function

A useful function is `unlist` which takes a bunch of lists and just smashes it all together into one vector

```
unlist( lst )
```

	B	inner_list1	inner_list2	inner_list3
1	3	10	20	30

It even keeps the names, if there are any!

44.3 Extra arguments to map

If your function takes extra stuff, then you can pass it along like so:

```
log( 10, base=3 )
```

```
[1] 2.095903
```

```
map( lst, log, base=3 )
```

```
[[1]]
```

```
[1] 0
```

```
$B
```

```
[1] 1
```

```
$inner_list
```

```
[1] 2.095903 2.726833 3.095903
```

45 Banging on Dataframes

If you have a dataframe that is a bunch of smaller dataframes, you can get a list of the smaller dataframes via `split()`. This will make a new dataframe with the rows corresponding to each unique value of a passed variable:

```
df = data.frame( A=1:6, B=c("A","A","A","B","B","C") )
df.list = split( df, df$B )
df.list
```

```
$A
```

```
  A B
1 1 A
2 2 A
3 3 A
```

```
$B
```

```
  A B
4 4 B
5 5 B
```

```
$C
```

```
  A B
6 6 C
```

```
class( df.list )
```

```
[1] "list"
```

```
map_dbl( df.list, nrow )
```

```
 A B C
3 2 1
```

See R for DS 25.5.1

45.1 Chaining map

You can chain maps together. Say we are going to fit a linear model for each group of a dataframe defined by different values of a categorical variable. For example, lets divide the famous “iris data” into groups by species, and then fit a linear model for each group, and then get the R squared from that model.

Before we dive into this, we need to note that you can use “.” as a pronoun for “the thing we are working on.” So, for example, the following two lines are equivalent:

```
iris.list = split( iris, iris$Species )
iris.list = iris %>% split( .$Species )
```

The “.” is what is “coming down the pipe.”

Now we call `lm()` on each part by writing a function to analyze our dataframe:

```
my.func = function( my.df ) {
  lm( Sepal.Length ~ Sepal.Width, data=my.df )
}
```

```
mods = iris.list %>% map( my.func )
```

Our `mods` is a list of results. Here is the first result (a linear regression output):

```
mods[[1]]
```

Call:

```
lm(formula = Sepal.Length ~ Sepal.Width, data = my.df)
```

Coefficients:

```
(Intercept)  Sepal.Width
      2.6390         0.6905
```

The above can be bundled as follows

```
mods = iris %>% split( .$Species ) %>% map( my.func )
```

But writing the wrapper function for `lm` is annoying. We can tell our `map` that we are “writing a function on the fly” with an abuse of the “~” notation:

```
mods = iris %>% split( .$Species ) %>%
  map( ~lm( Sepal.Length ~ Sepal.Width, data=. ) )
```

Here we are saying, inside the `map`, that we want to run the `lm()` function, and that the stuff we got from the map should be the data argument (`data = .`).

Finally, we map the above again to `summary` to calculate our R squared, and then we extract it:

```
names( summary( mods[[1]] ) )

[1] "call"          "terms"          "residuals"      "coefficients"
[5] "aliased"       "sigma"          "df"             "r.squared"
[9] "adj.r.squared" "fstatistic"     "cov.unscaled"
```

```
rs = iris %>% split( .$Species ) %>%
  map( ~lm( Sepal.Length ~ Sepal.Width, data=. ) ) %>%
  map( summary ) %>%
  map_dbl( ~.$r.squared )
rs

      setosa versicolor  virginica
0.5513756  0.2765821  0.2090573
```

The `~.$r.squared` line is a bit weird. We are taking our pronoun, and grabbing the “r.squared” variable from inside it (if there is no such value inside it we will get NA). This is a common activity, and thus the tidyverse provides a shortcut: just put in the name as a string:

```
rs = iris %>% split( .$Species ) %>%
  map( ~lm( Sepal.Length ~ Sepal.Width, data=. ) ) %>%
  map( summary ) %>%
  map_dbl( "r.squared" )
rs

      setosa versicolor  virginica
0.5513756  0.2765821  0.2090573
```

45.2 Dataframes are lists of the columns

Let’s calculate the unique number of values in each column of iris:

```
analyze_column = function( col ) {
  length( unique( col ) )
}
map_dbl( iris, analyze_column )
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
35	23	43	22	3

Or, by map chaining:

```
iris %>% map( unique ) %>% map_dbl( length )
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
35	23	43	22	3

45.3 The transpose() function

We can do some funky list manipulation. Say our map gives us back a list of pairs (note cool `select_if()` to grab only numeric columns):

```
my.func = function( col ) {
  c( mean=mean(col), sd=sd(col) )
}
res <- iris %>% select_if( is.numeric ) %>% map( my.func )
res
```

```
$Sepal.Length
  mean      sd
5.843333 0.8280661
```

```
$Sepal.Width
  mean      sd
3.057333 0.4358663
```

```
$Petal.Length
  mean      sd
3.758000 1.765298
```

```
$Petal.Width
  mean      sd
1.199333 0.7622377
```

But this is hard to use. `transpose()` to the rescue!

```
res2 <- transpose( res )
names(res2)
```



```
[1] "mean" "sd"
```

```
res2$mean
```

```
$Sepal.Length
```

```
[1] 5.843333
```

```
$Sepal.Width
```

```
[1] 3.057333
```

```
$Petal.Length
```

```
[1] 3.758
```

```
$Petal.Width
```

```
[1] 1.199333
```

We now have two lists, one for the means, one for the sds.

See R for DS 21.6 for more on this.

46 List columns and notes on R for DS 25 (25.2)

Chapter 25, while nominally about modeling, actually shows a lot of useful code for complex data wrangling. In particular it talks about list columns, which are ways of storing data in chunks and manipulating the data chunk-by-chunk. This connects heavily with the `map` concepts above.

46.1 List columns

A “list column” is a column of your dataframe where each entry in the column can be anything you want, including a list or entire dataframe. For example, we can make some on the fly using `enframe()` (See R for DS 25.4.4)

```
x <- list( a=c("a","b"),
          b = 3:7,
          c = iris
        )
```

```
df <- enframe(x)
df
```

```
# A tibble: 3 x 2
  name  value
<chr> <list>
1 a     <chr [2]>
2 b     <int [5]>
3 c     <df [150 x 5]>
```

There are several ways to make list columns to be aware of:

- Using `nest()`
- Using `mutate()` where your call gives a list back (e.g., due to `map()`)
- Using `summarise()` with a function that gives back multiple results (such as `quantile()`).

46.2 nest() and unnest()

In particular, you can group a dataframe by something, and then nest it to make a small dataframe with each row corresponding to a single dataframe slice of your original data

```
dfs <- iris %>% group_by( Species ) %>% nest()
dfs
```

```
# A tibble: 3 x 2
# Groups:   Species [3]
  Species    data
  <fct>     <list>
1 setosa   <tibble [50 x 4]>
2 versicolor <tibble [50 x 4]>
3 virginica <tibble [50 x 4]>
```

```
# look, our 2nd value in our dataframe's data variable is a
# dataframe!
head( dfs$data[[2]], 3 )
```

```
# A tibble: 3 x 4
  Sepal.Length Sepal.Width Petal.Length Petal.Width
      <dbl>         <dbl>         <dbl>         <dbl>
1           7           3.2           4.7           1.4
2          6.4           3.2           4.5           1.5
3          6.9           3.1           4.9           1.5
```

Now you can use map to make new columns!

```
dfs %>% mutate( n = map_dbl( data, nrow ) )
```

```
# A tibble: 3 x 3
# Groups:   Species [3]
  Species    data      n
  <fct>     <list>  <dbl>
1 setosa   <tibble [50 x 4]>  50
2 versicolor <tibble [50 x 4]>  50
3 virginica <tibble [50 x 4]>  50
```

```
dfs2 <- dfs %>% mutate( mod = map( data, ~lm( Sepal.Length ~ Sepal.Width, data = . )))
dfs2
```

```
# A tibble: 3 x 3
# Groups:   Species [3]
  Species    data      mod
  <fct>     <list>    <list>
1 setosa    <tibble [50 x 4]> <lm>
2 versicolor <tibble [50 x 4]> <lm>
3 virginica <tibble [50 x 4]> <lm>
```

```
map( dfs2$mod, "coefficients" )
```

```
[[1]]
(Intercept) Sepal.Width
  2.6390012   0.6904897
```

```
[[2]]
(Intercept) Sepal.Width
  3.5397347   0.8650777
```

```
[[3]]
(Intercept) Sepal.Width
  3.9068365   0.9015345
```

The `nest()` method has `unnest()` which will unstack an internal list, repeating the row the list came from over and over to make a full set. Witness

```
dfs3 <- dfs2 %>% mutate( coefs = map( mod, "coefficients" ) )
dfs3
```

```
# A tibble: 3 x 4
# Groups:   Species [3]
  Species    data      mod    coefs
  <fct>     <list>    <list> <list>
1 setosa    <tibble [50 x 4]> <lm>   <dbl [2]>
2 versicolor <tibble [50 x 4]> <lm>   <dbl [2]>
3 virginica <tibble [50 x 4]> <lm>   <dbl [2]>
```

```
unnest( dfs3, coefs )
```

```
# A tibble: 6 x 4
# Groups:   Species [3]
  Species    data      mod    coefs
  <fct>     <list>    <list> <dbl>
1 setosa    <tibble [50 x 4]> <lm>   <dbl>
2 setosa    <tibble [50 x 4]> <lm>   <dbl>
3 versicolor <tibble [50 x 4]> <lm>   <dbl>
4 versicolor <tibble [50 x 4]> <lm>   <dbl>
5 virginica <tibble [50 x 4]> <lm>   <dbl>
6 virginica <tibble [50 x 4]> <lm>   <dbl>
```

```

1 setosa      <tibble [50 x 4]> <lm>    2.64
2 setosa      <tibble [50 x 4]> <lm>    0.690
3 versicolor <tibble [50 x 4]> <lm>    3.54
4 versicolor <tibble [50 x 4]> <lm>    0.865
5 virginica   <tibble [50 x 4]> <lm>    3.91
6 virginica   <tibble [50 x 4]> <lm>    0.902

```

See Chapter 25 of R for DS for examples of getting `unnest()` to work, including the fiddly stuff to get the names right. It can be a frustrating pain.

46.3 Using `summarise()`

We can call `summarise()` with functions that give multiple things. We need to wrap the response in a list.

```

ir2 <- iris %>% group_by( Species ) %>% summarise( probs = list( c( 0, 25, 50, 75, 100 ) ),
                                                    quants = list( quantile( Sepal.Length ) ) )
ir2

```

```

# A tibble: 3 x 3
  Species    probs    quants
  <fct>      <list>    <list>
1 setosa    <dbl [5]> <dbl [5]>
2 versicolor <dbl [5]> <dbl [5]>
3 virginica <dbl [5]> <dbl [5]>

```

Note the wrapping of everything in the list. Now each row has a list of the probabilities and a list of the results. We can get back a long frame with `unnest()`:

```
unnest( ir2 )
```

```
Warning: `cols` is now required when using `unnest()`.
i Please use `cols = c(probs, quants)`.
```

```

# A tibble: 15 x 3
  Species    probs quants
  <fct>      <dbl> <dbl>
1 setosa      0    4.3
2 setosa     25    4.8
3 setosa     50    5

```

4	setosa	75	5.2
5	setosa	100	5.8
6	versicolor	0	4.9
7	versicolor	25	5.6
8	versicolor	50	5.9
9	versicolor	75	6.3
10	versicolor	100	7
11	virginica	0	4.9
12	virginica	25	6.22
13	virginica	50	6.5
14	virginica	75	6.9
15	virginica	100	7.9

See R for DS Chapter 25.4 for lots of examples and discussion.

47 Demonstration of the Cluster Bootstrap

48 The Cluster Bootstrap

The cluster bootstrap is a resampling technique commonly used in statistics and machine learning for estimating the variability of statistical estimators, such as mean, variance, or regression coefficients. It is especially useful when the underlying data have a complex dependence structure.

The main idea of the cluster bootstrap is to resample clusters of data points rather than individual data points. A cluster is a group of data points that are correlated with each other, for example, spatially adjacent data points or data points within the same experimental unit.

To perform the cluster bootstrap, we follow these steps:

1. Divide the original data set by the clustering variable, with each cluster containing correlated data points.
2. Randomly select clusters from the original data set with replacement, forming a bootstrap sample with the same number of clusters (but not necessarily the same number of individuals, if cluster sizes vary). If a cluster gets resampled twice, each time would be given a different, new, cluster id, so the data still has the right number of clusters and so clusters don't end up larger than in the original data.
3. Compute the statistical estimator of interest on the bootstrap sample.
4. Repeat steps 2 and 3 a large number of times, say 1000 times, to obtain a distribution of the estimator.
5. Use the distribution of the estimator to construct confidence intervals or calculate standard errors.

The cluster bootstrap is particularly useful when the underlying data have a complex correlation structure that cannot be easily accounted for using standard resampling methods, such as the simple random sampling bootstrap.

The following script is used to explore data from the Tenn Star education randomized controlled trial (RCT). It loads student-level and teacher-level data, merges them by the class ID variable, deletes observations with missing data, and creates an indicator variable for whether the student was in a small class (this is the treatment). The script then fits a linear regression model to estimate the impact of being in a small class on math scores after kindergarten, using small class status, student birth quarter, teacher education level, and teacher experience as predictors.

However, the model is not adjusted for clustering at the class level, so the standard errors may be wrong. To correct for this, we offer two solutions. First, the sandwich package is used to calculate cluster-robust standard errors. This is the classic econometric solution for clustering in a regression. Second, we demonstrate cluster bootstrapping for estimating the standard errors of the model coefficients.

48.1 The Tenn Star Data

The Tennessee Star Experiment was a large-scale randomized controlled trial (RCT) conducted in Tennessee, USA, in the mid-1980s. The experiment aimed to evaluate the effectiveness of class size reduction on student outcome. Students and teachers were randomized to different class sizes, making the treatment assignment effectively assigned at the cluster level when looking at student outcomes.

We load and prepare the data as so:

```
stud <- read.csv('data/tenn_stud_dat.csv') ## student-level data
teach <- read.csv('data/tenn_teach_dat.csv') ## teacher-level data
dat <- merge(stud, teach, by="clid")
dat <- na.omit(dat) ## for simplicity, we'll delete everyone missing any variables

## create an indicator for being in a small class
dat$small_class <- ifelse( dat$cltypek == 'small class', "yes", "no" )

# Drop some extra variables
dat <- dplyr::select( dat, -id, -cltypek, -sesk )

head( dat )
```

	clid	ssex	srace		sbirthq	sbirthy	treadssk	tmathssk
1	1	male	black	2nd qtr - april,may,june	1980		445	489
2	1	male	black	3rd qtr - july,aug,sept	1980		393	429
3	1	male	black	4th qtr - oct,nov,dec	1978		395	429
4	1	male	black	2nd qtr - april,may,june	1980		403	405
5	1	female	black	2nd qtr - april,may,june	1980		424	500
6	1	male	black	3rd qtr - july,aug,sept	1980		414	444
	hdegk	cladk	totexpk	tracek	small_class			
1	bachelors	apprentice	1	black	no			
2	bachelors	apprentice	1	black	no			
3	bachelors	apprentice	1	black	no			
4	bachelors	apprentice	1	black	no			
5	bachelors	apprentice	1	black	no			

```
6 bachelors apprentice      1 black      no
```

```
length(unique(dat$clid)) ## 196 classes
```

```
[1] 196
```

```
nrow(dat) ## 3219 students
```

```
[1] 3219
```

Our research question is whether student math achievement was higher for kids in small classrooms vs. large ones.

48.2 The Wrong Method

If we use simple regression we are not taking the correlation of students within a given teacher into account. I.e., say a teacher happened to be effective. Then this teacher being assigned to a small class would have a positive impact on a bunch of treated students due to the single teacher. The student outcomes are correlated with each other by the single teacher. We would need to take this into account when calculating uncertainty: the students are not independent.

```
mod <- lm(tmathssk ~ small_class + sbirthq + hdegk + totexpk,
          data = dat)
res_OLS <- tidy( mod )
knitr::kable( res_OLS, digits = 2 )
```

term	estimate	std.error	statistic	p.value
(Intercept)	486.49	2.33	208.38	0.00
small_classyes	6.39	1.69	3.79	0.00
sbirthq2nd qtr - april,may,june	-5.64	2.33	-2.42	0.02
sbirthq3rd qtr - july,aug,sept	-13.85	2.27	-6.11	0.00
sbirthq4th qtr - oct,nov,dec	10.19	2.49	4.08	0.00
hdegkmaster +	-2.71	5.44	-0.50	0.62
hdegkmasters	-5.25	1.82	-2.89	0.00
hdegkspecialist	16.98	7.62	2.23	0.03
totexpk	0.58	0.15	3.88	0.00

To be clear, the above regression gives a reasonable *estimate* of the *impact* of small class size, but the standard errors, and therefore p-values, etc., are wrong.

48.3 Cluster robust standard errors

One statistical way of handling this is with the `lm_robust` package that uses the `sandwich` package that does cluster robust standard errors:

```
library( estimatr )
mod_CRVE <- lm_robust(tmathssk ~ small_class + sbirthq + hdegk + totexpk,
                      clusters = dat$clid,
                      data = dat) ## another way to get the same result (more or less)
res_CRVE <- tidy(mod_CRVE) %>%
  dplyr::select( term, estimate, std.error, p.value )
knitr::kable( res_CRVE, digits = 2 )
```

term	estimate	std.error	p.value
(Intercept)	486.49	4.26	0.00
small_classyes	6.39	3.86	0.10
sbirthq2nd qtr - april,may,june	-5.64	2.27	0.01
sbirthq3rd qtr - july,aug,sept	-13.85	2.35	0.00
sbirthq4th qtr - oct,nov,dec	10.19	2.48	0.00
hdegkmaster +	-2.71	11.50	0.83
hdegkmasters	-5.25	4.15	0.21
hdegkspecialist	16.98	15.10	0.37
totexpk	0.58	0.35	0.11

48.4 Cluster Bootstrap

Another way is to use the cluster bootstrap. It is a versatile method for getting standard errors on data that is clustered, as it keeps clusters intact.

We write an analysis function as follows:

```
my_analysis <- function( the_dat ) {
  mod <- lm(tmathssk ~ small_class + sbirthq + hdegk + totexpk,
            data = the_dat)
  broom::tidy(mod)
}
```

```
my_analysis( dat )
```

```
# A tibble: 9 x 5
```

term	estimate	std.error	statistic	p.value
------	----------	-----------	-----------	---------

	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	"(Intercept)"	486.	2.33	208.	0
2	"small_classyes"	6.39	1.69	3.79	0.000156
3	"sbirthq2nd qtr - april,may,june"	-5.64	2.33	-2.42	0.0155
4	"sbirthq3rd qtr - july,aug,sept"	-13.9	2.27	-6.11	0.00000000110
5	"sbirthq4th qtr - oct,nov,dec"	10.2	2.49	4.08	0.0000452
6	"hdegkmaster + "	-2.71	5.44	-0.497	0.619
7	"hdegkmasters"	-5.25	1.82	-2.89	0.00393
8	"hdegkspecialist"	17.0	7.62	2.23	0.0259
9	"totexpk"	0.579	0.149	3.88	0.000108

We then nest our data so each row is an entire cluster:

```
dat_nst <- dat %>%
  group_by( clid ) %>%
  nest() %>%
  ungroup()
dat_nst
```

```
# A tibble: 196 x 2
  clid data
  <int> <list>
1     1 <tibble [20 x 11]>
2     2 <tibble [22 x 11]>
3     3 <tibble [21 x 11]>
4     5 <tibble [23 x 11]>
5     6 <tibble [21 x 11]>
6     7 <tibble [18 x 11]>
7     9 <tibble [23 x 11]>
8    11 <tibble [19 x 11]>
9    13 <tibble [17 x 11]>
10   14 <tibble [17 x 11]>
# i 186 more rows
```

We can then bootstrap our data as so:

```
one_cluster_boot <- function( ) {

  dat_nst_star = slice_sample( dat_nst, n = nrow(dat_nst), replace=TRUE )
  dat_nst_star$clid = 1:nrow(dat_nst_star)

  dat_star <- unnest( dat_nst_star, cols="data" )
```

```

  my_analysis( dat_star )
}

```

Note we are regenerating the cluster ID so if we have the same cluster multiple times, each time gets a new ID.

We bootstrap and analyze a bunch of times and get standard errors:

```
boots = map_df( 1:1000, ~ one_cluster_boot(), .id = "boot" )
```

```

res_boot <- boots %>% group_by( term ) %>%
  summarise( SE = sd( estimate ) )
res_boot

```

```

# A tibble: 9 x 2
  term                                SE
  <chr>                             <dbl>
1 "(Intercept)"                     4.38
2 "hdegkmaster + "                   12.7
3 "hdegkmasters"                     4.08
4 "hdegkspecialist"                 14.0
5 "sbirthq2nd qtr - april,may,june"  2.34
6 "sbirthq3rd qtr - july,aug,sept"   2.34
7 "sbirthq4th qtr - oct,nov,dec"     2.37
8 "small_classyes"                   3.77
9 "totexpk"                          0.363

```

We can compare to the original (WRONG) OLS estimates, the CRVE standard errors, and the bootstrap standard errors:

```

CRVE_sub <- res_CRVE %>%
  dplyr::select( -estimate, -p.value ) %>%
  rename( SE_CRVE = std.error )
OLS_sub <- res_OLS %>%
  dplyr::select( -estimate, -p.value ) %>%
  rename( SE_OLS = std.error )

```

```
# Make the table
```

```

tbl <- left_join( res_boot, OLS_sub,
                  by = "term" ) %>%
  left_join( CRVE_sub, by = "term" ) %>%
  relocate( term, statistic ) %>%

```

```
mutate( boot_v_OLS = SE / SE_OLS,
        boot_v_CRVE = SE / SE_CRVE )

knitr::kable( tbl, digits=2 )
```

term	statistic	SE	SE_OLS	SE_CRVE	boot_v_OLS	boot_v_CRVE
(Intercept)	208.38	4.38	2.33	4.26	1.88	1.03
hdegkmaster +	-0.50	12.75	5.44	11.50	2.34	1.11
hdegkmasters	-2.89	4.08	1.82	4.15	2.24	0.98
hdegkspecialist	2.23	13.96	7.62	15.10	1.83	0.92
sbirthq2nd qtr - april,may,june	-2.42	2.34	2.33	2.27	1.00	1.03
sbirthq3rd qtr - july,aug,sept	-6.11	2.34	2.27	2.35	1.03	1.00
sbirthq4th qtr - oct,nov,dec	4.08	2.37	2.49	2.48	0.95	0.95
small_classyes	3.79	3.77	1.69	3.86	2.23	0.98
totexpk	3.88	0.36	0.15	0.35	2.43	1.03

A ratio of 1 means the estimated SEs are about the same. More than 1 means the bootstrap is returning larger SEs.

Generally, we see that the bootstrap is increasing the SEs for the level-2 coefficients (those that are talking about how clusters are different). This is good: the OLS SEs are way too small since they are not taking clustering into account.

The CRVE is basically the same as the bootstrap here, with some mild differences. All these methods are for estimating standard errors; they do not change the estimated coefficients themselves.

Bootstrapping is a simple way of getting uncertainty when you don't know how to do that with math or a package. When we can do a mathematical approximation, the bootstrapping might not be worth it, due to the extra computation. But bootstrapping, by direct simulation, can also account for things like heteroskedasticity, outliers, or other weirdness that the mathematical approximations cannot. It is worth using in many cases due to this general applicability, versatility, and robustness.

Part V

On Machine Learning

49 A demonstration of different machine learning methods all fit to the same data

50 Overview

This document showcases a bunch of different machine learning tools, all used on the same data set. At the end we compare the different rmse on the validation set.

For context, we will use a classic data set (e.g., Almond et al., 2005) on child birthweight. This data set was originally constructed to estimate the causal effect of maternal smoking on child birthweight; that is not what we are up to now. Our goal is to instead *predict* child birthweight directly based on observable characteristics prior to birth.

The target of interest is the child's birthweight, stored as `child_birthwt`. This is a continuous outcome. All other variables are fair game as predictors.

While this is obviously a simplified data set, the prediction problem is very real – many medical insurers and providers target services based on algorithms similar to what you will put together.

Note: For the purposes of illustration we are going to use a proportionally small training set of 10,000 observations (so we need to regularize and use fancy stuff) and a very large validation set (so we see the real comparison of our different choices). In practice we would try to use as much of our data as possible for training.

51 Setup

Load our libraries, set our random seed for reproducibility:

```
library( MASS )
library( rattle )
library( glmnet )
library( tidyverse )
library( modelr )
library( broom )
library( caret )
library( ranger )
set.seed(8675309)

knitr::opts_chunk$set(echo = TRUE,
                      fig.width = 5,
                      out.width = "5in",
                      fig.align = "center")
options(list(dplyr.summarise.inform = FALSE))
theme_set( theme_classic() )
```

51.0.1 Load in the birthweight data.

We have two files: training and holdout. We keep our holdout set for our final validation after we pick our final models. This allows a fair comparison of all the methods at the end as we do not use the holdout set AT ALL for building our models.

```
# Load the data
ca_training = read.csv( "data/lbw_training.csv" )
ca_holdout = read.csv( "data/lbw_holdout.csv" )
```

Let's peek at our list of variables:

```
names( ca_training )
```

[1]	"child_birthwt"	"use_tobacco"
[3]	"born_in_hospital"	"MD_at_birth"
[5]	"any_drink_on_avg"	"mother_foreign_born"
[7]	"mother_age"	"father_age"
[9]	"num_prenatal_visits"	"mother_hispanic_other"
[11]	"mother_black"	"father_hispanic_other"
[13]	"father_black"	"mother_educ_less_than_hs"
[15]	"mother_educ_hs"	"father_educ_less_than_hs"
[17]	"father_educ_hs"	"adequate_care"
[19]	"birth_month"	"birth_weekday"
[21]	"child_female"	"mother_anemic"
[23]	"mother_cardiac_disease"	"mother_lung"
[25]	"mother_herpes"	"mother_diabetes"

52 Baseline linear model (OLS)

How do we do with simple linear regression models? We try two versions, one with just the main effects and one with all the pairwise interactions.

```
model_linear <- lm(child_birthwt ~ ., data = ca_training)
```

```
model_linear_int <- lm(child_birthwt ~ .^2, data = ca_training)
```

```
rmse( model_linear, ca_holdout )
```

```
[1] 559.436
```

```
rmse( model_linear_int, ca_holdout )
```

```
[1] 591.2767
```

Note the worse performance of the linear regression with interactions. We are likely overfitting.

53 Forward stepwise selection

Here we search for a good linear model by iteratively adding the “best” covariate until we are not improving on our model performance measure.

```
library(MASS)

# set up simplest and most complex to consider:
# `~ 1` is the "intercept only" model. Our max model
# has everything (but no interactions).
mod_simple <- lm(child_birthwt ~ 1, data = ca_training)
mod_max <- lm(child_birthwt ~ ., data = ca_training)

# use forward stepwise selection to pick an optimal model (in terms of AIC)
# `trace=0` makes it not print out a lot of stuff to the screen about what it
# is doing.
mod_forward <- stepAIC(mod_simple,
                      scope = list(lower = formula(mod_simple),
                                   upper = formula(mod_max)),
                      direction = "forward",
                      trace = 0 )

summary(mod_forward)
```

Call:

```
lm(formula = child_birthwt ~ use_tobacco + mother_black + num_prenatal_visits +
    child_female + adequate_care + mother_age + born_in_hospital +
    any_drink_on_avg + mother_foreign_born + mother_educ_less_than_hs +
    mother_anemic + MD_at_birth + father_hispanic_other + father_black,
    data = ca_training)
```

Residuals:

Min	1Q	Median	3Q	Max
-3038.05	-304.73	26.42	352.09	2399.93

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3254.332	60.917	53.423	< 2e-16 ***
use_tobacco	-225.548	15.441	-14.607	< 2e-16 ***
mother_black	-148.518	40.371	-3.679	0.000236 ***
num_prenatal_visits	27.124	2.088	12.989	< 2e-16 ***
child_female	-127.642	11.152	-11.445	< 2e-16 ***
adequate_careInadequate	208.301	33.101	6.293	3.25e-10 ***
adequate_careIntermediate	78.113	16.530	4.726	2.32e-06 ***
mother_age	4.934	1.065	4.634	3.63e-06 ***
born_in_hospital	-187.136	45.238	-4.137	3.55e-05 ***
any_drink_on_avg	-242.372	59.992	-4.040	5.38e-05 ***
mother_foreign_born	-63.806	27.741	-2.300	0.021466 *
mother_educ_less_than_hs	-54.475	18.572	-2.933	0.003363 **
mother_anemic	-132.998	56.122	-2.370	0.017817 *
MD_at_birth	-28.245	16.289	-1.734	0.082948 .
father_hispanic_other	-80.786	48.069	-1.681	0.092870 .
father_black	-62.116	38.409	-1.617	0.105865

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 556.9 on 9984 degrees of freedom

Multiple R-squared: 0.08228, Adjusted R-squared: 0.0809

F-statistic: 59.68 on 15 and 9984 DF, p-value: < 2.2e-16

`coef( mod_forward )`

(Intercept)	use_tobacco	mother_black
3254.331679	-225.548150	-148.518044
num_prenatal_visits	child_female	adequate_careInadequate
27.124442	-127.642279	208.301177
adequate_careIntermediate	mother_age	born_in_hospital
78.113446	4.934118	-187.136158
any_drink_on_avg	mother_foreign_born	mother_educ_less_than_hs
-242.371983	-63.806385	-54.475462
mother_anemic	MD_at_birth	father_hispanic_other
-132.998442	-28.244929	-80.785733
father_black		
-62.115737		

We can examine how many coefficients we zeroed out with this approach:

```
# Total number of coefficients (remember to subtract off intercept)  
length( coef( model_linear ) ) - 1
```

```
[1] 41
```

```
length(coef(mod_forward)) - 1
```

```
[1] 15
```

We have dropped several covariates.

Finally, how well do we do in terms of predictive performance?

```
rmse(mod_forward, ca_holdout)
```

```
[1] 558.9989
```

54 Ridge Regression

How do we do with ridge? To use `glmnet` we need to make our data into a matrix with no categorical covariates (i.e., we need to convert those to dummy variables). This “design matrix” or “model matrix” is used by those ML packages (in particular `glmnet`) that do not like formulae. We make this as follows:

```
x <- model.matrix(child_birthwt ~ ., ca_training)[,-1]
y <- as.numeric(ca_training$child_birthwt)

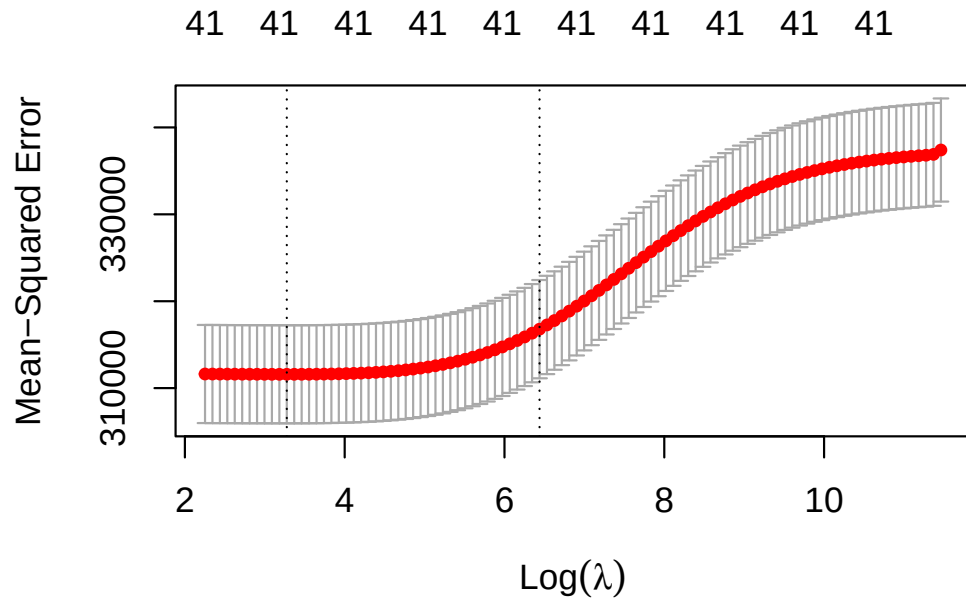
# how many covariates do we have now that we fleshed out our categorical
# ones?
dim( x )

[1] 10000    41
```

We are going to cross validate to pick the best tuning parameter.

```
model_ridge <- cv.glmnet(x = x, y = y, alpha = 0)

# Plot to see how our CV estimate of performane changes as our tuning parameter changes.
plot(model_ridge)
```

Now let's look at the model corresponding to the "1 SE" rule.

```
# Calculate coefficients at optimal lambda
predict(model_ridge, "coefficients", newx = x, s = "lambda.1se")[,1]
```

(Intercept)	use_tobacco	born_in_hospital
3369.117138	-109.333424	-110.752801
MD_at_birth	any_drink_on_avg	mother_foreign_born
-18.307264	-150.301619	-25.061873
mother_age	father_age	num_prenatal_visits
2.387578	1.087979	9.380462
mother_hispanic_other	mother_black	father_hispanic_other
-9.908412	-73.839709	-42.635734
father_black	mother_educ_less_than_hs	mother_educ_hs
-63.432580	-35.164414	3.169975
father_educ_less_than_hs	father_educ_hs	adequate_careInadequate
-17.728130	-11.615696	9.880843
adequate_careIntermediate	birth_monthM10	birth_monthM11
-4.005582	-12.728006	12.571847
birth_monthM12	birth_monthM2	birth_monthM3
-15.496326	15.208849	-10.498643
birth_monthM4	birth_monthM5	birth_monthM6
3.210140	13.952109	2.536463
birth_monthM7	birth_monthM8	birth_monthM9
7.638260	-3.250663	-11.412587

birth_weekdayD2	birth_weekdayD3	birth_weekdayD4
-2.340281	5.611307	17.808560
birth_weekdayD5	birth_weekdayD6	birth_weekdayD7
2.112503	8.272195	-8.147729
child_female	mother_anemic	mother_cardiac_disease
-62.913159	-72.776220	4.197639
mother_lung	mother_herpes	mother_diabetes
12.464810	14.774745	31.563450

```
# Re-fit model at optimal
model_ridge_optimal <- glmnet(x = x, y = y, alpha = 0, lambda = model_ridge$lambda.1se)
```

RMSE on holdout:

We first make our holdout data into a matrix just like the training data:

```
# Do the same for holdout.
x_holdout <- model.matrix(child_birthwt ~ ., ca_holdout)[-1]
y_holdout <- as.numeric(ca_holdout$child_birthwt)
```

Unfortunately, `glmnet` doesn't work with the `rmse()` method, so we have to write our own `rmse`. Sad!

```
# Function to calculate RMSE given our design matrix rather
# than original data.
```

```
rmse_by_hand <- function(this_model, x_holdout, y_holdout ){
  this_pred <- predict(this_model, newx = x_holdout)

  sqrt( mean( (this_pred - y_holdout)^2 ) )
}
```

```
# Calculate RMSE
```

```
rmse_by_hand(model_ridge_optimal, x_holdout, y_holdout)
```

```
[1] 565.0963
```

We can also use the helper function in `caret` that we saw in the random forest case study that takes observed and actual outcomes:

```
preds = as.numeric( predict( model_ridge_optimal, x_holdout ) )
rs = data.frame( obs = y_holdout,
                 pred = preds )
caret::defaultSummary( rs )
```

RMSE	Rsquared	MAE
565.09626065	0.07662296	420.94054253

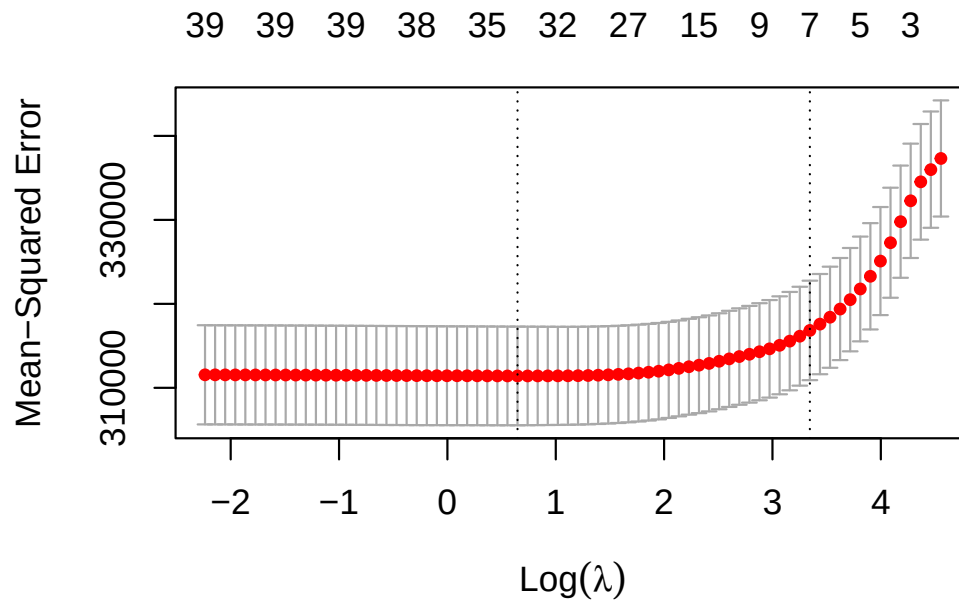
Note: The `predict()` method for `glmnet` gives back a matrix, not a vector of numbers. Annoying. We have to use `as.numeric` to get it to change to a list of numbers.

55 Lasso Regression

How do we do with Lasso? It is the same as ridge, except we need to set `alpha=1`.

```
model_lasso <- cv.glmnet(x = x, y = y, alpha = 1)
```

```
# Plot output  
plot(model_lasso)
```



```
# Calculate coefficients at optimal lambda. We only look at the non-zero ones.
```

```
coef <- predict(model_lasso, "coefficients", newx = x, s = "lambda.1se")[,1]
```

```
coef[ coef != 0 ]
```

(Intercept)	use_tobacco	born_in_hospital	mother_age
3288.089967	-167.324623	-13.895767	1.568242
num_prenatal_visits	mother_black	father_black	child_female
12.063002	-106.815386	-26.432899	-74.649746

```
# Re-fit model at optimal
model_lasso_optimal <- glmnet(x = x, y = y, alpha = 1, lambda = model_lasso$lambda.1se)

# Calculate RMSE for Lasso
rmse_by_hand(model_lasso_optimal, x_holdout, y_holdout)

[1] 564.7121
```

56 Single Tree (a CART)

We first do a single tree and prune it, and then we will do random forests later on.

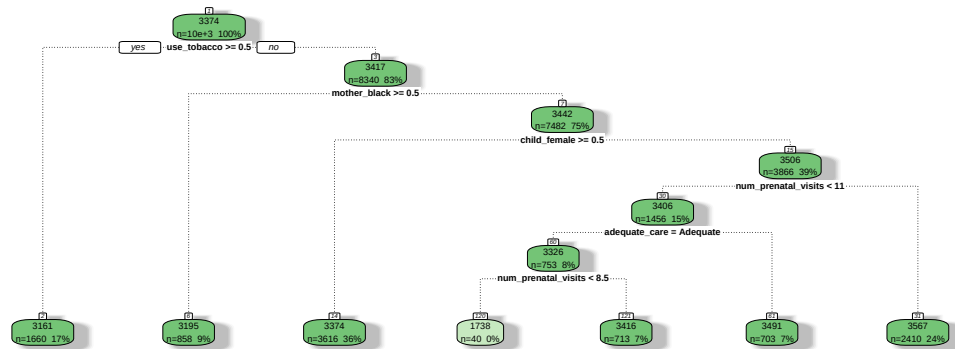
To fit a single tree we use `rpart` (Recursive Partition).

```
library(rpart)
```

```
model_tree <- rpart(child_birthwt ~ ., data = ca_training )
```

We can then plot it:

```
library( rattle )  
fancyRpartPlot(model_tree)
```



Rattle 2025-Mar-21 16:01:08 Imiratrix

Our RMSE:

```
# How well do we do on the test set?  
rmse(model_tree, ca_holdout)
```

```
[1] 558.7839
```

Let's use cross validation to pick our `cp` tuning parameter (which controls the depth and complexity of our tree):

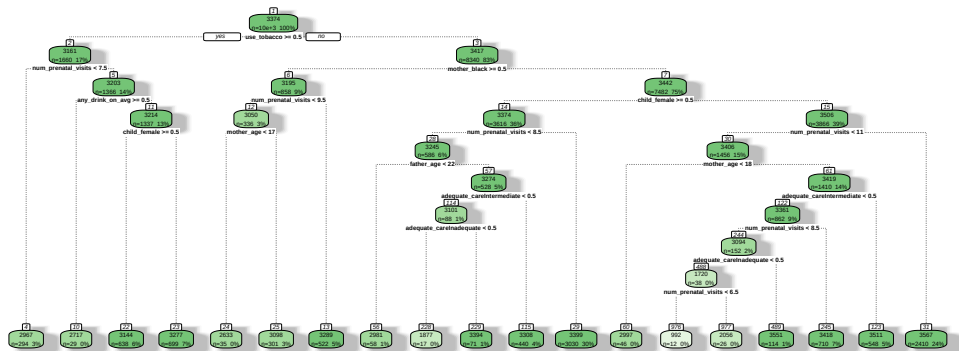
```
train_control <-
  trainControl(method = "cv",
              number = 10)

tune_grid <- data.frame(
  cp = exp( seq( log( 0.001 ), log( 0.5 ), length.out=100 ) ) )

rpart_cv <-
  train(child_birthwt ~ ., data = ca_training,
        method = "rpart",
        na.action = "na.omit",
        tuneGrid = tune_grid,
        trControl = train_control)
```

Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
: There were missing values in resampled performance measures.

```
# Plot the final model (fit using the best chosen tuning parameter automatically)
fancyRpartPlot(rpart_cv$finalModel)
```



Rattle 2024-Dec-26 17:42:49 Imiratrix

Now let's look at the RMSE of our final model. We use `predict` on the entire `rpart_cv` object since it will automatically give a tree refit to the entire training data using the best selected tuning parameter:

```
rmse(rpart_cv, ca_holdout)
```

```
[1] 548.8857
```

To get the predictions themselves we would use `predict`, handing over `rpart_cv` *not* `rpart_cv$finalModel`:

```
preds = predict( rpart_cv, ca_holdout )  
summary( preds )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
992	3289	3399	3373	3511	3567

57 Random Forests

We fit a random forest with the `ranger` package as so:

```
# fit a random forest model
model_rf <- ranger( child_birthwt ~ ., data = ca_training,
                    importance = "permutation" )
```

```
model_rf
```

Ranger result

Call:

```
ranger(child_birthwt ~ ., data = ca_training, importance = "permutation")
```

Type:	Regression
Number of trees:	500
Sample size:	10000
Number of independent variables:	25
Mtry:	5
Target node size:	5
Variable importance mode:	permutation
Splitrule:	variance
OOB prediction error (MSE):	302410.2
R squared (OOB):	0.1037941

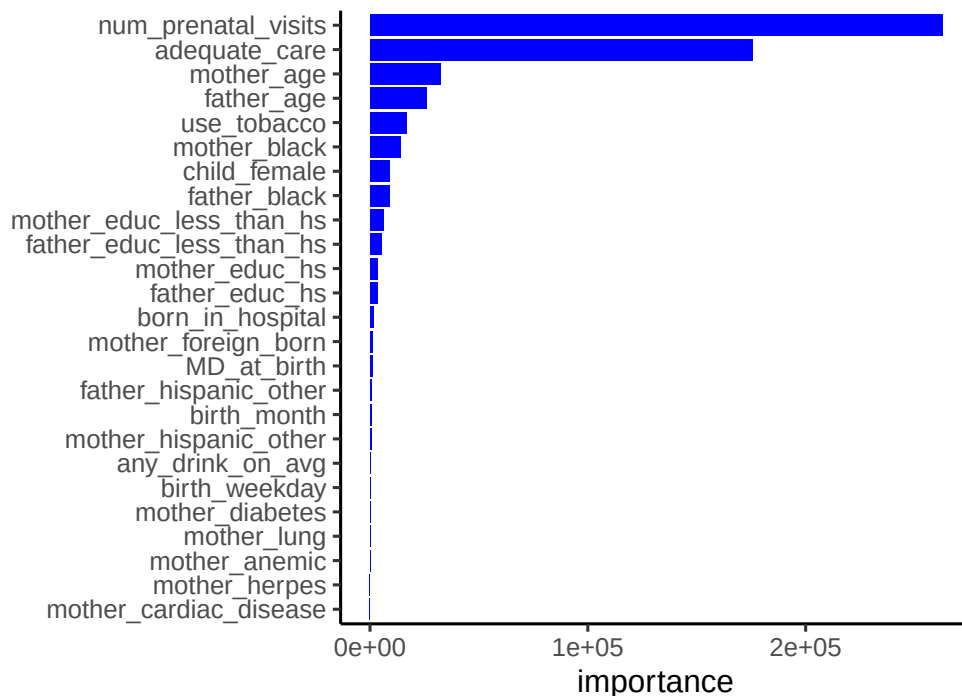
```
# Table of variable importance
importance(model_rf)
```

use_tobacco	born_in_hospital	MD_at_birth
16588.89005	1693.18649	1116.55033
any_drink_on_avg	mother_foreign_born	mother_age
448.65352	1328.86719	32323.92626
father_age	num_prenatal_visits	mother_hispanic_other
25935.76551	262591.71389	573.88371
mother_black	father_hispanic_other	father_black
13993.30014	792.15976	8945.04138

mother_educ_less_than_hs	6254.52673	mother_educ_hs	3419.14652	father_educ_less_than_hs	5492.35051
father_educ_hs	3398.24303	adequate_care	175591.38932	birth_month	614.54015
birth_weekday	380.24615	child_female	8977.13883	mother_anemic	105.88669
mother_cardiac_disease	-325.34716	mother_lung	144.81925	mother_herpes	-46.14257
mother_diabetes	313.14919				

Annoyingly, with the `ranger` package we have to make a variable importance plot by hand. (We could instead use the `randomForest` package, which has some default importance plot code. See prior scripts illustrating this if desired.) The following code makes our plot:

```
imps = importance(model_rf)
imps = tibble( var = names(imps),
               importance = imps )
imps = imps %>% arrange( importance )
ggplot( imps, aes(x = reorder( var, importance ), y = importance)) +
  geom_col(fill = "blue") +
  coord_flip() +
  labs( x = "" )
```



Note: The `reorder()` line in `ggplot` sets the order of our x variable so we get our variables from least to most important. The `coord_flip` makes our x-axis our y-axis and vice-versa, so we get nice horizontal bars.

The `ranger` package gets fancy with its `predict` which makes it a bit harder to calculate out of sample RMSE. We do it like so:

```
rf_preds = predict( model_rf, data=ca_holdout )
rf_preds
```

Ranger prediction

```
Type:                Regression
Sample size:         118805
Number of independent variables: 25
```

```
head( predictions( rf_preds ) )
```

```
[1] 3481.162 3594.208 3517.143 3432.828 3335.656 3394.036
```

```
RMSE( predictions( rf_preds ), ca_holdout$child_birthwt )
```

```
[1] 546.1326
```

Note how we get a package of predictions from `predict` and then need to get the actual predictions out of the package with `predictions()`. Also note the capital RMSE method in `caret` is different from the `rmse` from the `modelr` package.

Different packages are all slightly different. Keep reference code like this to easily remember how to do basic coding tasks.

57.1 Tuning the random forest

We should tune our random forest, to figure out which specification is best. We use `caret`'s `train()` to do this (borrowing code from the case-study walk-through from the Data Science in Education textbook):

```
# setting a seed for reproducibility
set.seed(2020)
```

```
# Create a grid of different values of mtry, splitrule, and min.node.size,
# the three tuning parameters for our random forest.
```

```
tune_grid <-
  expand.grid(
    mtry = c(2, 3, 7, 10),
    splitrule = c("variance"),
    min.node.size = c(10, 20, 50)
  )

# Fit a new model, using the tuning grid we created above. This will take
# awhile to run.
rf_tuned <-
  train(child_birthwt ~ ., data = ca_training,
        method = "ranger",
        tuneGrid = tune_grid)
```

```
rf_tuned
```

Random Forest

10000 samples
25 predictor

No pre-processing

Resampling: Bootstrapped (25 reps)

Summary of sample sizes: 10000, 10000, 10000, 10000, 10000, 10000, ...

Resampling results across tuning parameters:

mtry	min.node.size	RMSE	Rsquared	MAE
2	10	560.5576	0.08226263	421.6399
2	20	560.7701	0.08165209	421.7262
2	50	561.0561	0.08155258	421.8141
3	10	557.0580	0.08455732	420.1078
3	20	557.0095	0.08532631	419.9823
3	50	557.4392	0.08517044	419.9752
7	10	557.9592	0.08025062	423.9294
7	20	556.6819	0.08320896	422.4458
7	50	555.3160	0.08691731	420.4274
10	10	559.3643	0.07946736	426.1047
10	20	557.4722	0.08297020	424.0431
10	50	555.1987	0.08782704	421.2257

Tuning parameter 'splitrule' was held constant at a value of variance
RMSE was used to select the optimal model using the smallest value.

The final values used for the model were `mtry = 10`, `splitrule = variance` and `min.node.size = 50`.

We then take our final random forest, fit to all of our training data and using the winning tuning parameters, by simply using `predict` from the result of our `train()` call:

```
rmse( rf_tuned, ca_holdout )
```

```
[1] 547.0557
```

58 Comparing our machines

Now let's compare the RMSE for everyone! We add in a null model of predicting the grand mean to give a reference of how well we do when we do nothing. We can even compare all the RMSEs to the one of the mean as a point of reference:

```
model_null = lm( child_birthwt ~ 1, data=ca_training )
rmse_mean = rmse( model_null, ca_holdout )

RMSEs = c( mean = rmse_mean,
           OLS = rmse(model_linear, ca_holdout),
           OLS.quad = rmse( model_linear_int, ca_holdout ),
           forward = rmse(mod_forward, ca_holdout),
           ridge = rmse_by_hand(model_ridge_optimal, x_holdout, y_holdout),
           lasso = rmse_by_hand(model_lasso_optimal, x_holdout, y_holdout),
           CART = rmse(model_tree, ca_holdout),
           CART.prune = rmse(rpart_cv, ca_holdout),
           RF = RMSE( predictions( rf_preds ), ca_holdout$child_birthwt ),
           RF_tuned = rmse( rf_tuned, ca_holdout ) )
```

```
Warning in predict.lm(model, data): prediction from rank-deficient fit; attr(*,
"non-estim") has doubtful cases
```

```
results = tibble( model = names( RMSEs ),
                  RMSE = RMSEs,
                  perDecr = 100 * RMSE / rmse_mean )
knitr::kable( results, digits=1 )
```

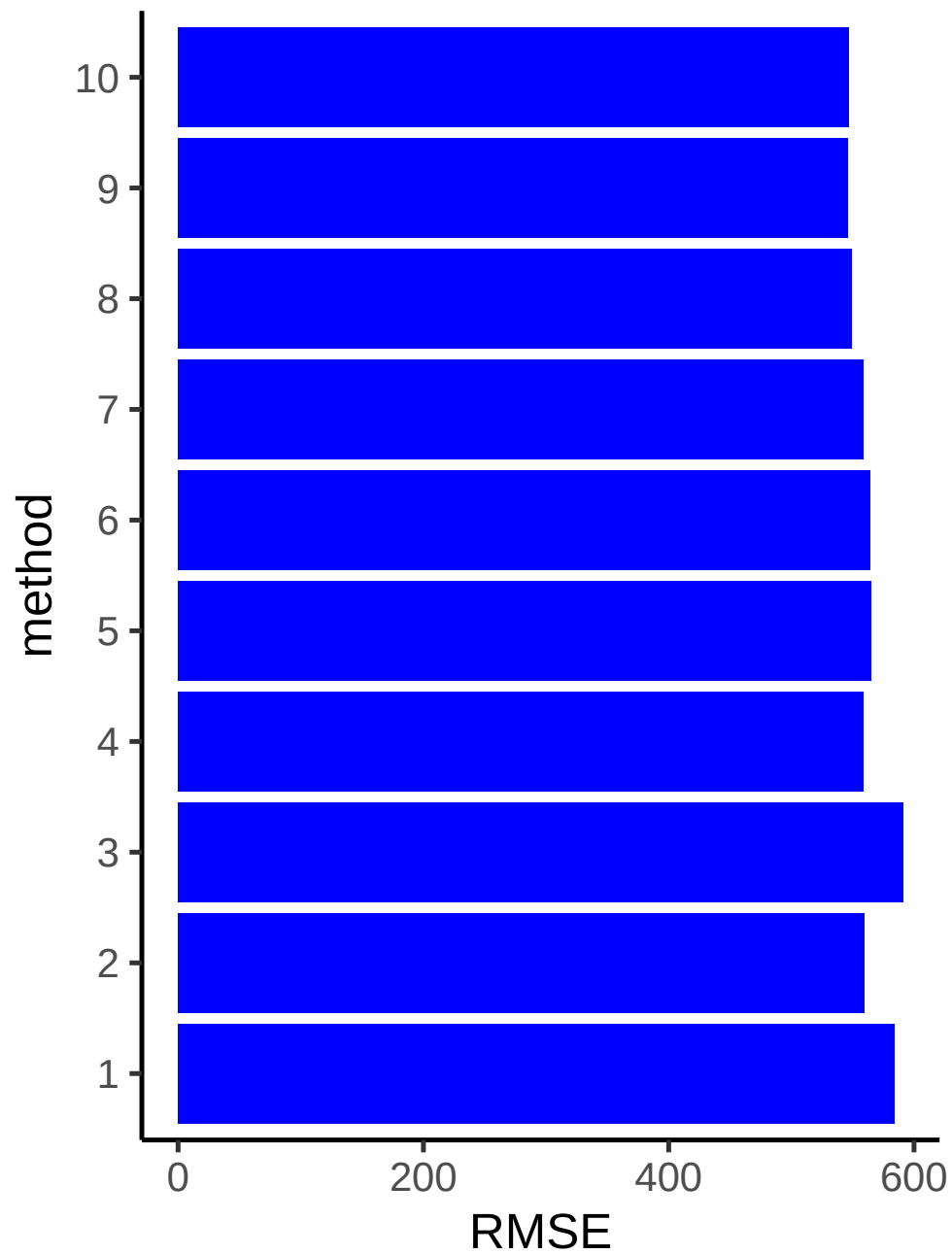
model	RMSE	perDecr
mean	584.1	100.0
OLS	559.4	95.8
OLS.quad	591.3	101.2
forward	559.0	95.7
ridge	565.1	96.8
lasso	564.7	96.7
CART	558.8	95.7

model	RMSE	perDecr
CART.prune	548.9	94.0
RF	546.1	93.5
RF_tuned	547.1	93.7

None of them are spectacular, but of them all, random forest wins, with a reduction of 6.5% in the RMSE over doing nothing. Interestingly, the tuned random forest was a tiny bit worse than the out-of-box random forest. The difference is small, so this is likely just due to instability in the tuning.

A plot of the comparisons follows. Note how the bars are similar heights, indicating that nothing is a slam dunk over, even, just predicting the mean. The interacted OLS model is so overfit it does *worse* than the mean!

```
results = rownames_to_column(results, var="method" )
results$method = reorder( results$method, 1:nrow(results) )
ggplot( results, aes(x = method, y = RMSE)) +
  geom_col(fill = "blue") +
  coord_flip()
```



A different way of thinking about the RMSE is that it is a measure of baseline variation:

```
sd( ca_holdout$child_birthwt )
```

```
[1] 584.053
```


This motivates why it is a reasonable baseline RMSE to compare the other RMSEs to.

59 Reflections on using machine learning (for final projects)

60 Test/train splits, test/train/validate, and cross validation

It is important to track what your machine learning methods are doing in terms of splitting your data inside the call. For example, the `caret` package's `train()` method will often be doing cross validation (or something similar) inside of it. You thus do not need to always do a test/train split outside of it!

Also, don't do a test/train split if you don't have a lot of data. It is too expensive! This is where cross validation is particularly useful, since it lets all of your data be used for testing, and it uses most of your data for fitting your models. It is kind of like a "have your cake and eat it too" situation.

Finally, the full test/train/validate trifecta is only if you are really, really concerned about predicting your future performance of a model. It turns out that if you use cross-validation to select the best set of tuning parameters, the final estimated accuracy from the cross-validation will generally be pretty close to what a final validation set would tell you.

Let's see how the case study in [Data Science in Ed, Chapter 14](#) went about things. They first split the data into a training set and a testing set. They then made a grid of tuning parameters related to tree size and so forth, and then used `train()` to identify what set of tuning parameters was best for the data. In particular, `train()` used this grid and a variant of cross validation (bootstrap resampling) to repeatedly divide the training data into a part used to fit one random forest for each tuning parameter combo and a part used to evaluate all of those random forests on out of sample data. The final table produced gives the estimated error rates, and finally `train()` selects the tuning parameter combo with the best rates.

The book then took the final model (via `rf_fit2$finalModel`), which is a last model fit by `train()` to all the training data using the best found parameter combo, as their answer. To evaluate the quality of their answer, they used `finalModel` to predict outcomes for the test data they set aside at the beginning of their case study.

In our language, their "test" data was used for the final validation step, and the `train()` method was doing little test/train splits inside of it to get the estimates of out of sample error as part of the tuning process.

Note that they also found the predicted error of the finally selected final model was about the same as reported by the out of sample error by `train()`: this is not unusual. If we are not

fitting too many different combos of tuning parameter and so forth, then the estimated error from this process will often be close.

60.1 Cross validation options for caret

The `caret` package does cross validation internally as part of `train()`, if you tell it. If you don't, it does a different kind of internal repeated test/train splitting that it calls bootstrap. This is not bootstrap for inference! What it is doing is resampling the data with replacement to get a training dataset of the same size as it was passed. It will then have about a third of the data not in the training set, and it will use this for out-of-sample testing to estimate the performance of all the models fit. Finally, it repeats this a bunch of times and averages: this means each observation will be in most training sets, but will be used for testing some of the time. This is just like cross validation (except more random)!

Also, `caret`'s cross-validation is, by default, a random cross-validation: repeatedly take 10% of the data as testing, and use the rest for training. Repeat 10 times. This means each observation will be used about once for testing and about nine times for training, but not exactly. Functionally, this will generally be nearly the same as the classic CV where we divide all the data into 10 parts systematically.

It basically does not matter which form of cross-validation or splitting you use. The number of iterations will impact running time: 25 iterations will be 2.5 times longer to run than 10!

60.2 Bootstrap vs. Cross validation

Don't confuse bootstrapping with cross-validation. Bootstrapping is a way of doing statistical inference: you use it to ask how much an estimate would change if you happened to get a different data set from the same source. This allows you to decide if, for example, a coefficient is "really" positive—if your bootstrapping doesn't really give you any negative estimates, then you can be sure that your estimate is probably not positive due to random chance.

Cross validation, by contrast, is a way of doing a lot of test/train splits because you want to know how a fit model would work on new data. It is focusing on estimating future predictive accuracy, not statistical inference.

61 Determining what variables are important

The Lasso is a sparse regression approach: as part of fitting a lasso model, you are given a subset of your original variables deemed important enough to include in the predictive model. Random forests, by contrast, allow for a second step of generating a variable importance plot, where you get a measure of how useful each variable was for making predictions.

Both these tools can be useful for identifying factors particularly tied to your outcome. In general, you would expect those variables kept by Lasso to also have high scores in a variable importance plot. Both approaches, however, have some caveats that one should think about when interpreting results.

First, the Lasso approach gives you a final complete model built out of only a few variables. Each kept variable comes with a coefficient, and some of those coefficients will be tiny and others larger. Before directly comparing coefficients, however, spend a moment to think about how the scale of each variable matters.

For example, in the following fake dataset each variable (other than the last) has the same impact on the outcome. Put another way, they are all about the same in terms of their correlation with the outcome, as shown by the last column of the output.

```
library( glmnet )

tb = data.frame( X1 = rnorm( 1000 ),
                 X2 = rnorm( 1000, sd = 10 ),
                 X3 = rnorm( 1000, sd = 100 ),
                 X4 = rnorm( 1000 ),
                 X5 = rnorm( 1000 ) )
tb$X4_proxy = tb$X4 + rnorm( 1000, sd=0.2 )
tb$Y = with(tb, X1 + 0.1 * X2 + 0.01 * X3 + X4 + rnorm( 1000 ) )

cor( tb ) %>%
  knitr::kable( digits=2 )
```

	X1	X2	X3	X4	X5	X4_proxy	Y
X1	1.00	-0.06	-0.03	-0.01	-0.06	-0.01	0.42
X2	-0.06	1.00	0.03	-0.04	0.03	-0.03	0.42
X3	-0.03	0.03	1.00	-0.04	-0.03	-0.03	0.45

	X1	X2	X3	X4	X5	X4_proxy	Y
X4	-0.01	-0.04	-0.04	1.00	0.01	0.98	0.42
X5	-0.06	0.03	-0.03	0.01	1.00	0.01	-0.03
X4_proxy	-0.01	-0.03	-0.03	0.98	0.01	1.00	0.42
Y	0.42	0.42	0.45	0.42	-0.03	0.42	1.00

We can fit a Lasso model as so:

```
Xmat = model.matrix( Y ~ ., data=tb )

mod <- cv.glmnet( x = Xmat[,-1], y = tb$Y )

coef( mod )
```

```
7 x 1 sparse Matrix of class "dgCMatrix"
      s1
(Intercept) 0.06167455
X1          0.86886193
X2          0.09107718
X3          0.00885584
X4          0.87535837
X5          .
X4_proxy    0.01155421
```

Note how we have the proxy, a little bit, and also X3. The proxy looks more important than X3 (but actually isn't important). X3 looks unimportant because the scale of X3 is so large. One fix is to standardize your covariates before putting them into your lasso:

```
tb2 <- tb %>%
  mutate( X2 = scale( X2 ),
          X3 = scale( X3 ) )
Xmat2 = model.matrix( Y ~ ., data=tb )
mod2 <- cv.glmnet( x = Xmat2[,-1], y = tb2$Y )
coef( mod2 )
```

```
7 x 1 sparse Matrix of class "dgCMatrix"
      s1
(Intercept) 0.06167455
X1          0.86886193
X2          0.09107718
X3          0.00885584
```

```

X4          0.87535837
X5          .
X4_proxy    0.01155421

```

If we use random forests and a variable importance plot, we get this:

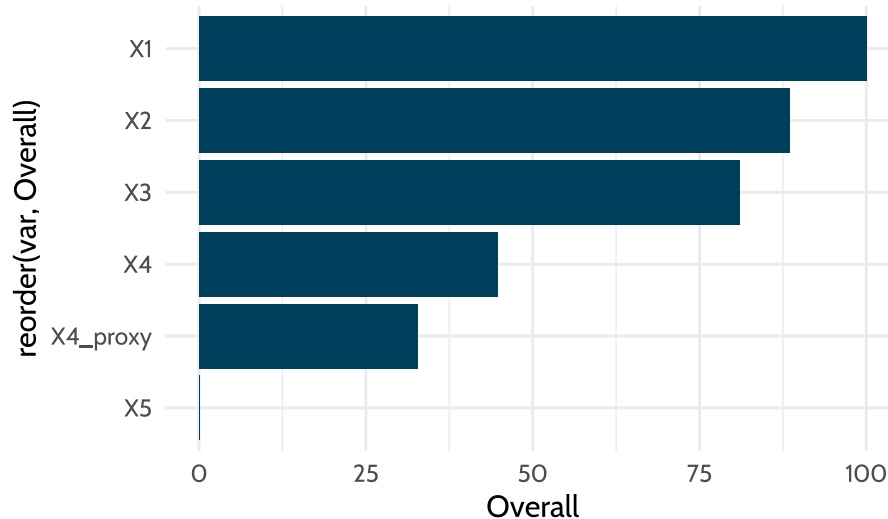
```

set.seed(2020)
library( caret )

rf_fit2_imp <- train( Y ~ .,
  data = tb,
  method = "ranger",
  importance = "permutation" )

varImp(rf_fit2_imp) %>%
  pluck(1) %>%
  rownames_to_column("var") %>%
  ggplot(aes(x = reorder(var, Overall), y = Overall)) +
  geom_col(fill = dataedu_colors("darkblue")) +
  coord_flip() +
  theme_dataedu()

```



The variable importance plot correctly puts X1, X2, and X3 all as about the same importance (they are: the larger variance perfectly balances out the smaller coefficient). The X4 variable gets dinged a bit because the X4_proxy variable can serve almost as well for prediction, so they both are deemed important.

62 Sensitivity check on trees

Say you want to present a tree in your final project to show how an outcome relates to different variables of interest. Trees, unfortunately, are notoriously unstable and thus you should see if you would get a very different tree with slight changes to your data.

Fortunately, it is easy to check this with the following:

Once you have your final tuning parameters and so forth set, repeat the following 5 times: * Bootstrap your data. * Fit a tree to the bootstrapped data, using the tuning parameters you selected. * Make a plot of your bootstrapped tree.

Now check to see if your five trees are basically similar, or entirely different, from each other. This is called a *stability check* and is a useful way of seeing if your results are particularly sensitive to mild changes in data.

63 A cool and easy way to do causal inference type stuff

Consider the problem of comparing Lyft vs Uber prices given a dataset of rides. You can't just compare the average price in each type because it is possible that, for example, Lyft rides are systematically shorter than Uber.

One direction to take is to fit two random forests, one for your Lyft data and one for your Uber data, to predict price given distance, time, weather, starting location, and anything else you have as a covariate. You then use each random forest to predict price for each observation in your entire dataset. You now have two predictions, one from your Lyft model and one from your Uber model, for each observation. Finally, you compare to see if the predictions are different on average. This gives you a “controlled comparison” where you are asking if there are systematic shifts in cost for otherwise similar things.

64 Interpreting LASSO models, and some other stuff

Back in Lab 10, or thereabouts, we walked through a LASSO model fit to student level data from the PISA exam. For our predictive task, we imagined that these students come from a single country, but in fact, we combined data from multiple nations. I sourced these data using the `learningtower` package.

65 The Story

You are newly hired analyst working in the State Ministry of Education. This morning, you open your email to find a request from your supervisor, your first project in the new job. How exciting!

Your boss writes that the Ministry has received funding to provide a one-on-one tutoring intervention for a modest number of struggling 8th grade math students. It's important that we select the right students for the intervention because we can't afford to give it to everyone. Unfortunately, we don't have previous math scores for students in the state. The first standardized math test they take is at the *end* of 8th grade.

Your boss proposes that you build a predictive model, using the other data the ministry has about students, to estimate who will perform poorly in math at the end of the 8th grade.

The only other requirement is that the model can be explained to policymakers and other higherups in the ministry. No black-box models.

Her email includes an attachment with the data on last year's 8th graders.

Our goal is to understand what is predictive of math scores. More specifically, we want to predict who the struggling students will be. There are a lot of things we could do. Remember whatever we do must be interpretable/explainable. Trees might work. Of course we'll settle on LASSO here. Generally speaking, LASSO is more interpretable than Ridge? (LASSO shrinks many predictors to zero, so we only have to explain the non-zero coefficients, which hopefully will be a small number.)

66 Fitting and interpreting LASSO

Here is a block of code that (without any actual exploration, etc.—please see the lab for that) fits the LASSO model.

```
stu_data <- read_csv('data/student_data.csv')
stu_data = stu_data[,-c(1:2)]
stu_data <- na.omit(stu_data)

x <- model.matrix(math ~ ., stu_data)[,-1]
y <- stu_data$math

set.seed(42120)

cross_validation_results_lasso <- cv.glmnet(x = x,
                                             y = y,
                                             alpha = 1, # 1 for LASSO, 0 for ridge
                                             nfolds = 10,
                                             type.measure = "mse")

print(cross_validation_results_lasso)
```

Call: `cv.glmnet(x = x, y = y, type.measure = "mse", nfolds = 10, alpha = 1)`

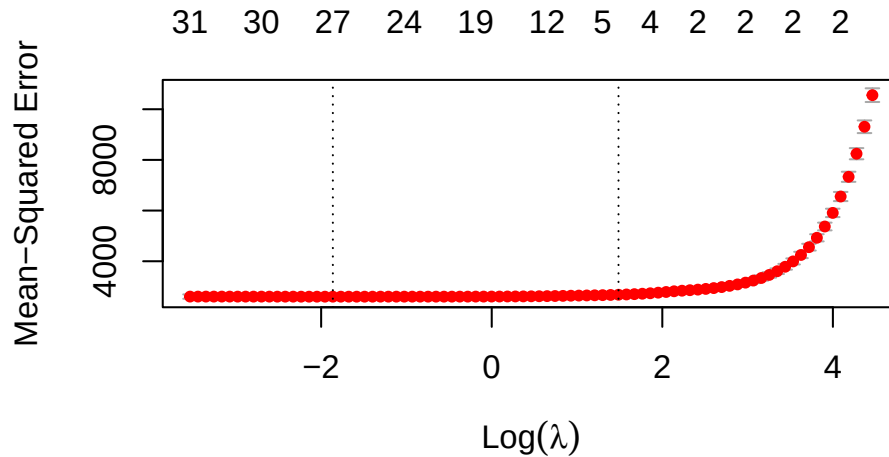
Measure: Mean-Squared Error

	Lambda	Index	Measure	SE	Nonzero
min	0.155	69	2601	85.10	27
1se	4.422	33	2678	87.96	5

In the above we loaded data, dropped anything with missing data (sad!) and fit LASSO, tuning with cross validation. We get an estimated Lambda of 4.422 using the 1SE rule.

We like plots, so we generate this one:

```
plot(cross_validation_results_lasso)
```



What does it tell us? Mainly, it says that small amounts of regularization are about the same as larger amounts—note the long flat part of the curve. This means that we don’t lose much if we regularize more (up to where it curves up). Regularizing more means fewer nonzero coefficients, and easier interpretation! If we regularize too much, however, our RMSE goes up quite fast.

But this plot is not particularly useful for our eventual consumers. They don’t want to know that you tuned your model; they want to know what your eventual model says! These plots, therefore, do not belong in a final report.

What we want is our final model:

```
best_lambda_lasso <- cross_validation_results_lasso$lambda.1se
final_model_lasso <- glmnet(x = x,
                             y = y,
                             alpha = 1,
                             lambda = best_lambda_lasso)

coef(final_model_lasso)
```

```
32 x 1 sparse Matrix of class "dgCMatrix"
              s0
(Intercept)  78.3051308
mother_educISCED 2      .
mother_educISCED 3A     .
mother_educISCED 3B, C  .
mother_educless than ISCED1 .
father_educISCED 2      .
father_educISCED 3A     .
father_educISCED 3B, C  .
```

father_educless than ISCED1	.
gendermale	9.0804121
computeryes	0.5051019
internetyes	.
read	0.3432561
science	0.4877308
deskyes	.
roomyes	.
television1	.
television2	.
television3+	.
computer_n1	.
computer_n2	.
computer_n3+	.
car1	.
car2	.
car3+	.
book101-200	.
book11-25	.
book201-500	.
book26-100	.
bookmore than 500	.
wealth	.
escs	3.0051742

We can print only the nonzero elements as so:

```
cc <- coef(final_model_lasso)
cc <- cc[ cc[,1] != 0, ]
cc
```

(Intercept)	gendermale	computeryes	read	science	escs
78.3051308	9.0804121	0.5051019	0.3432561	0.4877308	3.0051742

LASSO selected only a few of the many predictors available. In particular, the main predictors of math score are reading in science score, the student's economic, social and cultural status (ESCS), whether they have a computer, and whether they are assigned male in the data. Given this information, the Lasso model is saying that knowing how many televisions, cars, books, or computers are in the family is not useful for predicting math score. Neither is the mothers or fathers education level.

As an aside, it's generally hard to use categorical variables – continuous variables are easier to select because a single variable can describe a lot of variation. So this might be why these other variables are not being picked.

66.1 What variables to include?

We might not want to put all of our available variables into our model. For example, if we are trying to predict math score, we might not want to include other test scores from that same testing occasion, because we're looking for contextual factors that lead to a particular outcome. Thus, we might rerun everything we've done without including reading and science, which would force the model to use the contextual variables about the students rather than other measures of their academic ability.

66.2 Coefficient size

The SIZE of the coefficients is going to be too small, in general, because LASSO shrinks coefficients towards zero to stabilize. We should not, for example, expect that the increase in math for each unit increase in reading is 0.3432561. One trick some folks do is *refit* the selected coefficients using OLS to get a new, final model:

```
# Identify nonzero coefficients
coef_indices <- which( colnames(x) %in% names(cc) )
coef_indices
```

```
[1]  9 10 12 13 31
```

```
# Subset data to nonzero covariates
x_sub <- x[, coef_indices]
head( x_sub )
```

	gendermale	computeryes	read	science	escs
1	1	1	347.710	273.827	0.7018
2	0	1	427.958	460.357	0.2091
3	1	1	271.743	337.150	-1.1074
4	1	1	336.271	368.131	-0.4944
5	1	1	402.166	322.163	-0.1840
6	0	1	350.848	342.939	-0.6022

```
# Fit OLS model to nonzero covariates
fit_sub <- lm(y ~ ., data = as.data.frame(x_sub))
```

```
# Print summary of refit model
summary(fit_sub)
```

```

Call:
lm(formula = y ~ ., data = as.data.frame(x_sub))

Residuals:
    Min       1Q   Median       3Q      Max
-189.522  -32.514   -0.457   32.687  239.605

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  51.00114     5.30280   9.618 < 2e-16 ***
gendermale   19.31204     1.86074  10.379 < 2e-16 ***
computeryes  8.01858     2.71953   2.949 0.00322 **
read         0.38790     0.01883  20.602 < 2e-16 ***
science      0.47816     0.01944  24.596 < 2e-16 ***
escs         4.52125     0.97928   4.617 4.04e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 51.17 on 3309 degrees of freedom
Multiple R-squared:  0.7529,    Adjusted R-squared:  0.7525
F-statistic: 2016 on 5 and 3309 DF,  p-value: < 2.2e-16

```

We can compare how much shrinkage there was:

```

ols_coef = coef( fit_sub )
ctbl = tibble( coef = names(ols_coef),
               OLS = ols_coef,
               Lasso = cc )
ctbl

```

```

# A tibble: 6 x 3
  coef      OLS  Lasso
  <chr>    <dbl> <dbl>
1 (Intercept) 51.0  78.3
2 gendermale  19.3   9.08
3 computeryes  8.02  0.505
4 read        0.388 0.343
5 science     0.478 0.488
6 escs        4.52  3.01

```


66.3 Assessing stability (a kind of inference)

We might worry that the Lasso model is picking these variables only due to random chance. We could bootstrap this process to see if the same variables tend to get picked each time. Here is a single bootstrap iteration to illustrate:

```
datstar = slice_sample( stu_data, n = nrow(stu_data), replace=TRUE )
xstar <- model.matrix(math ~ ., datstar)[,-1]
ystar <- datstar$math

modstar <- glmnet(x = xstar,
                  y = ystar,
                  lambda = best_lambda_lasso,
                  alpha = 1 )

cc <- coef(modstar)
cc[ cc[,1] != 0, ]
```

(Intercept)	father_educless	than ISCED1
83.6514190		-2.6175776
gendermale		read
6.5947801		0.3550822
science		escs
0.4687787		3.3871490

We generally got the same answers, suggesting stability in what variables are selected. But we should do this a lot, and see how often each variable shows up. To do this, we need a function! And, for our function, life is always easier if we get a data frame back:

```
one_boot <- function( ) {
  datstar = slice_sample( stu_data, n = nrow(stu_data), replace=TRUE )
  xstar <- model.matrix(math ~ ., datstar)[,-1]
  ystar <- datstar$math

  modstar <- glmnet(x = xstar,
                    y = ystar,
                    lambda = best_lambda_lasso,
                    alpha = 1 )

  cc <- coef(modstar)
  cc <- cc[ cc[,1] != 0, ]
  cc = tibble( coef = names(cc), est = cc)
}
```

Then replicate:

```
rps = map_df( 1:100, ~ one_boot() )
rps %>% group_by( coef ) %>%
  summarise( n = n() ) %>%
  arrange( -n ) %>%
  knitr::kable()
```

coef	n
(Intercept)	100
gendermale	100
read	100
science	100
escs	99
computeryes	58
wealth	44
car2	26
internetyes	14
father__educless than ISCED1	13
deskyes	6
mother__educless than ISCED1	5
book201-500	2
computer__n3+	1
mother__educISCED 3A	1

The things at the top were *always* selected—they are clearly important. The things near the bottom are less so. Our initial “computer” variable seems somewhat important, but not as reliably selected as things such as reading and science.

66.4 Last words on writing up results

In writing up your results, consider the following elements:

1. Introduction: Begin by introducing the purpose of your analysis and the dataset you used. For example:

“We fit a Lasso regression model to predict the response variable [name of response variable] based on [list of predictor variables]. The dataset used was [name of dataset] and consisted of [number of observations] observations.”

2. Model Selection: Explain the process you used to select the best model, including the choice of the regularization parameter. For example:

“We used cross-validation to select the value of lambda that minimized the mean squared error. The selected lambda was [value].”

3. Model Performance: Report the performance of the Lasso model, including the model’s R-squared value, the root mean squared error (RMSE), and any other relevant metrics. For example:

“The Lasso model achieved an R-squared value of [R-squared value], indicating that [percentage of variation in response variable] of the variation in the response variable can be explained by the predictor variables. The RMSE of the model was [RMSE value], indicating an average error of [RMSE value] units in predicting the response variable.”

You can calculate R² by comparing the variance of the predictions to the variance of the original outcome:

$$R^2 = 1 - \frac{Var(predictions)}{Var(Y)}$$

4. Interpretation of Coefficients: Report the estimated coefficients. For example:

“Table XX shows the estimated coefficients for the predictor variables.”

You might include OLS values as well, on this table.

5. Conclusion: Summarize the key findings of your analysis and any insights gained from the Lasso model. For example:

“In conclusion, we found that [some variables] predicted our outcome, suggesting [something]. On the other hand, we expected [something else] to be selected, but it never was across the bootstrap samples, indicating that, given the other variables, there is no strong connection between [this thing] and the outcome.”

Remember to provide enough detail for the reader to understand your analysis, but keep your report concise and focused on the key points.

Part VI

Extra Stuff

67 Cleaning Excel files

How do you cleaning excel files? This “handout” (actually an R script) loads in a csv file made from a dirty excel file and cleans it. This is a data cleaning problem I have seen a few final project groups face. Download these files, look at the original csv file, and then source the script and see what you get!

Resources

- [The dataset](#)
- [The script](#)

Here is the code, for reference. But download the script!

```
#  
# This script walks through a common type of data prep process.  
#  
# If you have an excel file with merged cells and lots of tables in a big ugly  
# mass, then it can be hard to get those data "tidy". One way forward is to  
# first save the excel file as a CSV file, which breaks all the merged cells.  
# The content of the merged cell will usually show up in the leftmost cell,  
# which will give you a column of data that is a mix of table headers and table  
# content.  
#  
# At this point, you read the ugly table into R and start identifying which  
# tables start and stop where. You can then slice out the relevant rows, and  
# reformat those to get a final table that you could use.  
#  
# This script does the latter step, working with a sample csv file.  
  
library( tidyverse )  
dat = read_csv( "sample_ugly_data.csv", col_names = FALSE )  
  
names(dat)  
  
dat  
  
dat = filter( dat, !is.na( dat$X1 ) )
```

```

which( str_detect( dat$X3, "junk word" ) )
dat = filter( dat, is.na( X3 ) | !str_detect( X3, "junk word" ) )
dat

# Locate the rows that start a table, somehow.
#
table_rows = which( str_detect( dat$X1, "Table" ) )
table_rows
dat$X1[ table_rows ]

start = table_rows[[2]] + 1
stop = table_rows[[3]] - 1
subdat = dat[ start:stop, ]
subdat

# Note: You might also detect specific rows via targeted searching. E.g.,:
start_row = which( str_detect( dat$X1, "Alpha" ) )
start_row

# This will copy over the region names into a new "region" variable. We know a
# region name when the X2 data are missing (table header rows often do not have
# data in other cells in that row).
subdat = mutate( subdat,
                  region = ifelse( is.na( X2 ), X1, NA ) )
subdat

# This propagates the regions to all rows.
subdat = fill( subdat, region )
subdat

# Now we drop the region header rows since we have a new column of that variable!
subdat = filter( subdat, !is.na( X2 ) )
subdat

# We next start dropping empty columns. Any column with a lot of missing data
# will be considered junk. This counts the number of NAs in each column.
missings = apply( is.na( subdat ), 2, sum )
missings

# This keeps only those columns with fewer than 10 missing values.

```

```

subdat = subdat[ , missings < 10 ]
subdat

# We now name our columns by pulling off the first row that has the names. We
# have to put our hand-built rowname back in, and then we drop the row and set
# our column names!
names = as.character( subdat[ 1, ] )
names
names[ length(names) ] = "region"
names
names( subdat ) = names

subdat = subdat[ -1, ]
subdat

##### second table, just like above #####

# One might imagine making a function to do all this stuff...

start = table_rows[[3]] + 1
stop = nrow(dat)
subdat2 = dat[ start:stop, ]
subdat2

subdat2 = mutate( subdat2,
                   region = ifelse( is.na( X2 ), X1, NA ) )
subdat2 = fill( subdat2, region )
subdat2 = filter( subdat2, !is.na( X2 ) )
subdat2

missings = apply( is.na( subdat2 ), 2, sum )
missings

subdat2 = subdat2[ , missings < 10 ]
subdat2

names = as.character( subdat2[ 1, ] )
names
names[ length(names) ] = "region"
names

```

```

names( subdat2 ) = names

subdat2 = subdat2[ -1, ]
subdat2

#### Merge our two tables ####

# We merge by both region and location since we have duplicate location names
# within different regions.
final_dat = left_join( subdat, subdat2, by=c("region", "Location") )
final_dat

final_dat = relocate( final_dat, region )

# We finally convert our numeric columns to numbers.
final_dat = mutate( final_dat,
                    Grade = parse_number( Grade ),
                    Size = parse_number( Size ) )

final_dat

# Save your file and use the cleaned file for all your actual analysis!
write_csv( final_dat, file = "cleaned_data.csv" )

```


68 Main effects, interactions in linear models, and prediction

This document is designed to address a problem that I have seen several times in the final projects. In particular, people will fit a sequence of models, and noticed that the main effect changes massively when they include an interaction term. This document shows a possible solution for this “problem,” a solution where you fit a model, and predict two outcomes for each observation, one with a dummy variable of interest and one without. By then looking at the average difference between these predictions, we can recover what we should think of as a main effect.

68.1 Getting the data ready

We use a data set of a bunch of behavioral outcomes measured for how a mother interacts with a child. One question of interest is whether mothers systematically interact with girls differently than they do with boys.

We first load tidyverse and load the data.

```
qm = read.table( "data/KONG.txt" )
names( qm ) = c( "wealth", "age", "fed", "med", "bookread", "hwhelp", "talkchld", "female", "sibling" )
head( qm )
```

	wealth	age	fed	med	bookread	hwhelp	talkchld	female	sibling
1	11830	11.333330	9	9	2	2	2	0	1
2	8500	12.333330	2	5	2	2	2	1	1
3	20047	13.416670	9	8	2	2	2	1	1
4	18650	11.166670	0	12	2	2	2	1	1
5	6760	11.666670	5	0	1	2	1	0	1
6	4580	9.416667	8	8	2	2	2	1	1

```
nrow( qm )
```

```
[1] 1976
```

We make an aggregate measure of interaction by adding our three interview questions together. There are better ways of doing this, but this will work for now.

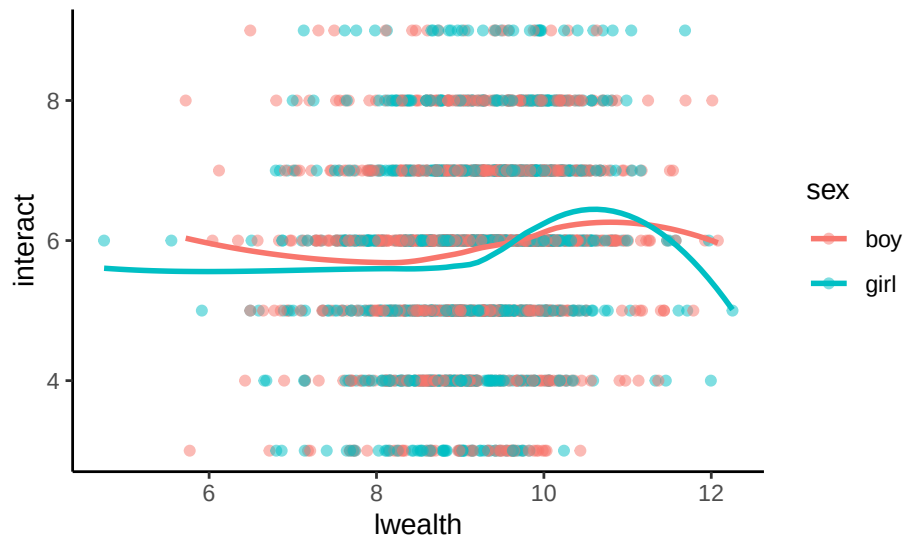
```
qm = mutate( qm, lwealth = log(wealth),
             interact = bookread + hwhelp + talkchld,
             sex = factor( female, levels=c(0,1), labels=c("boy","girl" ) ) )
```

To simplify things, we are going to take the middle 80% of the data based on wealth. (There is odd tail behavior that clouds the trends that I want to avoid for pedagogical purposes.)

```
qm = filter( qm, wealth >= quantile( wealth, 0.1 ), wealth <= quantile( wealth, 0.9 ) )
```

Lets see what we have:

```
ggplot( qm, aes( x=lwealth, y=interact, col=sex ) ) +
  geom_point( alpha=0.5 ) +
  stat_smooth( aes( group=sex ), method="loess", se=FALSE )
```



Lets look at a simple main effect of sex

```
M0 = lm( interact ~ sex + lwealth + sibling, data=qm )
summary( M0 )
```

Call:

```
lm(formula = interact ~ sex + lwealth + sibling, data = qm)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4280	-0.9432	0.0462	0.9583	3.7106

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.365853	0.311989	13.994	< 2e-16 ***
sexgirl	0.005984	0.062663	0.095	0.924
lwealth	0.197600	0.032674	6.048	1.75e-09 ***
sibling	-0.219954	0.043481	-5.059	4.62e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.367 on 1972 degrees of freedom

Multiple R-squared: 0.03351, Adjusted R-squared: 0.03203

F-statistic: 22.79 on 3 and 1972 DF, p-value: 1.679e-14

68.2 Fitting an interacted model

Now let's fit a model where we interact sex and wealth. The plot does not look particularly linear, but we will proceed for illustrative purposes.

```
M1 = lm( interact ~ sex * lwealth + sibling, data=qm )
summary( M1 )
```

Call:

```
lm(formula = interact ~ sex * lwealth + sibling, data = qm)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.3704	-0.9534	0.0387	0.9547	3.7785

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.80629	0.41588	11.557	< 2e-16 ***
sexgirl	-0.95389	0.60286	-1.582	0.113750
lwealth	0.14987	0.04422	3.389	0.000715 ***
sibling	-0.22096	0.04347	-5.083	4.06e-07 ***
sexgirl:lwealth	0.10468	0.06539	1.601	0.109569

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.366 on 1971 degrees of freedom
```

```
Multiple R-squared:  0.03476,    Adjusted R-squared:  0.0328
```

```
F-statistic: 17.74 on 4 and 1971 DF,  p-value: 2.543e-14
```

Notice the coefficient for sex changes a lot. This is because this is the estimated difference of girls and boys *for those with a lwealth of 0*.

To recover our “main effect” estimate we need to see what the average predicted difference would be for all the individuals in our dataset. We do this by doing the following steps:

- (1) Copy our dataset to make two new ones

```
qm.g = qm.b = qm
```

- (2) Now make all the folks in the `qm.g` dataset girls, and all folks in the other boys.

```
qm.g$sex = "girl"
```

```
qm.b$sex = "boy"
```

- (3) Predict interaction assuming *everyone* is a girl and *everyone* is a boy:

```
pred.g = predict( M1, qm.g )
```

```
pred.b = predict( M1, qm.b )
```

- (4) Then the difference of the predictions is our predicted difference in interaction for two kids with the same covariates except sex:

```
deltas = pred.g - pred.b
```

```
sex.diff = mean( deltas )
```

```
sex.diff
```

```
[1] 0.006737327
```

See? Now it matches more closely with the main effect estimate of `M0`. The core idea here is that the real interpretation of the main effect is a difference in average outcomes (when looking at a dummy variable) between two groups, when holding all else equal. When you have interactions, the main effect is the predicted difference *for those with zeros for the interaction terms*, which might not be very sensible. We can get back to the difference between the two groups by predicting for everyone and comparing the averages.

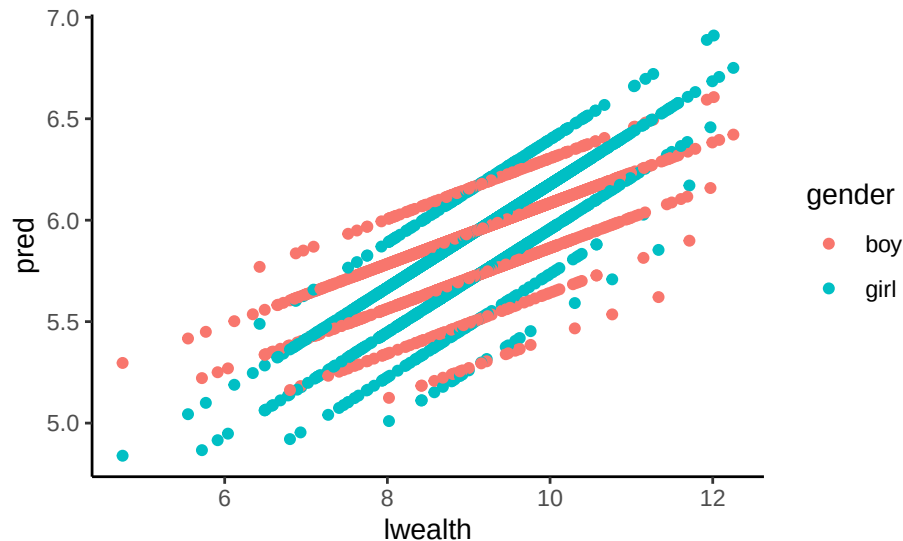
68.3 Plotting

As a reminder, we can plot with our predictions in a nice way. The cleanest would be to put all our predictions into a dataframe, convert to long form, and plot.

(Note we use 'gender' as our key, in the following, to avoid colliding with the 'sex' variable name.)

```
qm$girl = pred.g
qm$boy = pred.b
qml = qm %>%
  pivot_longer( cols = c(girl, boy),
                names_to = "gender",
                values_to = "pred" )

ggplot( qml, aes( x=lwealth, y= pred, col=gender ) ) +
  geom_point()
```



We see different stripes for the different numbers of siblings.

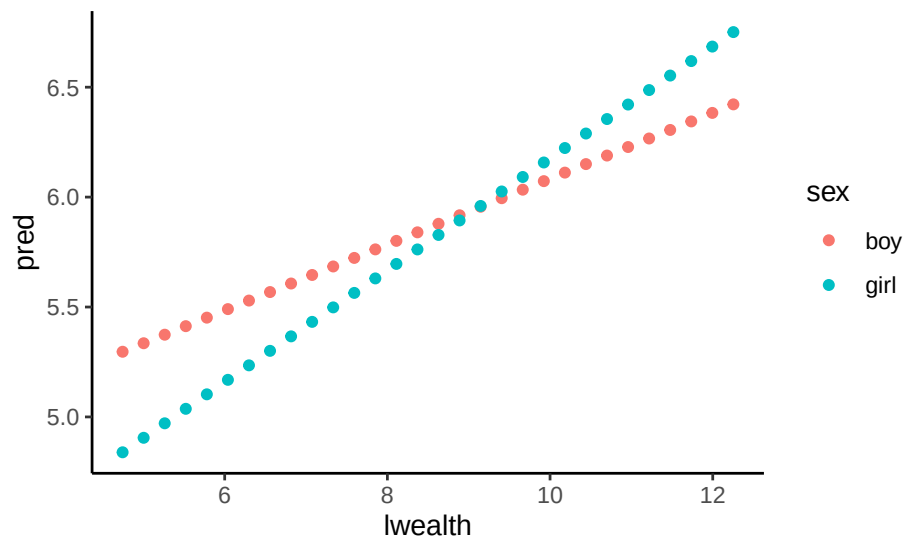
As a better way, we can get fancier with the data grid stuff.

```
library( modelr )
grid = data_grid( qm, lwealth = seq_range( lwealth, 30 ), sex, .model = M1 )
head( grid )
```

```
# A tibble: 6 x 3
  lwealth sex  sibling
  <dbl> <fct>   <dbl>
1  4.74 boy     1
2  4.74 girl    1
3  5.00 boy     1
4  5.00 girl    1
5  5.26 boy     1
6  5.26 girl    1

grid = add_predictions( grid, M1, "pred" )

ggplot( grid, aes( x=lwealth, y= pred, col=sex ) ) +
  geom_point()
```



I used `geom_point()` to show the individual predictions. Normally we would use `geom_line()`

68.4 Centering, an alternative approach

You can also center your continuous variable, which means your coefficient for your main effect on your dummy variable will correspond to the average value of the continuous. Much more interpretable!

```
qm = mutate( qm, lwealth.cent = lwealth - mean(lwealth) )
M1b = lm( interact ~ sex * lwealth.cent + sibling, data=qm )
```

```
coef( M0 )
```

```
(Intercept)      sexgirl      lwealth      sibling
4.365852906  0.005984015  0.197600022 -0.219953588
```

```
coef( M1b )
```

```
(Intercept)      sexgirl      lwealth.cent
6.181712978      0.006737327      0.149873199
sibling sexgirl:lwealth.cent
-0.220960207      0.104675040
```

See? Much more interpretable!

68.5 Confidence intervals for the gap?

A bootstrap is the easiest here.

```
reps = replicate( 1000, {
  qm.star = mosaic::sample( qm, replace=TRUE )

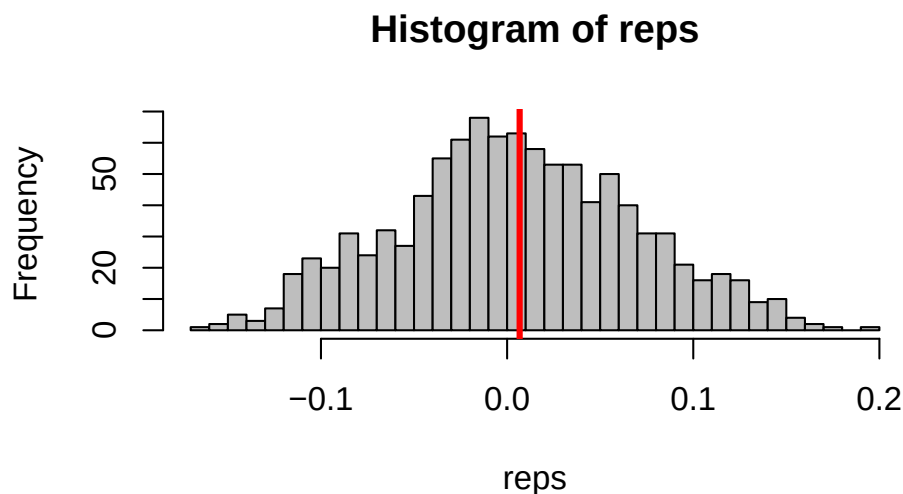
  # refit the original model with the bootstrap data
  M1.star = update( M1, data=qm.star )

  pred.g = predict( M1.star, qm.g )
  pred.b = predict( M1.star, qm.b )

  deltas = pred.g - pred.b
  mean( deltas )
} )
```

Our bootstrap confidence interval: quantile(reps, c(0.025, 0.975))

```
hist( reps, breaks=30, col="grey" )
abline( v=sex.diff, col="red", lwd=3 )
```



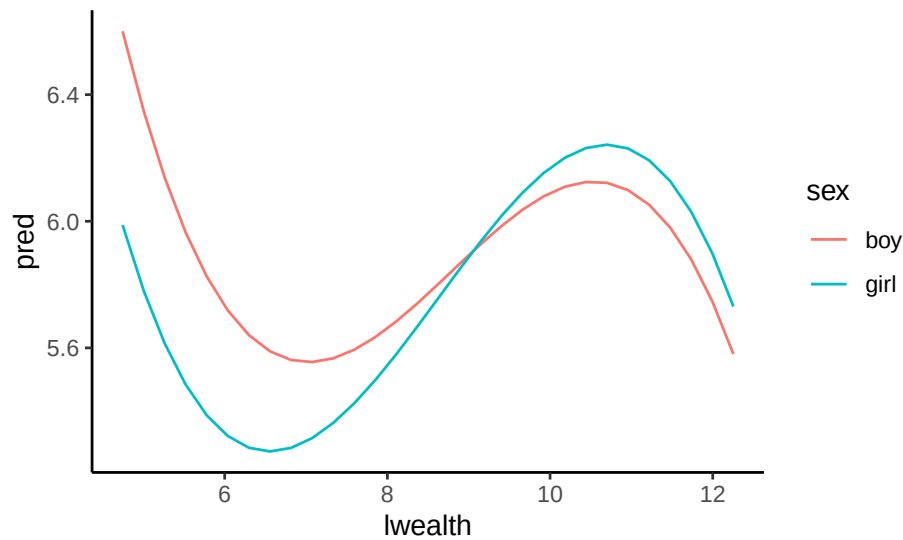
68.6 A final investigation

I don't like the linear relationship. Better to have, perhaps a cubic for both girl and boy?

```
M3 = lm( interact ~ ( lwealth + I((lwealth-mean(lwealth))^2) +
                    I((lwealth-mean(lwealth))^3) ) * sex + sibling,
        data=qm )
coef( M3 )
```

(Intercept)	lwealth
3.9677413630	0.2376064190
I((lwealth - mean(lwealth))^2)	I((lwealth - mean(lwealth))^3)
0.0226463368	-0.0264912833
sexgirl	sibling
-1.0082023177	-0.2159902502
lwealth:sexgirl	I((lwealth - mean(lwealth))^2):sexgirl
0.1117568699	-0.0122453285
I((lwealth - mean(lwealth))^3):sexgirl	
-0.0007258967	

```
grid = add_predictions( grid, M3, "pred" )
ggplot( grid, aes( x=lwealth, y= pred, col=sex ) ) +
  geom_line()
```

In truth, splines would be the best. A topic for another day (or prior handout).

68.7 Disclaimer

The coefficients of the above are not significant, and we are not finding a real difference between girls and boys in this case. But the code is designed to illustrate the core concept of using predict to get access to a real estimate of a main effect when you are fitting a model with an interaction term.

To check if a complex model is an improvement, use `anova()`

```
anova( M0, M1, M3 )
```

Analysis of Variance Table

Model 1: `interact ~ sex + lwealth + sibling`

Model 2: `interact ~ sex * lwealth + sibling`

Model 3: `interact ~ (lwealth + I((lwealth - mean(lwealth))^2) + I((lwealth - mean(lwealth))^3)) * sex + sibling`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	1972	3684.0				
2	1971	3679.2	1	4.7838	2.5691	0.10913
3	1967	3662.7	4	16.5032	2.2157	0.06503 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Nope, the interactions (or the more flexible cubic) are not helping us (other than helping our understanding of how to model interactions).

69 Finding and Merging Data Online

An Example with IPEDS Data

69.1 Datasets Posted on the Web

Downloading and merging publicly posted datasets is a very important skill. This handout walks through an example case of downloading datasets posted on a government website, cleaning them, merging them together, and using the merged dataset for an analysis. It addresses some of the common challenges of getting acquainted with a data portal.

No guide can cover all cases, but this example should give you a sense of what to look for/think about when wrangling data posted on websites. Note that this is *not about scraping*. What we're considering here are datasets posted for download, often - but not always - found on government sites.

69.2 IPEDS

This example uses data from IPEDS, the Integrated Postsecondary Education Data System. These data are made available by NCES, the National Center for Education Statistics, a great place to start looking for education-related data in the US. The data represent a set of surveys, conducted every year, of all colleges, universities, and other post-secondary institutions that take federal student aid money (which is like, many of them). Data are available going back to 1986 (some scattered earlier years also have data). The surveys cover enrollments, graduation rates, prices, and many other things.

There are 12 total survey components that cover 9 major topics:

- Academic Libraries
- Admissions
- Completions

- Enrollment (Fall and 12-Month)
- Finance
- Graduation Rates and Outcome Measures
- Human Resources
- Institutional Characteristics
- Student Financial Aid

It's worth noting that some info is only collected every other year. In addition, the data are not consistent over time. New survey items get added. Old items get dropped. Definitions change. Identifying codes, like the Classification of Instructional Program (CIP) codes or the Standard Occupational Classification (SOC), are periodically reviewed and updated. Being aware of (and dealing with) these issues is part of our job as data scientists. When you're working with new data, take some time to identify these kinds of challenges before you dig too deeply into analysis.

69.3 Finding IPEDS Data

The landing page for IPEDS is [here](#). When you open the page, you'll see many links. NCES provides a lot of information about their data (and different ways to access it). Sorting through surfeits of info like this is often the biggest challenge of working with public data sources. My description of IPEDS (in the above section) came from digging around these links.¹

On the IPEDS page, there's a link to their [data explorer](#). When you hit this page, you'll see links to the most recently released surveys. You can filter the datasets by year or by which of the 12 survey components you're interested in. These data are aggregated, but it's a good way to get a sense of what information is available in the individual survey components.

To *get* the survey data, look for this section of the landing page:

¹Along the way, I encountered some broken pages, like [this link to the IPEDS data release calendar](#). Maybe they'll fix it, but you can often find the information you need on some other page. When that doesn't work, ask someone who's familiar with the data. When *that* doesn't work, you can email someone who works for the agency responsible for the data. In my experience, people get back to you reasonably quickly and are happy to be helpful, as long as you write politely to them.



Survey Data

Data are available starting with the 1980-81 collection year for the *Complete data files* and *Custom data files* functions, which zip the data into comma separated value (*csv). Beginning with the 2004-05 collection year, data for each collection year are compiled into an *Access database*.

Select download option

In the dropdown menu, select “Complete Data Files”. This will take you to a page with download links for each survey component. The files we want are found in the “Data File” column. Let’s download this admissions and test scores file,

2021	Admissions and Test Scores	Admission considerations, applications, admissions, enrollees and test scores, fall 2021	ADM2021
------	----------------------------	--	-------------------------

When we download the file, we should put it in a folder on our computer that’s dedicated to this project. That’ll make things easier once we start writing code. On my laptop, I’ve made a folder called “ipeds_handout”. Inside *that* folder, I made another folder called “data”. I put the downloaded file into that “data” folder. When I start writing code, I’ll store it in the parent folder (“ipeds_handout”).

The downloaded file is a zipped csv. We know how to work with a csv in R, but before we can do that, we have to unzip the file. If you haven’t done this before, [here’s a page](#) that walks through how to unzip files on PC or Mac. Once you have the unzipped csv, you can delete the original zip file to tidy things up.

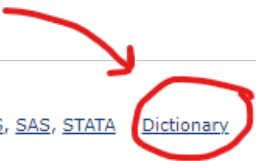
While we’re at it, let’s grab one more file, data on instructional staff salaries,

2021	Instructional Staff/Salaries	Number and salary outlays for full-time nonmedical instructional staff, by gender, and academic rank: Academic year 2021-22	SAL2021 IS
------	------------------------------	---	----------------------------

Like before, unzip the file and put the csv in a project-specific folder.² You should also download and unzip the dictionary files for both datasets.

²To be clear, this should be the same folder as the first dataset. Also, if you’re working with many datasets for a project, it’s a good idea to rename them at this stage. Give the datasets clear, descriptive names so you know what they contain. This will make it easier to write and read code later.

2021	Admissions and Test Scores	Admission considerations, applications, admissions, enrollees and test scores, fall 2021	ADM2021	ADM2021 STATA	SPSS, SAS, STATA	Dictionary
------	----------------------------	--	-------------------------	-------------------------------	----------------------------------	----------------------------



The dictionaries contain codebooks that explain what each variable and value in the dataset means. These usually aren't big files, so we can open and browse them directly in Excel. I'm going to use them soon to figure out how to find the information I'm interested in.

Now we're ready to get started.

69.4 Our Example Case

Let's start by loading our libraries and data.

```
library(tidyverse)
library(ggplot2)
```

```
admit <- read_csv('data/adm2021.csv')
```

```
Rows: 1981 Columns: 68
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (29): XAPPLCN, XAPPLCNM, XAPPLCNW, XADMSSN, XADMSSNM, XADMSSNW, XENRLT, ...
```

```
dbl (39): UNITID, ADMCON1, ADMCON2, ADMCON3, ADMCON4, ADMCON5, ADMCON6, ADMC...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
sal <- read_csv('data/sal2021_is.csv')
```

```
Rows: 15382 Columns: 110
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (54): XSAINSTT, XSAINSTM, XSAINSTW, XSA_9MCT, XSA_9MCM, XSA_9MCW, XSATOT...
```

```
dbl (56): UNITID, ARANK, SAINSTT, SAINSTM, SAINSTW, SA_9MCT, SA_9MCM, SA_9MC...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

With the datasets we've downloaded, we can explore the relationship between assistant professor salaries and the math ability of incoming Freshmen. To begin, we narrow down each dataset to the variables of interest. To identify these, I looked at the dictionaries we downloaded from IPEDS.

```
# Keep only school IDs and average salary for 9-month
# Contract instructors
sal <- sal %>% filter(ARANK==3) %>%
  # The dictionary tells us that these are assistant profs
  select(UNITID, SA09MAT)

# Keep school IDs, # of applicants, % that submit SATs,
# and 75th pctile of SAT scores
admit <- admit %>% select(UNITID, APPLCN, SATPCT, SATMT75)
```

In other cases, you might have to do much more cleaning before you merge these datasets together. In general, the process will be to clean each dataset so they can be easily merged. (Of course, more data cleaning may be necessary, post-merge.)

69.4.1 Merging the Admissions and Salary Data

It's a good idea to do a full join. This will keep all rows from both datasets and allow us to explore which rows did and did not match, post-merge. That said, I often start with a left or right join just as a quick check of how many records match.

```
df <- left_join(sal, admit, by='UNITID')
sum(is.na(df$SATPCT))
```

```
[1] 1298
```

About half (around 1,300 institutions) don't have a match. This should be explored. Maybe it's from particular types of institutions (e.g. Vocational/Technical Schools). We might have to merge in additional data to explore the matching behavior. We might figure it out by reading carefully through the dictionaries or other documentation on IPEDS. You can take these steps by following the framework laid out above.

For now, we can run a preliminary regression and call it a day.

```
# First let's mean-center both variables
df <- df %>% mutate(across(c(SA09MAT, SATMT75), ~.x-mean(.x, na.rm=TRUE)))

summary(lm(SA09MAT ~ SATMT75, df))
```

```

Call:
lm(formula = SA09MAT ~ SATMT75, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-49118  -8410   -889    8346   52056

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2829.703    459.431   6.159 1.18e-09 ***
SATMT75      164.646      6.186  26.616 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12740 on 771 degrees of freedom
(1529 observations deleted due to missingness)
Multiple R-squared:  0.4788,    Adjusted R-squared:  0.4782
F-statistic: 708.4 on 1 and 771 DF,  p-value: < 2.2e-16

```

It looks like assistant professors make more at schools with higher SAT math scores (as measured by the 75th percentile of the distribution). Assistant profs at schools with 100 points higher scores earn about 10k more on average. There are lots of reasons we shouldn't put much weight on this result. Lots of rows didn't merge. We haven't investigated missingness in the data. But the above steps walk through the process of a first, rough-cut analysis with IPEDS data.

70 Applied Prediction Papers

Below is a list of papers related to predictive data science and machine learning in public policy (including education). These are organized into: (1) surveys and reviews and (2) original applications. You may find many of these papers helpful as you are working on your final project and interpreting results - please peruse them with this use case in mind as well as for your general understanding of how these methods are used in the real world.

71 *Surveys and reviews*

The following list includes review articles, textbooks, and textbook chapters that provide an overview of the application of machine learning in various fields related to public policy. These fields include education, public health, urban planning, economics, and urban planning, among many others. Many of these works discuss or cite specific applications of these methods within these fields as well (including some of the references that follow after this section).

71.0.0.1 Athey, Susan. “Beyond Prediction.” *Science* (American Association for the Advancement of Science), vol. 355, no. 6324, 2017, pp. 483–85, <https://doi.org/10.1126/science.aal4321>.

Discussion of the challenges of using machine learning predictions to improve public policies. Includes many references to real-world uses of predictive data science.

71.0.0.2 Athey, Susan. The Impact of Machine Learning on Economics. In: *The Economics of Artificial Intelligence: An Agenda* [Internet]. University of Chicago Press; 2018 [cited 2023 Mar 31]. p. 507–47. Available from: <https://www.nber.org/books-and-chapters/economics-artificial-intelligence-agenda/impact-machine-learning-economics>

This chapter discusses use of machine learning in economics and includes a discussion of “prediction policy” applications for informing economic decision-making.

71.0.0.3 Casali, Ylenia, et al. Machine learning for spatial analyses in urban areas: a scoping review. *Sustainable Cities and Society*. 2022 Oct 1;85:104050.

Authors review the use of machine learning in city/urban planning. Specific attention is given to machine learning applications involving spatial data.

71.0.0.4 Hu, Xindi C., et al. “The Utility of Machine Learning Models for Predicting Chemical Contaminants in Drinking Water: Promise, Challenges, and Opportunities.” *Current Environmental Health Reports*. 2023 Mar;10(1):45–60.

Review of the use and challenges of machine learning models for predicting chemical contaminants in drinking water. Also discusses their frequent use to guide sampling efforts by prioritizing at-risk areas.

71.0.0.5 Payedimarri, Anil Babu, et al. Prediction Models for Public Health Containment Measures on COVID-19 Using Artificial Intelligence and Machine Learning: A Systematic Review. *International Journal of Environmental Research and Public Health*. 2021 Jan;18(9):4499.

A brief review of the use of machine learning (and artificial intelligence) methods to evaluate public health interventions to contain the spread of SARS-CoV-2.

71.0.0.6 Perry, Walt L. 2013. Predictive Policing: The Role of Crime Forecasting in Law Enforcement Operations. Rand Corporation.

Open access book gives an overview of common predictive policing practices.

71.0.0.7 Williamson, Ben. 2016. “Digital education governance: data visualization, predictive analytics, and ‘real-time’ policy instruments.” *Journal of Education Policy* 31(2):123-141.

Broad overview of uses of modern education data. Includes a section on predictive analytics. See first full paragraph on p. 136.

71.0.0.8 Baker, R. S., Martin, T., & Rossi, L. M. (2016). Educational Data Mining and Learning Analytics. In *The Wiley Handbook of Cognition and Assessment* (pp. 379–396). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9781118956588.ch16>

This chapter written by experts in the field provides a nice overview of educational data mining and learning analytics. Data obtained through naturally occurring log data (e.g., from learning management systems) or specifically procured sources (e.g., eyetracking) can be used to both make inferences and predictions on learning behavior and outcomes.

72 *Original applications*

The following list includes papers that apply machine learning methods to specific research questions. All are related to prediction and public policy, but span a similarly wide range of disciplines. While reading these, focus on the descriptions of the methods and results. This may be useful for your final project (and, of course, more generally!).

72.0.0.1 Bansak, Kirk, et al. “Improving Refugee Integration through Data-Driven Algorithmic Assignment.” *Science* (American Association for the Advancement of Science), vol. 359, no. 6373, 2018, pp. 325–29.

In this paper, the authors propose a model that aims to predict where refugees will integrate best. They suggest governments take up their approach to make their refugee programs more efficient.

72.0.0.2 Goel S, Rao JM, Shroff R. Precinct or prejudice? Understanding racial disparities in New York City’s stop-and-frisk policy. *The Annals of Applied Statistics*. 2016 Mar;10(1):365–94.

The authors of this paper used machine learning to estimate the probability that a detained individual truly has a weapon for stops related to suspicion of criminal weapon possession in New York City. Disproportionate stops by racial/ethnic groups, and the factors resulting in these disparities, are also discussed.

72.0.0.3 Hino, M, et al. Machine learning for environmental monitoring. *Nature Sustainability*. 2018 Oct;1(10):583–8.

Demonstration of how machine learning methods can help allocate resources to more efficiently conduct inspections for violations of the Clean Water Act.

72.0.0.4 Kelly, Sean, et al. “Automatically Measuring Question Authenticity in Real-World Classrooms.” *Educational Researcher*, vol. 47, no. 7, 2018, pp. 451–64, Available from: <https://doi.org/10.3102/0013189X18785613>.

This team uses regression trees to predict which teacher questions are “authentic”.

72.0.0.5 Lee Kwang-Sig, et al. Association of Preterm Birth with Depression and Particulate Matter: Machine Learning Analysis Using National Health Insurance Data. *Diagnostics*. 2021 Mar;11(3):555.

Demonstrates a use of machine learning to predict and identify the majors determinants of preterm birth in South Korea. Authors discuss that strategies to reduce air pollution could be an effective intervention based on these findings.

72.0.0.6 Yoo, Sanglim. Investigating important urban characteristics in the formation of urban heat islands: a machine learning approach. *Journal of Big Data*. 2018 Jan 24;5(1):2.

The author presents and discusses use of random forest to predict the formation of urban heat islands in Indianapolis, Indiana.

72.0.0.7 <https://chicago.github.io/food-inspections-evaluation/>

The city of Chicago uses a predictive model to decide which food establishments are inspected first.

72.0.0.8 Romero, C., López, M. I., Luna, J. M., & Ventura, S. (2013). Predicting students' final performance from participation in on-line discussion forums. *Computers & Education*, 68, 458-472.

The researchers used classification and clustering to predict student performance based on a collection of features from an online discussion forum. This paper is a good example of the complicated data cleaning and feature pre-processing usually required for learning analytics (LA) work. It also demonstrates the various models that can be used in the process; plus, the paper actually explains what the models are and how they work before going into the results. Overall, it is a good entry paper into the field.

72.0.0.9 Bozick, Robert and Dalton, Benjamin. Balancing Career and Technical Education With Academic Coursework: The Consequences for Mathematics Achievement in High School. *Educational Evaluation and Policy Analysis*. 2013 Jun 1;35(2):123-38.

These researchers assessed the ability of career and technical education courses to improve student learning. Their work provides an example of the use of bootstrapping methods to calculate standard errors of regression coefficients.